Changes in Employment and Wages in the United States, 1980–2019: Skills, Occupations, and Technology *

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Abstract

Changes in U.S. employment and wages since 1980 pose a joint puzzle: routine employment contracted, but relative wage growth was strongest among high-skill workers and stronger still among managers. Unlike other high-paying worker groups, managers experienced rising relative wages alongside a declining employment share after 2000. We construct a tractable model of skills, occupations, and sectors in which individuals differ in worker and managerial skills, and an additive occupational requirement governs comparative advantage among workers. The equilibrium connects worker assignment, wage differences among workers, managerial earnings relative to workers, and sectoral reallocation. Under explicit sufficient conditions, we derive analytical comparative statics for these margins. In the 1980–2019 quantitative exercise, task-productivity changes account for much of employment reallocation but produce too little high-skill and managerial wage growth and too much managerial employment growth. Changes in requirements and skill supply help reconcile employment and wages. Occupational task characteristics support the distinction between productivity and effective-skill changes. An illustrative AI scenario shows that abstract relative-wage growth can be higher even when abstract employment grows more slowly than under a continuation benchmark.

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Job polarization, structural change, and rising wage inequality are well-documented features of the U.S. economy since 1980. But their relationship is less straightforward than the long-run trends suggest. Low-skill jobs expanded without comparable relative wage gains; high-skill workers gained relative to other workers, and managers' relative wages continued to rise after their employment share began to fall. The fact that employment and wages trace different paths does not establish that distinct forces drove them; one change could shape both. The empirical question is which measured channels can account for the joint pattern. We develop a framework in which task-specific productivity explains much of employment reallocation, while changing occupational requirements alter skill-task complementarity and help account for the wage pattern. An additive task requirement provides a tractable specification of this comparative advantage for our requirement comparisons and quantitative model. We call changes that favor higher-skill workers in relative terms upskilling. O*NET task measures provide descriptive support for this mechanism, though they do not establish it as a unique cause.

The two ends of the occupational distribution expose different margins of the same equilibrium. At the bottom, employment can expand without relative-wage gains when the task does not benefit from upskilling. In our quantitative benchmark, manual services use a fixed amount of effective skill per worker: more workers can perform these tasks, but additional worker skill does not raise their productivity. Their wages respond to equilibrium prices, while skill-sensitive occupations can gain from upskilling. At the top, managerial earnings depend on a different selection margin. Individuals enter management according to their managerial skill z relative to the earnings they could obtain as workers with skill h. Managerial mean earnings therefore depend on both the price of managerial services and the selected mean of z, not simply on the number of managers or their position in the worker-skill ranking. This permits managerial relative wages to rise without employment expansion. The quantitative restriction on manual skill exposure and the separate managerial selection margin let us study both patterns without requiring them to arise from independent shocks.

The distinction between managers and other high-paying occupations is central to this question. Professionals and management-support workers gained both employment share and wages relative to manual workers. Managers instead experienced continued relative-wage growth after their employment share peaked: between 2000 and 2019 their share fell from 14.0 to 13.0 percent, while their mean wage relative to manual workers rose from 2.51 to 3.31. Over the same period, professionals' employment share rose from 14.5 to 18.2 percent and their relative wage from 1.96 to 2.34. Managers' employment share did rise over 1980–2019 as a whole; the contrast is particularly clear after 2000 (Table 1). We therefore distinguish the return to high worker skill from the scarcity and selection of managerial skill, rather than treating management as the top worker task.

In our model, individuals are heterogeneous in two dimensions of skill—managerial skill and worker skill—based on which they choose to become a manager or a worker. Workers then select into occupations that differ in how much of their skill can be used in production. Managers combine occupation-level outputs to produce sectoral goods, and sectors differ in how intensively they use different occupations. The model separates task-specific productivity from occupational skill requirements and the supply of skills. Productivity determines output for given inputs, while requirements govern comparative advantage and the return to worker skill. Managerial selection simultaneously determines the supply of management, allowing us to study the wage premium of managers alongside wage differences among workers and occupational employment. Despite these different layers, the model remains tractable enough to characterize equilibrium, including unique relative service prices, and to derive analytical comparative statics.

A simple modelling device gives the requirement channel its economic interpretation. Subtracting an occupational requirement $\chi(j)$ from worker skill makes higher-skill workers relatively more productive in more demanding tasks, generating log-supermodularity. Changes in this requirement profile therefore

affect both occupational sorting and wage differences among workers. We refer to changes that favor higher-skill workers in relative terms as *upskilling*. Whether a particular change has this effect depends on how requirements change across occupations and on the equilibrium reassignment of workers; we give explicit sufficient conditions below.

Our theoretical analysis shows how these forces can have different implications for employment and wages. Under the stated local conditions, faster productivity growth among middle-skill tasks polarizes worker assignment and wage growth, and raises fixed-type managerial earnings relative to workers. A class of requirement changes instead raises the relative wages of higher-skill workers, including the earnings of an upper interval of worker skills relative to fixed managerial types. Changes in relative managerial productivity or the supply of managerial skills provide a further force by changing the scarcity of management. Under additional skill-dependence conditions, this force also affects wage differences among workers. These effects need not have the same sign, and occupational mean wages also depend on who enters and leaves each occupation. The comparative statics distinguish the separate mechanisms; the quantitative exercise determines their combined magnitudes.

Armed with these results, we calibrate the model to the U.S. economy in 1980 and 2019. Our first quantitative result is that task-productivity changes account for much of the reallocation of employment. Manufacturing uses routine tasks more intensively than services, so sector-neutral productivity changes can also generate its declining employment share without exogenous sector-specific productivity growth. Second, unlike employment changes, wage changes are poorly explained by productivity growth alone: the model produces too much manual wage growth, too little abstract and managerial wage growth, and too much managerial employment growth. Changes in requirements and the skill distribution help reconcile these outcomes in the full model. Since wages and employment both discipline the calibration, the contribution of each force is assessed through the counterfactuals rather than through the wage fit alone.

We then relate the recovered changes to occupational task characteristics. Task-productivity growth is associated with routine-manual content and employment contraction, while changes in effective skill relative to mean skill are associated with nonroutine analytical content, educational attainment, and wage growth. These relationships provide an empirical basis for distinguishing changes in the productivity of work from changes in how skills are used, beyond the employment and wage moments targeted in the calibration.

The distinction is especially relevant for artificial intelligence, whose occupational incidence need not resemble routine-task changes in the past. The 2010–2019 evidence associates the AI exposure measure of Felten et al. (2019) with wage growth after controlling for the preceding wage change, but finds no statistically significant employment-share association; other measures give different results. In two illustrative scenarios, using the broad-AI measure of Felten et al. (2019) and the task-level LLM exposure measure of Eloundou et al. (2024), abstract employment grows more slowly than in the continuation benchmark, while abstract wages relative to the economy-wide mean grow faster. The LLM measure gives smaller abstract effects. Relative to the same continuation benchmark, it gives lower routine employment and higher relative wages, whereas the broad-AI measure gives higher routine employment and lower relative wages. The regressions are descriptive and the scenarios are not forecasts, but both illustrate why employment exposure alone gives an incomplete account of a technology's distributional consequences. Robot taxes, manufacturing subsidies, and a short work-from-home illustration examine other changes in task demand.

Related literature Our model combines the assignment of heterogeneous skills to tasks in Costinot and Vogel (2010) with the occupational and sectoral structure of Stokey (2016). We extend this framework by distinguishing task productivity from occupational requirements and by introducing a separate managerial skill. The additive requirement generates comparative advantage across worker tasks, while selection

between management and worker occupations connects changes in skill prices to occupational mean earnings. Together, these features allow us to study employment and wage changes arising from the same technological forces.

Alonzo and Gallipoli (2026) also studies occupational employment and earnings jointly, emphasizing how latent non-wage job values shift labor supply and affect equilibrium wages. In our model, labor supply across occupations changes through productive skills, occupational requirements, and selection between management and worker tasks. We relate these changes to task content and educational attainment. The two frameworks thus offer different economic explanations for the joint movement of wages and employment.

The managerial technology extends the span-of-control setting of Lucas (1978). Managers choose the composition of worker tasks and the effective skills used in them, with a CES relationship between managerial and worker inputs. This separates the managerial wage premium from wage differences among workers. Goos et al. (2014) studies task-specific capital costs and employment polarization; our equilibrium adds the wage, selection, and aggregate-output responses that also discipline the interpretation of task-productivity changes.

Occupation-specific technological change also generates structural transformation in Duernecker and Herrendorf (2022). Related accounts connect skill demand, polarization and structural change, including Buera and Kaboski (2012), Buera et al. (2015), Bárány and Siegel (2018), and Bárány and Siegel (2020). Our sectoral productivity differences arise through occupational choices and task composition, connecting production-driven structural change as in Ngai and Pissarides (2007) and Acemoglu and Guerrieri (2008) to the changing rewards to worker and managerial skills. Structural change is one consequence of the joint occupational mechanism.

Section 1 documents wage and employment changes. Section 2 develops the model and analytical comparisons. Sections 3 and 4 calibrate the 1980–2019 economy, compare the mechanisms, and relate the recovered changes to O*NET characteristics. Section 5 applies the framework to AI, robot taxes, manufacturing subsidies, and WFH. Section 6 concludes.

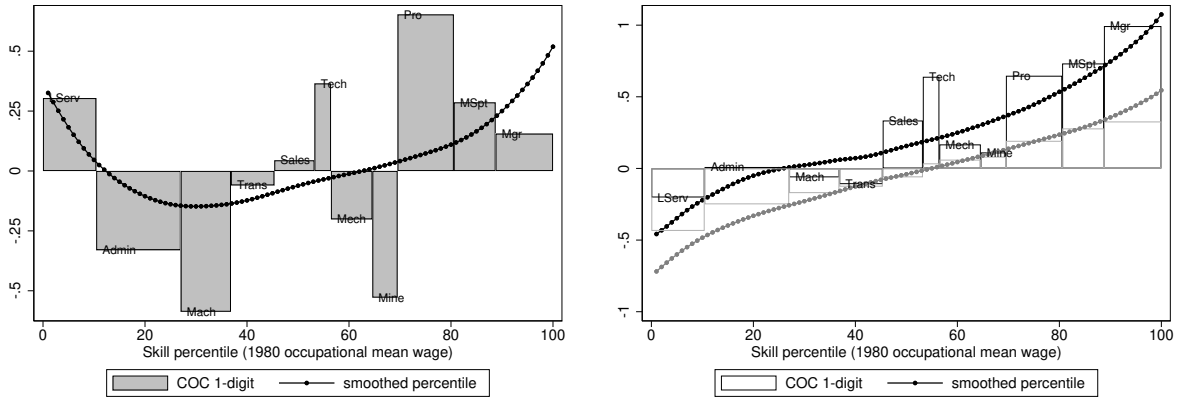
1 Facts

We document wage changes alongside employment reallocation across occupations and sectors. The central wage facts are the larger gains at the top of the worker distribution and the still larger gains for managers. Employment polarization and structural change provide the accompanying quantity changes, but their timing and incidence do not fully explain the wage pattern. We then relate both margins to task content and educational attainment.

Employment and Wage Change by Occupation Figure 3 shows nominal value-added and employment share trends of three broadly defined sectors: agriculture, manufacturing and services, from 1947 to 2019. Following convention, e.g. Herrendorf et al. (2014), "manufacturing" is the aggregation of manufacturing, mining and construction, and "services" the sum of all service and government sectors. The data are from the National Accounts published by the Bureau of Economic Analysis (BEA). In particular, employment is based on National Income and Product Accounts (NIPA) Table 6 (persons involved in production), counted in terms of full-time equivalent workers.[1]

Two facts are well documented in the literature. First, starting from even before 1970, agriculture was a negligible share of the U.S. economy. So for the remainder of this paper, we drop agriculture, and all moments are computed as if the economy consisted only of manufacturing and services.

[1]Computing employment shares from the decennial census yields more or less the same trend.



(a) Employment Polarization

(b) Occupational Average Wage, 1980 and 2019

Fig. 1: Employment and Wage Change by Occupation, 1980–2019.

From the US Census (5%), occupations are ranked by their 1980 mean wage for 11 one-digit groups and smoothed across 310 three-digit groups, separately. The widths of the bars are the employment shares of the one-digit occupation groups in 1980. Panel (a): The vertical axis measures the 39-year changes in employment shares, of which units are in percentage points per percentile. Panel (b): Average wage of occupations as log-deviations from the aggregate average wage in 1980. Gray for 1980 and black for 2019. A.

Second, structural change—the shifting of GDP and employment from manufacturing to services—exhibits a smooth trend starting from at least the 1970s, as noted in Herrendorf et al. (2014). The nominal output and employment shares move broadly together. This comparison does not identify sectoral labor income shares, which also depend on earnings per worker. We set the capital income share separately in Section 3.2.

In Figure F.13, we show TFP of manufacturing and services from 1947 assuming a capital income share of 0.363.[2] Note that manufacturing's TFP relative to services was more or less constant from the mid-1960s to early 1980s, after which it quickly increased.[3] In our quantitative analysis, we relate this to faster task-specific technological progress among middle-skill jobs, which began to show a distinct declining trend around the same time (Autor and Dorn, 2013).

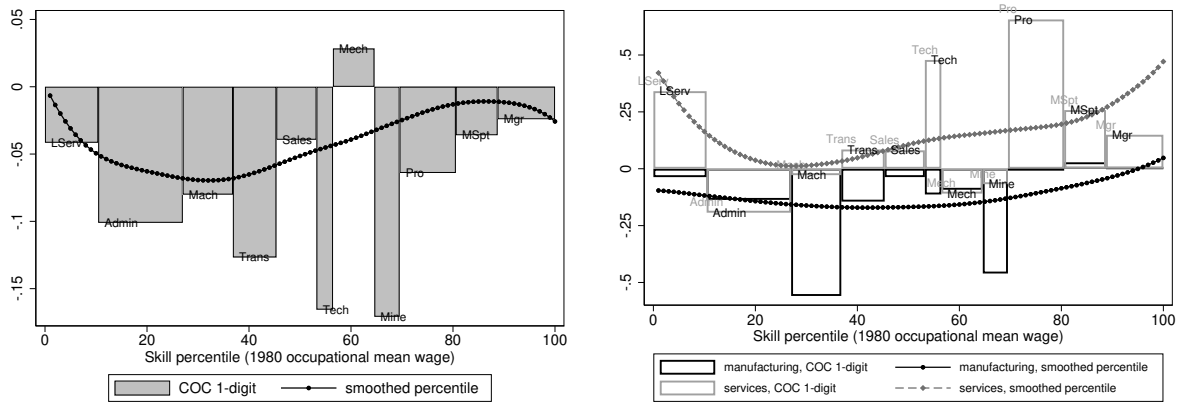
Note that Figure 3 exhibits a slowdown in structural change in the 2010s, especially in terms of employment shares. This also coincides with a slowdown in manufacturing's relative TFP growth, as seen in Figure F.13. While this may mark a break from previous trends, it will be hard to determine whether that was the case especially with the subsequent Covid-19 shock from 2020 onward.

We now ask whether polarization and structural change are interrelated. In Figure 2, we plot the same data but along two different dimensions. In Panel (a), occupations are ordered along the x-axis in the same way as we did in Figure 1. For each occupation, we compute the employment share of manufacturing in 1980 and 2019, and plot the difference.[4] The bars measure the percentage point change in the share of manufacturing employment within each COC occupation group, and the dots the smoothed percentage point change for each skill percentile. The entire plot is negative, which represents structural change from manufacturing to services. More important, manufacturing shrank the most in the middle (except

[2]This is the longest time period allowed by the industry accounts, which we need to compute real GDP and capital at the sector level. Herrendorf et al. (2015) argues that Cobb-Douglas sectoral production functions with equal capital income shares can quantitatively capture the effect of differentially evolving sectoral TFP's. See Section 3.2 for details.

[3]Herrendorf et al. (2014) also notes that manufacturing's relative TFP has grown quicker than services post 1970s, but that such trends may not be stable over a longer horizon.

[4]Appendix Figure F.6(a) shows the share of manufacturing employment within each occupation in 1980, which shows manufacturing's reliance on middle-skill jobs.



(a) Change in Manufacturing Employment Share

(b) Employment Polarization by Sector

Fig. 2: Employment Change by Sector and Occupation, 1980–2019.
Source: U.S. Census (5%) and the 2019 ACS. Panel (a): Change in the manufacturing employment share within each occupation. Panel (b): Change in occupational employment by sector. Occupations are ranked by their 1980 mean wage for 11 one-digit groups and smoothed across 310 three-digit groups. The horizontal axis measures 1980 employment percentiles. Changes are measured over 1980–2019; in Panel (b), units are percentage points per percentile. Further details are in Appendix A.

mechanics). In Panel (b) we plot the same changes as in Figure 1(a), but separate manufacturing (in dark) and services (in light).[5] Manufacturing lost employment across all jobs (except managerial support), which again is structural change. This loss is more pronounced in the middle, especially machine operators and miners. In contrast, services gained employment overall, but mostly among occupation-skill percentiles in the far left and right (e.g., low-skill services and professionals, respectively).

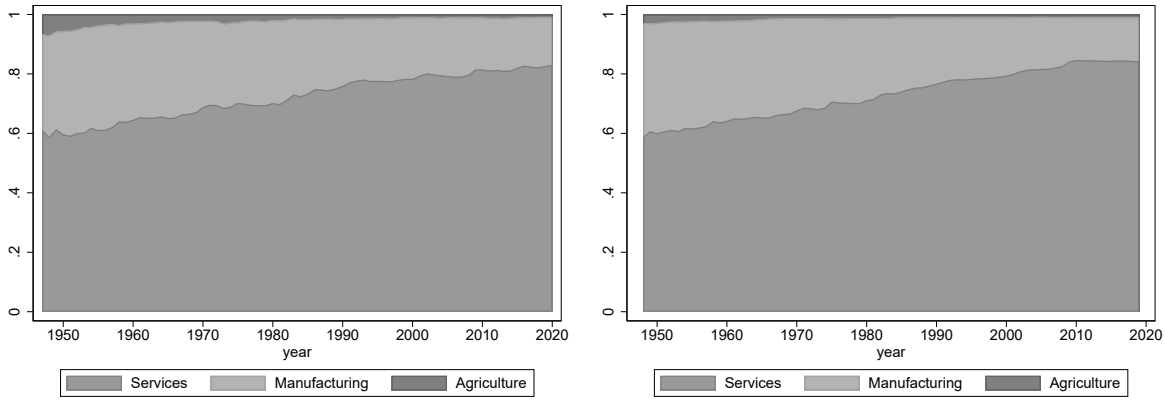
Polarization occurs within both sectors, so aggregate polarization is not solely an accounting consequence of employment shifting between sectors with fixed occupational compositions. These patterns motivate task-specific forces, although sectoral shocks may also induce changes within sectors. Moreover, as shown later in Figure 9, within-sector polarization was faster in manufacturing than in services: Between 1980 and 2019, the share of middle-skill jobs out of all manufacturing jobs fell by 13 percentage points, while the corresponding number in services was only 7 percentage points.

Wages across Occupations The rest of our empirical analysis is primarily based on the decennial U.S. Censuses 1980–2010, and the 2019 American Community Survey (ACS), for which we closely follow Autor and Dorn (2013). We restrict our sample to 16–65 year-old non-farm employment. Figure 1 plots employment and wage changes by occupation from 1980 to 2019, extending Figure 1 in Autor and Dorn (2013) who considered changes up to 2005. Occupations are sorted into employment share percentiles by skill along the x-axis, where skill is proxied by the mean (log) hourly wage of each occupation in 1980. We follow the three-digit `occ1990dd` occupation coding convention in Dorn (2009), which harmonizes the `occ1990` convention in Meyer and Osborne (2005). This results in 310 occupation categories for which employment is positive in all 4 censuses and the 2019 ACS.[6] Employment is defined as the product of weeks worked times usual weekly hours.

The data are presented in two ways. First, following Autor and Dorn (2013), each dot in Figure 1 represents one percent of employment in 1980. The y-axis in Panel (a) measures each skill percentile's

[5]The area of all bars for one sector represents structural change, while adding them across both yields 0. Similarly, the integrals of the smoothed graphs should sum to 0, modulo locally weighted regression errors.

[6]The resulting number of occupation codes from harmonizing the censuses with the 2019 ACS are slightly different from Dorn (2009), who harmonized census occupation codes with the 2005 ACS.



(a) Nominal Value-added Shares

(b) Sectoral Employment Shares

Fig. 3: Sectoral Output and Employment Shares, 1947–2019.
Source: BEA NIPA accounts. Manufacturing combines manufacturing, mining, and construction; services combines service and government sectors. Employment is measured in full-time equivalents. See Section 3.2.

employment change from 1980 to 2019 in percentage points, and in Panel (b) its mean wage in each year. The changes are smoothed by percentiles across neighboring occupations using a locally weighted smoothing regression. Despite the Great Recession happening between 2005 and 2010, the long-run patterns are virtually the same as in their study: employment has shifted from the middle toward both lower- and higher-skill jobs. Likewise, wages have risen the least in the middle, and much more at the top.

Second, we group all occupations into 11 broad categories, roughly (but not exactly) corresponding to the one-digit Census Occupation Codes (COC). These groups are ordered by the mean wage of each broadly defined group. To represent the groups in skill percentiles, the width of each group along the x-axis is set equal to its 1980 employment share.[7] In Figure 1(a), the percentage point change of a group's employment share is represented by the area of the bar,[8] while the height of each bar in Panel (b) measures the change in its mean wage. Except for sales and technicians, the patterns of polarization emphasized by Acemoglu and Autor (2011) and Autor and Dorn (2013) remain intact till 2019.[9,10]

Managers and other high-paying occupations Managers earned the highest wages in 1980 and experienced the largest subsequent wage gains, yet their employment share increased only modestly in the aggregate and barely at all in services. The relevant comparison is not only with workers as a whole, but also with other high-paying occupations. Table 1 separates professionals and management support—the abstract-worker groups in our quantitative model—from managers. "Management support" is a worker occupation group, distinct from the managerial category.

Both abstract-worker groups gained employment share and relative wages between 1980 and 2019, and again between 2000 and 2019. Management combines a much larger relative-wage gain with a smaller

[7] See Appendix A for more details on wage, employment, and occupation definitions. The three-digit occupations in each group do not necessarily correspond to those used for the smooth graphs by percentile.

[8] By construction, the area of all bars must add up to 0. In theory, the smoothed graph should also integrate to 0, but does not due to the locally weighted regression errors.

[9] The occupation groups form the basis for the calibration in Section 3. Appendix Table F.1 reports the 1980 and 2019 relative-wage targets.

[10] Technicians include software engineers and programmers, paralegals, and health technicians, which grew rapidly during this time period. Indeed, many of the smooth graphs are flattened due to occupations in this group. But technicians comprise a very small fraction of the U.S. economy throughout the sample period. Note also that most of such an increase is due to a sharp rise of technicians in the service sector, and not manufacturing (Figure 2(b)).

	Employment share (%)			Mean wage / manual wage		
	1980	2000	2019	1980	2000	2019
Professionals	11.0	14.5	18.2	1.87	1.96	2.34
Management support	8.2	8.5	10.5	2.04	2.06	2.55
Managers	11.3	14.0	13.0	2.15	2.51	3.31

Table 1: Managers and Other High-Paying Occupations

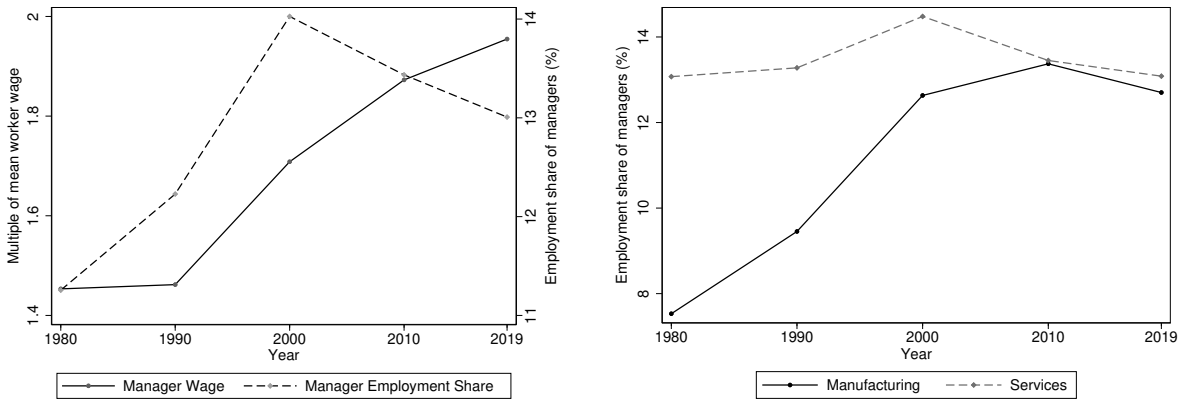
Source: 1980 and 2000 Census and 2019 ACS, using the retained occupational aggregates underlying Figures 1 and 4. Employment shares use annual hours and Census sampling weights and include the self-employed. Mean hourly wages exclude self-employment income and are divided by the low-skill-service (manual) mean wage in the same year. See Appendix A for the sample and occupation definitions.

long-run employment increase and a decline after 2000. Thus the high-paying end contains two different wage–employment patterns. Task productivity and upskilling provide a natural account of expanding high-h worker occupations; the managerial pattern supplies a separate margin against which to assess that account. The contrast motivates modeling managerial services and their endogenous supply separately, without identifying a unique cause from the two observed series.

In our model and quantitative analysis, we treat managers as a special occupation that organizes all other occupations. While many studies emphasize the organization of production (Garicano and Rossi-Hansberg, 2006), most focused on top CEO's of publicly traded companies (Tervio, 2008; Gabaix and Landier, 2008) or certain industries (Caliendo et al., 2015). We treat managers as a broader group including CEOs, middle managers, and the self-employed, and also connect them to a notion of establishments.[11]

Previous papers have shown top CEO wages rising astronomically compared to the median worker's, and Figure 4(a) shows that even with our broader definition, the "manager wage premium" over all other workers rose from 45 percent in 1980 to more than 95 percent in 2019. At the same time, the employment share of managers also rose from 11 to 14 percent between 1980 and 2000, but then dropped back to 13 percent from 2000 to 2019.[12] Important for our purposes, managerial wages continued to rise at a much higher rate than workers despite the drop in their employment share, suggesting they play a qualitatively different role in production.

Also important for our purposes, the rise of managers was only pronounced in manufacturing. Figure 4(b) shows that when separated by sector, managers' employment share rose by more than 4 percentage points in manufacturing but barely at all in services. Appendix Figure F.7(a) reconfirms what we already saw in Figure 2(b), that structural change was barely noticeable among managers despite its prevalence among all other workers. The differing occupational patterns motivate a role for task-specific forces; they do not by themselves rule out sectoral shocks operating through occupational composition and equilibrium prices.[13]



(a) Relative Employment and Wages

(b) Manager Employment Share

Fig. 4: Managers vs Workers

Source: U.S. Census (5%). Left: Relative wage and employment share of managers in aggregate. Right: Employment share of managers within manufacturing and services. See Appendix A for how we define managers in the census and Figure A.1 for a detailed breakdown of the manager group.

1.1 Explaining Long-Run Trends

Changes in the education and skill composition of the workforce have long been central to explanations of employment and wage changes (Katz and Murphy, 1992; Acemoglu and Autor, 2011). The routinization literature relates employment shifts to technologies that replace repetitive tasks, using DOT or O*NET measures of occupational content (Autor et al., 2003; Acemoglu and Autor, 2011; Autor and Dorn, 2013). Complementary evidence comes from occupational software and robot exposure (Webb, 2020) and observed industrial robot adoption (Acemoglu and Restrepo, 2020). AI studies use exposure measures and vacancy data to examine another direction of technological change (Felten et al., 2019; Webb, 2020; Acemoglu et al., 2022).

Tables 2–3 show different task-content associations for employment and wages. Routine-manual content is associated with employment declines, while nonroutine analytical content is associated with wage growth. In both tables, we consider consolidated O*NET measures from Acemoglu and Autor (2011), robot and software exposure indices constructed by Webb (2020), and the change in the college share of workers in each occupation between 1980-2019, as explanatory variables. Since all such indices have no meaningful units, we normalize them so that they are mean zero with a standard deviation of 1.

The O*NET measures considered are "Nonroutine-Cognitive Analytical", "Nonroutine-Cognitive Interpersonal", "Routine-Cognitive", "Routine-Manual", and "Nonroutine-Manual Physical." When all measures are included as independent variables in Table 2, none is individually statistically significant for the change in the three-digit occupation employment share. But when we only include as a regressor the routine-manual task content, it has significant explanatory power. So does Webb's robot measure, but it

[11]Appendix Figure A.1 shows the employment shares of the managerial and self-employed categories used in this definition. Employment includes the self-employed; the wage measure uses labor income and excludes self-employment income, as explained in Appendix A.

[12]A separate analysis of the annual ACS from 2001 onward, not shown here, shows that managers' employment share continued to rise up to 14.5 percent by 2005, but then continually dropped since the Great Recession.

[13]Appendix Figure F.8(b) plots the wage ratio between managers working in the manufacturing sector and in the services, as well as the same ratio for all other workers. Both ratios change by less than 10 percentage points between sectors, and we do not seek to explain why wages differ across sectors within the same occupation. The different wage patterns for managers and workers provide an additional dimension against which to assess the model.

	(1)	(2)	(3)	(4)	(5)	(6)
NR-Cog Anal	0.128					
	(0.237)					
NR-Cog Pers	0.258					
	(0.255)					
R-Cognitive	-0.035					
	(0.128)					
R-Manual	-0.251	-0.601**		-0.555**		-0.589**
	(0.134)	(0.150)		(0.160)		(0.119)
NR-Man Phys	-0.204					
	(0.140)					
Robot (Webb)			-0.487*	-0.071		
			(0.227)	(0.138)		
Software (Webb)			0.087			
			(0.209)			
ΔCollege					2.761	0.268
					(1.982)	(1.839)
Constant	0.609**	0.609**	0.609**	0.609**	0.297	0.579**
	(0.128)	(0.129)	(0.131)	(0.129)	(0.204)	(0.210)
R^2	0.089	0.067	0.035	0.068	0.015	0.067
Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$.						

Table 2: Routinization and Employment

The dependent variable is the change in the employment share of three-digit COC occupations from 1980 to 2019. The first 5 independent variables are consolidated O*NET measures from Acemoglu and Autor (2011), while the next two are taken from Webb (2020). ΔCollege is the change in the share of college workers in each three-digit occupation.

becomes insignificant once we also include jobs' routine-manual content.

In Table 3, nonroutine-cognitive analytical task content is positively associated with wage growth even when the other O*NET measures are included. Nonroutine-manual physical content and Webb's robot measure have negative coefficients. Wage growth is also positively associated with the rise in an occupation's college-educated share.

The contrast between Tables 2 and 3 motivates separate productivity and skill channels in the model. Routine-task measures are associated with employment reallocation, while analytical content and educational attainment help account for wage growth. These are descriptive relationships, rather than causal estimates of technological change. Section 5.1 asks whether the same relationships extend to the most recent decade and examines the distinct wage and employment associations of AI exposure.

2 Model

We construct a model in which individuals with worker and managerial skills choose their occupations, and managers combine worker-task outputs to produce sectoral goods. The same allocation determines who becomes a manager, where workers are employed, and the wages paid to different skill types. Task productivity, occupational requirements, and the distribution of skills have distinct roles in production and occupational choice, as summarized in Figure 5. This structure allows us to study wage differences among workers and the earnings of managers relative to workers while also accounting for occupational and sectoral reallocation. We first characterize equilibrium and then derive comparative statics for employment and wages. The latter have explicit sufficient conditions, distinct from the general equilibrium assumptions. Proofs are in the appendix.

	(1)	(2)	(3)	(4)	(5)	(6)
NR-Cog Anal	0.087**	0.097**		0.089**		0.073**
	(0.015)	(0.013)		(0.014)		(0.013)
NR-Cog Pers	0.005					
	(0.017)					
R-Cognitive	0.019					
	(0.011)					
R-Manual	-0.036*					
	(0.017)					
NR-Man Phys	-0.039**	-0.060**		-0.033*		
	(0.014)	(0.012)		(0.016)		
Robot (Webb)			-0.134**	-0.044**		-0.036**
			(0.016)	(0.016)		(0.012)
Software (Webb)			0.041*			
			(0.016)			
ΔCollege					1.316**	0.913**
					(0.117)	(0.151)
Constant	0.270**	0.270**	0.270**	0.270**	0.121**	0.167**
	(0.011)	(0.011)	(0.012)	(0.011)	(0.017)	(0.020)
R^2	0.344	0.333	0.231	0.350	0.336	0.468
Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$.						

Table 3: Skills, Task Content, and Wage Growth

The dependent variable is wage growth in three-digit COC occupations from 1980 to 2019. The first 5 independent variables are consolidated O*NET measures from Acemoglu and Autor (2011), while the next two are taken from Webb (2020). ΔCollege is the change in the share of college workers in each three-digit occupation.

2.1 Technology

There is a continuum of individuals endowed with two types of skill, $(h, z) \in \mathcal{H} \times \mathcal{Z} \subset \mathbb{R}^2_+$. Worker human capital, h, is used to produce worker tasks. Managerial talent, z, is a skill for organizing tasks. The distinction is economic rather than a relabeling of the highest worker skill: h governs comparative advantage across worker tasks, while z governs productivity in management. Individuals compare the earnings from these two uses of skill, so managerial supply depends on selection as well as the joint skill distribution. This permits the reward to organizing production to rise differently from the rewards to performing high-skill worker tasks. Without loss of generality, we assume that the mass of individuals is 1, with associated cumulative distribution function $\mu(h, z)$.

There are two sectors $i \in \{m, s\}$.[14] In each sector, goods are produced by teams. A team is led by a manager who uses his managerial skill and physical capital to organize a continuum of worker tasks $j \in \mathcal{J} = [0, J]$, where J is finite.

We will refer to the manager's job as "task z," which is qualitatively different (in a hierarchical sense) from tasks $j \in \mathcal{J}$, which are horizontally differentiated among workers. Each worker task requires both physical and worker human capital, and their allocations are decided by the manager. Specifically, we assume that

$$y_i(z) = \left[\eta_i^{\frac{1}{\omega}} x_z(z)^{\frac{\omega-1}{\omega}} + (1-\eta_i)^{\frac{1}{\omega}} x_h(z)^{\frac{\omega-1}{\omega}} \right]^{\frac{\omega}{\omega-1}}, \tag{1a}$$

$$x_z(z) = M(z) k_{iz}(z)^{\alpha} z^{1-\alpha}, \qquad x_h(z) = \left[\int_{j=0}^{J} \nu_i(j)^{\frac{1}{\sigma}} \tau_i(j; z)^{\frac{\sigma-1}{\sigma}} dj \right]^{\frac{\sigma}{\sigma-1}}, \tag{1b}$$

[14] In our application, the two sectors indexed by m and s stand for "manufacturing" and "services," respectively. However, the analysis can be extended to any finite number N of sectors.

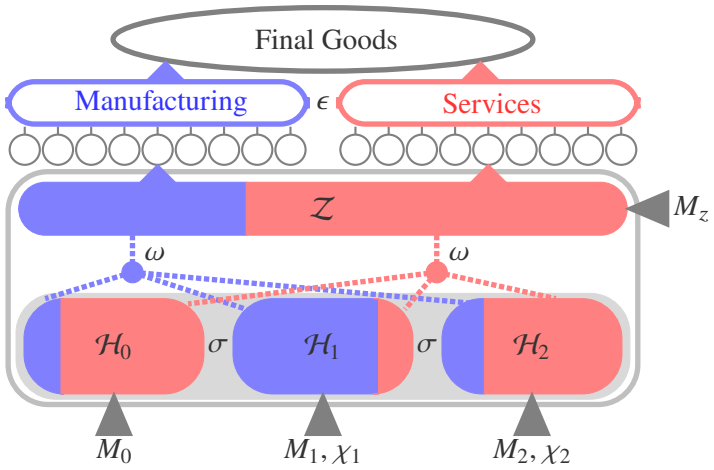


Fig. 5: Model

Individuals sort into becoming a manager or worker according to their skill (z, h), and workers further sort into tasks. While the model has a continuum of skills and tasks, in the figure we depict the latter as three discrete groups. Tasks are combined with elasticity σ, and worker output is combined with managerial output with elasticity ω. Each team is led by a manager, and the collection of team output is sectoral output. The sectoral outputs form final output according to an elasticity parameter ϵ. The shaded areas show that sectors differ in how intensively they use each task in production. (M_j, χ_j) denote task-specific productivity and technology used in task j, respectively.

$$\tau_i(j;z) = M(j)k_{ih}(j;z)^{\alpha}\left[\int_{\mathrm{h}_i(j;z)} b(h,j)d\mu\right]^{1-\alpha}, \tag{1c}$$

with $\eta_i \in (0,1)$, $\nu_i(j) \geq 0$, and $\sum_i \nu_i(j) > 0$ almost everywhere, and $\int_j \nu_i(j)dj = 1$.[15] It is important to note that sectors are different in terms of how intensively they use each task in production—i.e., ν and η have subscript i. The quantity $\tau_i(j;z)$ is the amount of task j output produced for a manager of skill z in sector i. This manager uses physical capital $k_{iz}(z)$ for himself, and allocates capital $k_{ih}(j;z)$ and a set of workers $\mathrm{h}_i(j;z)$ to task j. The function $b(h,j)$ is the productivity of a worker with human capital h assigned to task j.[16] Without loss of generality, normalize $b(0,0) = 1$. This fixes productivity at one skill–task pair; it does not make $b(h,0)$ constant in h. The general analytical model permits skill-dependent productivity even in the lowest task. For unique sorted worker assignment, assume that

Assumption 1 (Log-supermodularity) *The function $b : \mathcal{H} \times \mathcal{J} \mapsto \mathbb{R}_+$ is strictly log-supermodular. That is, for all $h' > h$ and $j' > j$:*

$$\log b(h',j') + \log b(h,j) > \log b(h',j) + \log b(h,j'). \tag{2}$$

Assumption 1 ensures that high-h workers sort into high-j tasks in equilibrium. Integrating $b(h,j)$ over h of the workers in the set $\mathrm{h}_i(j;z)$ yields the total productivity of all workers assigned to task j by a manager of skill z in sector i.

The economic content of log-supermodularity can be seen in a simple example with different skill loadings. At fixed positive task prices p_j, suppose earnings in task j are $y_j(h) = p_j h^{\lambda_j}$ for $h > 0$. For

[15] A useful mnemonic is index i for industry (sector) and j for job (task).

[16] Except Costinot and Vogel (2010), most task-based models assume that either tasks or skills are a continuum, but not both (Acemoglu and Autor, 2011; Stokey, 2016). The worker side of our model is similar to theirs, but includes capital as a production factor and is extended to multiple sectors. We add two-dimensional skills and distinguish managers from workers, allowing managerial selection to interact with assignment across worker tasks.

two tasks $j' > j$,

$$\log \frac{y_{j'}(h)}{y_j(h)} = \log \frac{p_{j'}}{p_j} + (\lambda_{j'} - \lambda_j) \log h.$$

With a common loading, skill raises earnings in both tasks by the same proportion and does not affect their relative return. If $\lambda_{j'} > \lambda_j$, higher-skill workers instead have a comparative advantage in task j'. Comparing earnings in logs preserves this skill gradient.

Our requirement specification gives this comparative advantage an interpretation in terms of the skills used by a task. In the additive benchmark $b(h, j) = c + h - \chi(j)$, c is a common baseline capacity and $\chi(j)$ is the task requirement. A higher requirement reduces the skill available for production, proportionately more for a low-skill worker. On the domain where both effective skills are positive,

$$\frac{\partial}{\partial h} \log \frac{p_{j'}[c + h - \chi(j')]}{p_j[c + h - \chi(j)]} = \frac{\chi(j') - \chi(j)}{[c + h - \chi(j')][c + h - \chi(j)]} > 0 \quad \text{if } \chi(j') > \chi(j).$$

Thus ordered requirements implement log-supermodularity without assigning a different power to each task. The additive form is simple to use analytically, with explicit skill derivatives, and quantitatively, with occupational assignment described by skill thresholds. It also gives changes in comparative advantage an economic interpretation: requirement changes that favor higher-skill workers in relative terms are upskilling. Log-supermodularity describes the ordering at a given date; upskilling describes a change in relative returns. Proposition 4 states sufficient conditions for the equilibrium effects of such a change. The general model allows other log-supermodular technologies, while the quantitative specification in Section 3 uses additive requirements for nonmanual tasks and a separate flat manual task.

The elasticity parameter σ captures substitutability among tasks, while ω captures the elasticity between the composite worker output x_h and managerial output x_z. The $M(j)$'s, $j \in \mathcal{J}^z \equiv \mathcal{J} \cup \{z\}$, are task-specific TFP's, which are sector-neutral.

Let $\mathcal{Z}_i$ denote the set of individuals working as managers in sector i. Aggregating over the output from all managers in sector i yields sectoral output

$$Y_i = A_i \int_{\mathcal{Z}_i} y_i(z) d\mu, \quad i \in \{m, s\}, \tag{3}$$

where A_i is an exogenous, sector-level productivity parameter. Final goods are produced by combining output from both sectors according to a CES aggregator:

$$Y = \left[\gamma_m^{\frac{1}{\epsilon}} Y_m^{\frac{\epsilon-1}{\epsilon}} + \gamma_s^{\frac{1}{\epsilon}} Y_s^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}}, \tag{4}$$

where $\gamma_m + \gamma_s = 1$. The equilibrium result allows any finite positive ϵ; the quantitative analysis uses $\epsilon < 1$.[17]

Assumption 2 (Technology and task weights) *The common capital share satisfies $0 < \alpha < 1$, capital supply is $K > 0$, and σ, ω, ϵ are finite and positive. Sector and managerial productivities are positive and finite. Worker-task productivity $M(j)$ is positive, finite, and measurable almost everywhere, with $0 < \int \nu_i M^{\sigma-1} < \infty$ for $\sigma \neq 1$; at $\sigma = 1$, require $\int \nu_i |\log M| < \infty$. The weights satisfy $\gamma_i > 0$, $\sum_i \gamma_i = 1$, $0 < \eta_i < 1$, and $\nu_i(j) \geq 0$ with $\int \nu_i(j) dj = 1$ and $\sum_i \nu_i(j) > 0$ almost everywhere. Zero-weight tasks are omitted from a sector's aggregator. The function b is positive and measurable, with $0 < \mathbb{E}b(h, 0) < \infty$.*

[17] The estimated ϵ between the manufacturing and the service sectors (broadly defined) is close to 0, as we show in section 3.2. As usual, (4) can be viewed as a representative consumer's preferences over the two sources of output.

Its relative-task productivity $r_b(h, j) = b(h, j)/b(h, 0)$ is uniformly bounded above and away from zero, and is uniformly continuous in j, uniformly over h. These bounds follow automatically from a positive continuous b on a compact worker/task domain, but also allow unbounded worker skills with finite productivity moments. Unit elasticities use Cobb–Douglas limits.

2.2 Planner's Problem

Since our model has no frictions, the planner's allocation coincides with a competitive equilibrium allocation.[18] A planner allocates aggregate capital K and all individuals into sectors $i \in \{m, s\}$ and tasks $j \in \mathcal{J}^z$. Formally, define $h_i(j; z)$ as the amount of human capital the planner allocates to task j in sector i under a manager with skill z. Also define $l_{ih}(s, j)$ as the number of individuals with skill $s = (h, z)$ that the planner assigns to task j, and $l_{iz}(s)$ the number of individuals with skill s the planner assigns as managers, in sector i. Then the planner's problem is to choose factor allocation rules $\{k_{iz}(z), k_{ih}(j; z), h_i(j; z)\}$ and assignment rules $\{l_{ih}(s, j), l_{iz}(s)\}$ to maximize current output (4) subject to

$$
\begin{aligned}
&Y_i = A_i \int y_i(z) l_{iz}(s) ds, \quad \forall i \in \{m, s\}, \\
&K = K_m + K_s \equiv \sum_{i \in \{m,s\}} \int \left\{ \left[k_{iz}(z) + \int_j k_{ih}(j; z) dj \right] \cdot l_{iz}(s) \right\} ds \qquad (5) \\
&H_i(j) \equiv \int h_i(j; z) \cdot l_{iz}(s) ds = \int b(h, j) l_{ih}(s, j) ds \quad \forall i \text{ and } \forall j \in \mathcal{J}, \\
&d\mu(s) = \sum_{i \in \{m,s\}} \left[\int l_{ih}(s, j) dj + l_{iz}(s) \right] ds, \quad \forall s \in \mathcal{H} \times \mathcal{Z}, \qquad (6)
\end{aligned}
$$

where K_i is the capital allocated to sector i, and $H_i(j)$ the total productivity of workers allocated to task j in sector i. These density-form expressions describe the regular case. For skill distributions with atoms or singular parts, type allocations are measures adding up to μ; tied types can be split across activities. The price proof in Appendix C.4 covers that general case.

For the equilibrium result we impose the following skill regularity:

Assumption 3 *Worker skill may have bounded or unbounded nonnegative support, and managerial skill satisfies $0 < \mathbb{E}z < \infty$.*

No density or common-support restriction is imposed for existence or uniqueness of service prices. For unique sorted worker assignment, the selected worker-skill measure $F(w_h(h)/w_z \mid h) dG(h)$ is atomless; atomlessness of the full marginal G is a sufficient example. Unique worker–manager choice additionally requires that the indifference curve have zero joint probability; conditional managerial atomlessness is a sufficient example, but atoms away from the cutoffs need not be excluded.

We can write the joint distribution as

$$
\mu(\tilde{h}, \tilde{z}) = \int_{[0,\tilde{h}]} F(\tilde{z} \mid h) dG(h).
$$

When density representations exist, write g for the marginal worker density and $f(\cdot \mid h)$ for the conditional managerial density. The differential characterization below concerns a regular interior equilibrium on a compact worker interval $[0, h_M]$: b is twice continuously differentiable, g and composite task-cost

[18] Nancy Stokey shared with us the insight that solving for the competitive equilibrium is more intuitive than solving for the planner's problem. While we agree in principle, the equilibrium approach necessitates introducing several implicit price functions due to our model's nested structure of production and inclusion of capital, as shown in Appendix B. In the main text we solve the planner's problem and present how to solve for prices in Section 2.5.

weights $\nu_i(j)M(j)^{\sigma-1}$ are continuous and positive, and worker participation $F(\tilde{z}(h) \mid h)$ is continuous and positive along the selection curve. Density smoothness is needed only where the comparative-static proofs differentiate selected supplies. These regularity conditions are not assumptions for the service-price existence result.

The optimal factor allocation rules across managers, $\{k_{iz}(z), k_{ih}(j;z), h_i(j;z)\}$, are straightforward: They must equalize marginal products across managers with different z's. Since team production is constant returns to scale in managers' and workers' composite output, we can aggregate over teams (each led by a single manager) in (1) to write sectoral output as

$$Y_i = A_i \left[\eta_i^{\frac{1}{\omega}} X_{iz}^{\frac{\omega-1}{\omega}} + (1-\eta_i)^{\frac{1}{\omega}} X_{ih}^{\frac{\omega-1}{\omega}} \right]^{\frac{\omega}{\omega-1}}, \tag{7a}$$

$$X_{iz} = M(z) K_{iz}^{\alpha} Z_i^{1-\alpha}, \qquad X_{ih} = \left[\int_j \nu_i(j)^{\frac{1}{\sigma}} T_i(j)^{\frac{\sigma-1}{\sigma}} dj \right]^{\frac{\sigma}{\sigma-1}}, \tag{7b}$$

where K_{iz} is the total amount of capital allocated to managers and $Z_i \equiv \int z l_{iz}(s) ds$. Similarly, the sectoral task composite X_{ih} combines sectoral task aggregates

$$T_i(j) = M(j) K_{ih}(j)^{\alpha} H_i(j)^{1-\alpha}, \tag{8}$$

where $K_{ih}(j)$ is the total amount of capital allocated to task j in sector i.

In the remainder of this section, the solution to the planner's problem is solved by characterizing: (i) the optimal allocation of physical capital across tasks within a sector; (ii) the optimal worker assignment across tasks within a sector; and (iii) the optimal allocation of managers vs. workers within a sector. We then solve for the within-sector solution in Section 2.3, which allows us to express the sectoral production function (3) only in terms of the optimal assignment rules and sectoral aggregates. Given this, we show in Section 2.4 that the two-sector equilibrium is unique, which enables comparative statics in Section 2.6.

Capital allocation within sectors The planner equalizes the marginal product of capital across tasks. Given the technologies in (7)-(8), this means that all capital decisions can be expressed as a linear function of the capital used in task 0. Specifically,

$$\pi_{ih}(j) \equiv \frac{K_{ih}(j)}{K_{ih}(0)} = \left[\frac{\nu_i(j)}{\nu_i(0)} \right]^{\frac{1}{\sigma}} \cdot \left[\frac{T_i(j)}{T_i(0)} \right]^{\frac{\sigma-1}{\sigma}}, \tag{9}$$

with which we can express the worker task composite X_{ih} in (8) as

$$X_{ih} = \nu_i(0)^{\frac{1}{\sigma-1}} \Pi_{ih}^{\frac{\sigma}{\sigma-1}} T_{i0}, \quad \text{where } \Pi_{ih} \equiv \int_j \pi_{ih}(j) dj. \tag{10}$$

Of course, marginal products must also be equalized across the managerial task and the rest. Using (10) we can define

$$\pi_{iz} \equiv \frac{K_{iz}}{K_{ih}(0)} = \left(\frac{\eta_i}{1-\eta_i} \right)^{\frac{1}{\omega}} \cdot \left(\frac{X_{iz}}{X_{ih}} \right)^{\frac{\omega-1}{\omega}} \cdot \Pi_{ih}, \tag{11}$$

which does not vary with j. Equations (9) and (11) subsume the capital allocation decisions into the labor allocation rules through $\pi_{ih}(j)$ and π_{iz}.

Sorting workers across tasks within sectors Under strict log-supermodularity and the regular interior characterization just stated, there is a continuous assignment function $\hat{j} : [0, h_M] \mapsto \mathcal{J}$ s.t. $\hat{j}(h)$ is strictly increasing, and $\hat{j}(0) = 0$, $\hat{j}(h_M) = J$.[19] That is, there is positive sorting of workers into tasks, and workers of skill h are assigned to job $\hat{j}(h)$. Since $\hat{j}(h)$ strictly increasing, we can also define its inverse $\hat{h} : \mathcal{J} \mapsto [0, h_M]$.

It should be clear that $\hat{j}(h)$ and $\hat{h}(j)$ are identical across sectors, and hence not indexed by i. Otherwise, the planner would be able to reallocate h across sectors and increase output. So the planner's problem of choosing $l_{ih}(s, j)$ has two parts: One of choosing $l_h(s, j)$, the number of individuals with skill s assigned to task j *regardless of sector*, and the other of choosing $q_{ih}(j)$, the fraction of task j workers allocated to sector i, which satisfies $\sum_{i \in \{m,s\}} q_{ih}(j) = 1$. That is, we can write $l_{ih}(s, j) = q_{ih}(j) \cdot l_h(s, j)$.

Sorting managers vs. workers within a sector Standard Roy selection with two-dimensional heterogeneity implies a cutoff rule $\tilde{z}(h)$ such that for every h, individuals with z above $\tilde{z}(h)$ become managers and those below become workers; at equality either occupation is optimal. Since we know that workers sort positively into j, we also define $\hat{z}(j) = \tilde{z}(\hat{h}(j))$ for all $j \in \mathcal{J}$. As with workers, the manager selection rule must be identical across sectors.

Similar to Lucas (1978), $\hat{z}(j)$ is chosen so that the marginal product of the threshold manager is equalized between task z and j, so

$$\hat{z}(j) \big/ b(\hat{h}(j), j) = \pi_{ih}(j) Z_i \big/ \pi_{iz} H_i(j), \quad \forall j \in \mathcal{J} \tag{12}$$

and in particular, with some abuse of notation,

$$\hat{z} \equiv \hat{z}(0) = \tilde{z}(0) = Z_i \big/ \pi_{iz} H_i(0), \tag{13}$$

since we normalized $b(0, 0) = 1$. Equation (13) is the counterpart of (11): it normalizes the total productivity of managers in sector i by task-0 productivity $H_i(0)$. In the next subsection, we normalize all other workers' productivities by $H_i(0)$ as well.

2.3 Within-Sector Solution

First consider the rule $l_{iz}(s)$. The occupation-choice cutoff $\tilde{z}(h)$ determines the economy-wide pool of managers. For sectoral headcount identities, we use allocations with the same mean managerial skill $\bar{z}$ in both sectors. Equality of higher moments or entire skill distributions is unnecessary. For local comparative statics, equality of the baseline means and their first-order responses suffices. A convenient sufficient implementation gives each sector a constant fraction of every selected managerial type. Writing $l_z(s) = \sum_i l_{iz}(s)$, that implementation is

$$l_{iz}(s) = q_{iz} l_z(s), \qquad q_{iz} = Z_i / Z, \qquad \sum_{i \in \{m,s\}} q_{iz} = 1. \tag{14}$$

In this implementation, q_{iz} is independent of (h, z) within the selected pool; it need not be one half. More generally, $q_{iz} = Z_i / Z$, and common sectoral means alone imply the same aggregate headcount identity. In particular,

$$\int l_{iz}(s) ds = q_{iz} \int [1 - F(\tilde{z}(h) \mid h)] dG(h), \tag{15}$$

[19] For a more formal proof, refer to Lemma 1 in Costinot and Vogel (2010).

with

$$Z = \int \int_{\tilde{z}(h)}^{\infty} z \, dF(z \mid h) dG(h), \qquad Z_i = q_{iz} Z.$$

The planner requires the sectoral totals Z_i but is indifferent among assignments of selected managers delivering those totals. The constant-fraction convention selects an allocation and makes sectoral managerial headcounts proportional to effective managerial services. It applies separately in each equilibrium. It does not fix who becomes a manager: changes in $\tilde{z}(h)$ alter the selected pool before it is divided between sectors.

Next consider the planner's choice of $l_h(s, j)$. The characterization is similar to Costinot and Vogel (2010) and summarized in Lemma 1.

Lemma 1 *Define*

$$B_j(j; \hat{h}) \equiv \exp\left[\int_0^j \frac{\partial \log b(\hat{h}(j'), j')}{\partial j'} dj'\right]. \tag{16}$$

At the optimum, the productivity of all workers assigned to task j can be expressed as

$$H(j) = b(\hat{h}(j), j) \cdot F(\hat{z}(j)|\hat{h}(j)) g(\hat{h}(j)) \cdot \hat{h}'(j), \quad \forall j \in \mathcal{J}, \tag{17}$$

and in each sector sector i,

$$H_i(j) = q_{ih}(j) H(j) = H_i(0) \cdot \pi_{ih}(j) B_j(j; \hat{h}), \tag{18}$$

Equation (17) simply states that total worker productivity in task j is the product of the infinitesimal mass of individuals assigned to task j, times their effective productivity. Equation (18) is the counterpart of (9): the marginal products of labor are equated at every point along $\mathcal{J}$. Similarly, all manager allocations can also be normalized by $\hat{z}$.

Corollary 1 *Define the counterpart of B_j in* (16)*:*

$$B_h(h; \hat{j}) \equiv \exp\left[\int_0^h \frac{\partial \log b(h', \hat{j}(h'))}{\partial h'} dh'\right]. \tag{19}$$

At the optimum, the productivity of the cutoff rules $\hat{z}(j)$ and $\tilde{z}(h)$ can be expressed as

$$\hat{z}(j) = \hat{z} \cdot b(\hat{h}(j), j) \big/ B_j(j; \hat{h}) = \hat{z} \cdot B_h(\hat{h}(j); \hat{j}) \quad \Leftrightarrow \quad \tilde{z}(h) = \hat{z} \cdot B_h(h; \hat{j}). \tag{20}$$

The corollary makes all manager cutoff rules except $\hat{z}$ (the threshold at $h = 0$) redundant, since they can be expressed only in terms of $\hat{z}$ in (13) and $\hat{h}(j)$. In what follows, we suppress the dependence of B_j and B_h on $\hat{h}$ and $\hat{j}$ unless necessary.

We can now show that the within-sector solution is characterized only by $\{\hat{h}(j), \hat{z}\}$:

Proposition 1 *Suppose $[q_{ih}(j)]_{j=0}^J$ and q_{iz} are given. Under the regular interior characterization above, the within-sector equilibrium characterization depends only on $\left\{[\hat{h}(j)]_{j=0}^J, \hat{z}\right\}$, and does not depend on sectoral aggregates K_i and L_i. Existence of an equilibrium with endogenous sectoral weights is established in Proposition 2.*

A within-sector solution yields a simple expression for sectoral output:

Corollary 2 *At a planner's optimum, sectoral output can be expressed as*

$$Y_i = \Phi_i \cdot K_i^{\alpha} L_i^{1-\alpha}, \quad \text{where } \Phi_i \equiv M(0) \cdot \psi_i \cdot \Pi_{ih}^{\frac{\sigma-\omega}{(\sigma-1)(1-\omega)}} \Pi_{K_i}^{\frac{\omega}{\omega-1}-\alpha} \Pi_{L_i}^{\alpha-1} \tag{21}$$

is the sectoral TFP, endogenously determined by the optimal allocation rules. Sectoral TFP can be decomposed into 3 parts:

1. $M(0)$*, which is common across both sectors and exogenous;*
2. $\psi_i \equiv A_i(1-\eta_i)^{\frac{1}{\omega-1}} \nu_{i0}^{\frac{1}{\sigma-1}}$*, which is also exogenous but sector-specific;*
3. *the part determined by* $(\Pi_{ih}, \Pi_{K_i}, \Pi_{L_i})$*, which is sector-specific and endogenously determined by the allocation rules* $\hat{h}(j)$ *and* $\hat{z}$*, where*

$$\Pi_{K_i} \equiv \Pi_{ih} + \pi_{iz} = K_i/K_i(0) \text{ and } \Pi_{L_i} \equiv \Pi_{il} + (\hat{z}/\bar{z})\pi_{iz} = L_i/H_i(0) \tag{22}$$

are the total amounts of capital and labor in sector i *in units of the task-0 capital and labor allocated to sector* i*, respectively;* $\Pi_{il} \equiv \int \left[\pi_{ih}(j)/B_h(\hat{h}(j))\right] dj$*, and* $\bar{z} \equiv Z/\int [1 - F(\tilde{z}(h)|h)] dG(h)$ *is the mean of managers' productivity.*

Sectoral TFP depends on occupational composition. Let W_i denote the labor wage bill and $\bar{w}_i = W_i/L_i$ sectoral average earnings. Common capital shares imply $RK_i = \alpha p_i Y_i$ and $W_i = (1-\alpha) p_i Y_i$, hence

$$\kappa \equiv \frac{L_s}{L_m} = \frac{K_s}{K_m} \frac{\bar{w}_m}{\bar{w}_s} = \frac{\gamma_s}{\gamma_m} \left(\frac{\Phi_s}{\Phi_m}\right)^{\epsilon-1} \left(\frac{\bar{w}_s}{\bar{w}_m}\right)^{-[\alpha+\epsilon(1-\alpha)]}. \tag{23}$$

Capital and wage-bill ratios coincide with the sectoral revenue ratio; headcount ratios also depend on relative average earnings. Thus relative TFP alone does not determine employment shares. The common managerial-skill allocation convention used in (22) is retained when reporting sectoral managerial headcounts.

Lemma 2 *Under Assumptions 2 and 3, the one-sector equilibrium has unique service prices. Under Assumption 1, atomless selected worker skills, and zero probability on the worker–manager indifference curve, occupational choices are also unique almost everywhere. This statement concerns equilibrium prices, not arbitrary imposed sectoral allocation weights.*

2.4 Existence and Uniqueness of Equilibrium

The equilibrium conditions above can be collected into a price problem. This separates the assumptions needed for equilibrium from the stronger restrictions used for particular comparative statics. Let $p_h(j)$ be the price of a unit of effective worker service in task j, and let p_z be the corresponding managerial price. These prices are common across sectors; they are distinct from the sectoral allocation weights $q_{ih}(j)$ and q_{iz}. Define

$$\mathcal{G}(p_h, p_z) = \int \max\left\{\max_{j \in \mathcal{J}} p_h(j) b(h,j),\ p_z z\right\} d\mu(h,z). \tag{24}$$

This is aggregate labor earnings at occupational choices induced by the prices. It is convex and homogeneous of degree one in prices.

The common capital share allows the rental rate to be factored out of the nested unit cost. Write

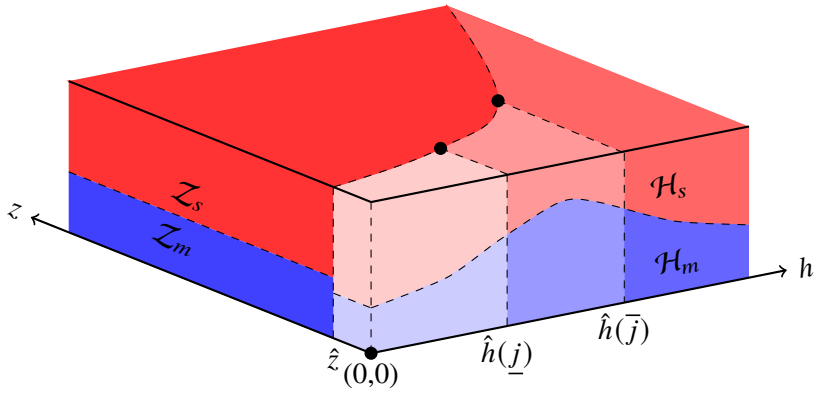


Fig. 6: Equilibrium

$\Sigma = \alpha + (1-\alpha)\sigma$, $\Omega = \alpha + (1-\alpha)\omega$, and $\Xi = \alpha + (1-\alpha)\epsilon$. The remaining service-price index is

$$\begin{aligned} C_{ih}(p_h) &= \left[\int \nu_i(j) M(j)^{\sigma-1} p_h(j)^{1-\Sigma} dj\right]^{1/(1-\Sigma)}, \\ C_i(p_h, p_z) &= \left[\eta_i M(z)^{\omega-1} p_z^{1-\Omega} + (1-\eta_i) C_{ih}^{1-\Omega}\right]^{1/(1-\Omega)}, \\ C(p_h, p_z) &= \left[\sum_i \gamma_i A_i^{\epsilon-1} C_i^{1-\Xi}\right]^{1/(1-\Xi)}. \end{aligned} \tag{25}$$

Expressions with a unit elasticity are interpreted by their Cobb–Douglas limits. The full output unit cost is $\alpha^{-\alpha}(1-\alpha)^{-(1-\alpha)} R^{\alpha} C^{1-\alpha}$. For finite positive elasticities, C is concave and strictly concave between nonproportional positive price vectors. Relative equilibrium prices solve

$$\max_{p_h \geq 0,\ p_z \geq 0} C(p_h, p_z) \quad \text{subject to} \quad \mathcal{G}(p_h, p_z) \leq 1. \tag{26}$$

The normalization fixes the common price scale; it imposes no restriction on the equilibrium allocation.

Proposition 2 (Existence and Uniqueness) *Suppose Assumptions 2 and 3 hold. Then a competitive equilibrium exists with unique relative service prices. Sectoral output and aggregate capital and effective-service allocations are also unique, and the final-good numeraire fixes absolute prices. Under Assumption 1, atomless selected worker skills, and zero probability on the worker–manager indifference curve, occupational choices and worker assignments to tasks are unique up to sets of individuals of measure zero. Individuals may remain indifferent between assignments to sectors or teams that deliver the same aggregate services.*

The proof is in Appendix C.4. Existence uses positivity and regularity of technology and finite skill moments; log-supermodularity identifies the sorted worker assignment. Neither existence nor uniqueness requires MLR, a particular Pareto shape, log-concavity, or a vanishing cross-partial. Those restrictions, when used below, belong to individual comparative-static results.

For illustrative purposes, the equilibrium skill allocation with a uniform $\mu(z, h)$ is depicted in Figure 6. Individuals in $\mathcal{Z}$ are managers, and those in $\mathcal{H}$ workers, while the subscripts s and m denote services and manufacturing. Workers sort into tasks indexed by j according to $\hat{h}(j)$. The different masses of sectoral employment across tasks are due to the task intensity parameters $\nu_i(j)$ and η_i.

2.5 Equilibrium Wages and Prices

The solution $\hat{h}(j)$ and $\hat{z}$ give all the information needed to derive equilibrium prices (which are unique). The price of the final good can be normalized to 1:

$$P = 1 = \left[\gamma_m p_m^{1-\epsilon} + \gamma_s p_s^{1-\epsilon}\right]^{\frac{1}{1-\epsilon}}, \qquad p_i = [Y_i/\gamma_i Y]^{-\frac{1}{\epsilon}} \tag{27}$$

and sectoral output prices p_i are obtained by plugging in the sectoral production function (21). Let R denote the rental rate of capital and $w_h(h)$ the wage of a worker with skill h. Equal task-service prices imply that $w_h(0)$ can be found from either sector:

$$w_h(0) = \frac{1-\alpha}{\alpha} \cdot \frac{RK_i(0)}{H_i(0)} \cdot b(0,0) = \frac{1-\alpha}{\alpha} \cdot \frac{\Pi_{L_i}}{\Pi_{K_i}} \cdot R\frac{K_i}{L_i},$$

where the second equality follows from (22) and the normalization $b(0,0) = 1$. Similarly, all workers earn their marginal product, so we can write

$$w_h(h) = w_h(0) \cdot B_h(h). \tag{28}$$

To sign the slope of worker earnings and the selection curve, assume

Assumption 4 *Along the baseline assignment,* $\partial_h \log b(h, \hat{j}(h)) \geq 0$.

This assumption says that increasing a worker's skill does not reduce productivity at the task actually assigned in equilibrium. No sign is imposed at unassigned skill–task combinations, and since

$$\frac{d \log B_h(h; \hat{j})}{dh} = \frac{\partial \log b(h, \hat{j}(h))}{\partial h}, \tag{29}$$

Assumption 4 implies that $w(h)$ is (weakly) increasing in h; Likewise from (20), it also implies that the threshold $\tilde{z}(h) = \hat{z} \cdot B_h(h)$ is increasing in h. And since for each $h \in \mathcal{H}$, threshold managers with skill $z = \tilde{z}(h)$ are indifferent between becoming a worker or manager, we can determine a managerial wage rate or rental rate of z (i.e., w_z) that satisfies

$$w_z \tilde{z}(h) = w_h(h) \quad \Rightarrow \quad w_z = w_h(0)/\hat{z}, \tag{30}$$

and (20) and (28) guarantee that the first equality holds for all h.

Productivity, requirements, and relative wages. We now consider how changes in task productivity, requirements, and skill supply affect employment and wages. Our wage comparisons concern both managers relative to workers and wage differences among worker skills; sorting and managerial selection connect these fixed-type comparisons to occupational employment and mean earnings. These are the two empirical margins in Table 1. A result about higher-h workers alone need not explain managerial earnings, and a result about managerial scarcity alone need not explain wage growth within worker occupations. We therefore track both h-related assignment and wage differences and z-related selection and relative earnings in the propositions. A change in worker technology or requirements can affect management through equilibrium even when the managerial skill distribution is held fixed. We begin with automation of middle-skill tasks, then turn to skill requirements and relative managerial earnings, before considering the implications for sectoral reallocation. Each proposition gives sufficient conditions for the effect of one force in isolation. These conditions are not imposed on the historical calibration, in which the forces change jointly.

The skill-supply and requirement comparisons below concern a one-sector equilibrium. They use finite positive ω, σ and a regular, differentiable interior assignment with $\partial_{hj} \log b > 0$. Conditional densities and the selected-supply derivatives used in Appendix D.2 are smooth and finite near the baseline selection curve. Strict managerial employment changes additionally require a positive selection density on a set of worker skills with positive measure. These local regularity conditions are additional to the general equilibrium assumptions; they impose no ordering of ω and σ and do not require vanishing log-supermodularity. The managerial relative-gain result uses $\omega < 1$; the requirement wage-gradient result does not, while its managerial wage-bill conclusion is stated separately for that case. Weak inequalities refer to first-order responses; a zero derivative alone does not sign higher-order finite changes.

2.6 Task Automation and Reallocation

Given the equilibrium allocation, we can extend Costinot and Vogel (2010) to conduct several comparative statics, such as the effect of a change in task-specific TFP's $M(j)$.[20] In particular, we are interested in the effect of an increase in the TFP of middle-skill tasks. As shown in Section 4, these tasks are "routine jobs," and hence the productivity growth specific to them is called "routinization."

The following restrictions are sufficient conditions for this local task-automation result. They are not imposed on general equilibrium, the other skill comparative statics, or the quantitative calibration.

Assumption 5 (Regular weak comparative advantage) *On the compact worker and task domains, consider*

$$b_\delta(h, j) = b_0(h)k_0(j) \exp\{\delta u(h, j)\}, \qquad \delta > 0,$$

where b_0, k_0, u are C^3, $b_0, k_0 > 0$, and $u_{hj} > 0$ along the reference assignment in (49)*. Strict log-supermodularity holds for the baseline family, but the cross-partial need not be bounded away from zero at unassigned skill–task combinations. All other primitives are fixed. The worker density g and the baseline unit-cost task weight $v(j)M(j)^{\sigma-1}$ are smooth and strictly positive. Conditional worker participation is smooth and positive near the equilibrium cutoffs, with smooth conditional densities there and a finite total managerial first moment. No smoothness restriction is imposed on the conditional distribution away from these cutoffs. Use the regular interior characterization for sufficiently small δ.*

The family specifies how log-supermodularity becomes weak while holding preferences, skill distributions and task productivities fixed. Its smoothness is used for the local response and continuation argument, not for the general equilibrium theorem.

Assumption 6 (Participation ordering at the reference selection curve) *At the separable reference equilibrium $\delta = 0$, normalize $p_z = 1$, write $t_0(h) = q_0 b_0(h)$, and define*

$$e_0(h) = \frac{t_0(h) f(t_0(h) \mid h)}{F(t_0(h) \mid h)}, \qquad \bar{e}_0 = \frac{\int t_0(h) F(t_0(h) \mid h) e_0(h) dG(h)}{\int t_0(h) F(t_0(h) \mid h) dG(h)}.$$

The elasticity $e_0(h)$ is nondecreasing, and the limiting managerial wage-bill share satisfies

$$(1 - \omega)\tau_{z0}[\bar{e}_0 - e_0(0)] < (1 - \sigma)[\bar{e}_0 + \Omega\tau_{z0}], \qquad \Omega = \alpha + (1 - \alpha)\omega. \tag{31}$$

[20] A change in factor-neutral task-TFP, $M(j)$, is different from an increase in the amount of skill working in any given task. Since we model worker skill as human capital, the qualitative effect of a change in task-TFP is the same as if it were only capital-augmenting—e.g., a fall in the price of task-specific capital.

These restrictions apply only along the reference selection curve. In terms of $G(x,h) = \log F(e^x \mid h)$, the ordering is $G_{xh} + G_{xx}\partial_h \log b_0 \geq 0$ there; neither density MLR nor a global dominance restriction is needed. A simpler, more restrictive sufficient check is

$$\tau_{z0} \leq \frac{1-\sigma}{1-\omega}. \tag{32}$$

The bounds concern managerial labor earnings, not managerial headcounts. They are not imposed on the quantitative calibration.

Polarization Consider the comparative statics for $\hat{h}(j)$ and $\hat{z}$ within a sector i, ignoring sectoral reallocation. Suppose that there is an increase in the exogenous productivity, $M(j)$, of middle-skill j-tasks.[21]

Proposition 3 (Routinization and Polarization) *Consider a one-sector equilibrium and* $0 < \underline{j} < \overline{j} < J$. *Suppose task productivity rises uniformly on* $\mathcal{J}_1 = (\underline{j}, \overline{j})$*:*

$$M_1(j) = M(j)\exp\{m\mathbf{1}_{\mathcal{J}_1}(j)\}, \qquad m > 0,$$

with all other primitives held fixed and jumps permitted at the interval boundaries. If $0 < \omega < 1$ *and* $0 < \sigma < 1$*, with no ordering between them, then under Assumptions 1, 2, 3, 5 and 6 along the baseline family, for every sufficiently small positive* δ *and then every sufficiently small positive* m*:*

1. *There is exactly one* $j^* \in \mathcal{J}_1$ *such that* $\hat{h}_1(j) > \hat{h}(j)$ *for* $0 < j < j^*$ *and* $\hat{h}_1(j) < \hat{h}(j)$ *for* $j^* < j < J$.
2. *The managerial cutoff falls at every fixed worker skill* h*:* $t_1(h) < t(h)$*. Thus every fixed managerial type gains relative to every fixed worker type. In particular* $\hat{z}_1 < \hat{z}$ *and* $w_{z1}/w_{h1}(0) > w_z/w_h(0)$.

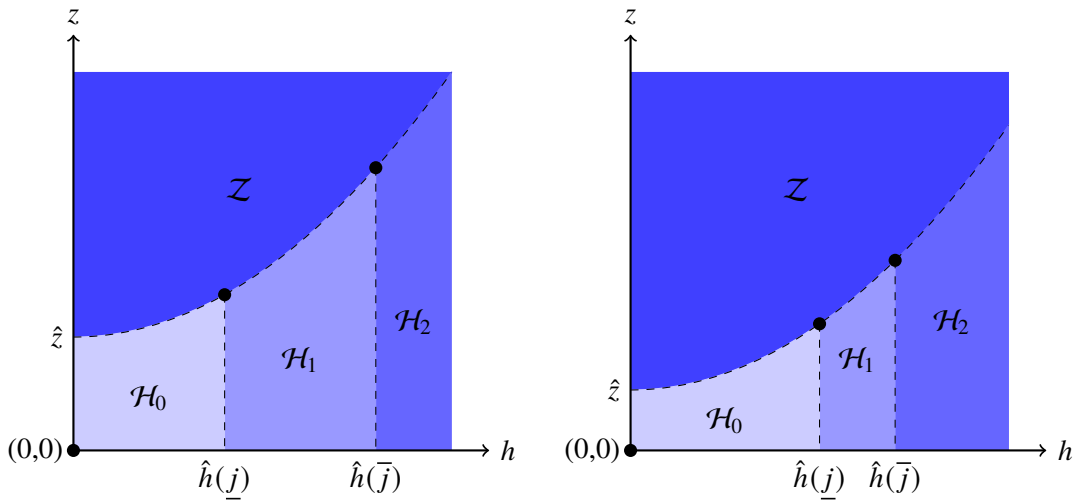
Part 1 describes polarized task assignment among workers: below the crossing, fixed worker types move toward less skill-intensive tasks; above it they move toward more skill-intensive tasks. It also gives polarized worker wage growth. If h^* is the worker type whose assignment is unchanged, then

$$\log\frac{w_1(h)}{w_1(h^*)} - \log\frac{w(h)}{w(h^*)} = \int_{h^*}^{h}\left[\partial_h \log b(h, \hat{j}_1(h)) - \partial_h \log b(h, \hat{j}(h))\right] dh > 0 \quad (h \neq h^*).$$

Part 2 isolates a complementary managerial wage channel. Task automation raises managerial earnings relative to each fixed worker type, and, when the selection margin is active, management entry changes occupational composition. The mean managerial premium still requires the wage-bill and employment terms in (36). Likewise, the assignment crossing alone does not sign occupational headcounts or capital shares when workers can enter management. Figure 7 illustrates the within-sector assignment comparison.

Complementarity across worker tasks generates reallocation away from the improved middle interval, as in Costinot and Vogel (2010), while endogenous managerial selection can oppose that force. The share restriction ensures that selection does not overturn the polarized assignment in this local experiment. It is sufficient rather than necessary and does not claim that every calibrated automation shock must polarize employment. The quantitative exercise assesses finite changes and occupational composition. The next comparison changes skill requirements while holding task productivities fixed.

[21] This *within-sector* exercise is similar to Lemma 6 in Costinot and Vogel (2010), except that we have capital and, more important, the second dimension of skill that qualitatively separates managers from workers.



(a) Equilibrium with low $M(j),\ j \in \mathcal{J}^1$

(b) Equilibrium with high $M(j),\ j \in \mathcal{J}^1$

Fig. 7: Comparative Static, Within-Sector.

2.7 Upskilling and Worker Wage Growth

We next change the requirement function while holding task productivities fixed. For an analytically positive benchmark, let

$$b(h, j) = c + h - \chi(j), \qquad \chi' > 0, \quad c > \max_j \chi(j), \quad c - \chi(0) = 1.$$

A change in requirements is a direction in function space:

$$\chi_\varepsilon(j) = \chi(j) + \varepsilon d(j), \qquad A(j) = \frac{d(j)}{\chi'(j)}.$$

Here $d(j)$ specifies how the entire requirement profile changes. The upskilling interpretation concerns whether that profile change favors higher-skill workers in relative terms, including the equilibrium response. The function $A(j)$ measures the change in units of the task index: to first order, $\chi_\varepsilon(j) = \chi(j + \varepsilon A(j))$. A proportional index stretch is only the special case $A(j) = j$; it is not the definition of a rise in requirements. A parallel level shift, a steepening, and a change concentrated in high-skill tasks are different experiments. We restrict ε so that $b_\varepsilon > 0$ and $\chi'_\varepsilon > 0$. The normalization $c - \chi(0) = 1$ applies at the baseline; if $d(0) > 0$, the perturbed bottom productivity is $1 - \varepsilon d(0)$, and occupational comparisons use the actual earnings ratio $w_h(h)/w_z$.

At a fixed task, the return to worker skill changes by $d(j)/b(h, j)^2$. Including reassignment, the exact local worker wage-gradient response is

$$\partial_\varepsilon \partial_h \log w(h) = \frac{d(j) + \chi'(j)\partial_\varepsilon \hat{j}(h)}{b(h, j)^2}, \qquad j = \hat{j}(h). \tag{33}$$

Thus the sign of $d(j)$ alone, or a steeper requirement profile by itself, does not establish the equilibrium wage ranking. The following proposition gives a sufficient class of profile changes with an upskilling interpretation: wage growth tilts toward higher-skill workers even after reassignment. This is a partial effect to be combined with the relative-productivity change.

Assumption 7 (Selection elasticity for requirement changes) *Along the regular baseline selection curve, worker participation has constant elasticity*

$$\frac{t(h)f(t(h) \mid h)}{F(t(h) \mid h)} = k \geq 0.$$

Only the elasticity at the selected cutoffs enters the local wage-gradient proof. No common scale family, global power law, or upper-tail shape is required. A primitive sufficient example is $z = S(h)U$, with U independent of h, finite mean, and $F_U(u) = \rho u^k$ on a lower body containing the selected arguments, where $k > 0$ and $0 < \rho \leq 1$. Outside that region the distribution can be any finite-mean continuation. Appendix D.3 retains a primitive interior-selection bound and a smooth common-support example as illustrations, rather than additional model assumptions.

Proposition 4 (Skill Requirements and Wage Growth) *Consider a one-sector equilibrium satisfying Assumption 7 and the local regularity conditions stated above, with any finite positive ω, σ. Let $A_0 = \log \nu + (\sigma - 1) \log M$ and A be twice continuously differentiable. Suppose requirements change smoothly to $\chi_\varepsilon = \chi + \varepsilon d$, with task productivities and the skill distribution held fixed, where*

$$d(j) = \chi'(j)A(j), \qquad A(0), A(J) \geq 0, \qquad A \not\equiv 0, \qquad -[A' + AA_0']' \geq 0.$$

Then:

1. *Let $y(h) = \partial_\varepsilon \log[w_h(h)/w_z]$ denote relative wage growth at a fixed worker skill. The worker-wage-bill-weighted average of y has the sign of $1 - \omega$. If $0 < \omega < 1$, workers in a nonempty upper interval have $y(h) > 0$ and gain relative to every fixed managerial type, while the managerial share of the labor wage bill falls. In that case, lower-skill workers may gain or lose relative to managers; if any lose, they lie below a unique zero of y. At $\omega = 1$ the managerial wage-bill share is unchanged, and for $\omega > 1$ it rises. The effect on managerial headcount is ambiguous.*
2. *For every $h_2 > h_1$,*

$$\partial_\varepsilon \log \frac{w_h(h_2)}{w_h(h_1)} > 0.$$

Thus wage growth is strictly increasing in worker skill, including the equilibrium reassignment response.

The joint restriction is $-A'' - A'A_0' - AA_0'' \geq 0$; it concerns the direction of the requirement change together with baseline composite task weights. It implies $A > 0$ in the interior under the stated endpoint and nontriviality conditions. The separate restrictions $A \geq 0$, $A' \geq 0$, $A'' \leq 0$, $A_0' \leq 0$, and $A_0'' \leq 0$ are an easy sufficient example, but are not each necessary. The index stretch $A = j$ is one example under those weight restrictions. This result does not assert that d itself is increasing or that every upward requirement change has the same wage effect. All restrictions are specific to this comparative static.

This class of requirement changes raises the relative wages of higher-skill workers among workers and, when $\omega < 1$, raises wages for an upper skill interval relative to managers. The second effect can work against the managerial gains in Proposition 5. The quantitative exercise below determines how these forces combine and how selection affects occupational mean wages.

The wage ranking does not itself sign reassignment. Writing $y_J(j) = y(\hat{h}(j))$ on the baseline assignment,

$$\partial_\varepsilon \hat{j}_\varepsilon(h) = \frac{y_J'(j)}{\partial_{hj} \log b(h, j)\, \hat{h}'(j)} - A(j), \qquad j = \hat{j}(h). \tag{34}$$

For $A = j$, Appendix D.3 gives an additional sufficient restriction for upward sorting. This allocation result is distinct from either wage comparison.

The exact power body is a useful proof benchmark; it need not be a general assumption on skills. A smooth common-support example is

$$f_{U,m}(u) = C_m u^{k-1}(1+u^m)^{-(k+q)/m}, \qquad u > 0, \quad q > 1,$$

with $C_m = m/\mathrm{B}(k/m, q/m)$. It has a Pareto upper tail. For $S(h) = S_0 e^{\beta h}$ with $\beta > \sup \partial_h \log b$, it satisfies density MLR and a global conditional-distribution dominance inequality sufficient for the reference participation ordering used in routinization. For the index-stretch direction $A = j$ and all sufficiently large finite m, the requirement wage result holds around a regular limiting power-body equilibrium with the stated selection margin. The same family gives nonzero managerial sorting under Proposition 5; the proof and scope of this extension are in Appendix D.3.

2.8 Relative Productivity, Skills, and Wages

We now distinguish managerial earnings from the earnings of other high-skill workers. Relative productivity and managerial skill supply can change the scarcity value of management even when task requirements are held fixed.

Write worker-task productivity as $M(j) = M_h \bar{M}(j)$, with a fixed task profile $\bar{M}(j)$. We orient the relative-productivity change toward faster managerial earnings growth:

$$a = \log \frac{M_h}{M(z)}.$$

A rise in a means that managerial productivity grows less than the common worker-productivity component, not necessarily that managerial productivity falls in levels. Because the two inputs are complementary, this relative productivity lag raises the scarcity value of managerial services. The reverse change is stated separately as a corollary.

The same experiment has a skill-supply interpretation. Write $z = s_z z^0$ for a uniform scaling of each person's managerial skill, with the joint distribution of (z^0, h) fixed. Then managerial technology depends on $M(z)s_z^{1-\alpha}$, and the effective relative change is

$$da = d\log M_h - d\log M(z) - (1-\alpha) d\log s_z. \tag{35}$$

If μ_z denotes population mean managerial skill, $s_z = \mu_z/\mu_z^0$ under this scale change. A lower μ_z therefore has the same relative effect as lower $M(z)$, with factor $1-\alpha$. An arbitrary mean change that alters dispersion or dependence on h is not covered. Comparisons follow the same individuals and express selection cutoffs in original z^0 units. Uniform worker productivity is also distinct from changing the skill argument inside $b(h, j)$.

Assumption 8 (Participation elasticity for managerial sorting) *Along the baseline selection curve $t(h) = w_h(h)/w_z$, the elasticity*

$$e(h) = \frac{t(h) f(t(h) \mid h)}{F(t(h) \mid h)}$$

is nondecreasing in worker skill h.

This restriction concerns participation at the selected cutoffs, not the whole joint skill distribution. A primitive sufficient example is $z = S(h)U$, with U independent of h, finite mean, and nonincreasing

lower-tail elasticity $r_U(u) = u f_U(u)/F_U(u)$ over the selected range. It suffices that $S'(h)/S(h) \geq \partial_h \log b(h, \hat{f}(h))$ along the baseline assignment. Restrictions away from that curve are unnecessary. Positive correlation alone would not establish the required elasticity ordering.

Proposition 5 (Relative Managerial Scarcity) *Consider the one-sector equilibrium described above, with $0 < \omega < 1$ and any finite $\sigma > 0$. Suppose $a = \log[M_h/M(z)]$ increases by a small amount, while the skill distribution, b, and $\bar{M}(j)$ remain unchanged. Then:*

1. *For every fixed managerial skill $z > 0$ and worker skill h,*

$$\partial_a \log \frac{w_z z}{w_h(h)} > 0.$$

The entire selection curve $\tilde{z}(h)$ falls, and the managerial share of the labor wage bill rises. Managerial employment weakly rises; its first-order response is strictly positive when the selection density is positive on a set of worker skills with positive measure.

2. *Under Assumption 8, for any $h_2 > h_1$,*

$$\partial_a \log \frac{w_h(h_2)}{w_h(h_1)} \geq 0.$$

Workers move weakly toward more skill-intensive tasks: $\partial_a \hat{j}(h) \geq 0$ and $\partial_a \hat{h}(j) \leq 0$. If participation elasticity is nonconstant, wage differences strictly widen and interior task responses are strict. If it is constant, worker wage growth is identical across skills and the within-worker inequality effect is zero.

Uniform changes in managerial-skill scale have these same effects for the effective change in (35)*, with original skill units used for the selection curve.*

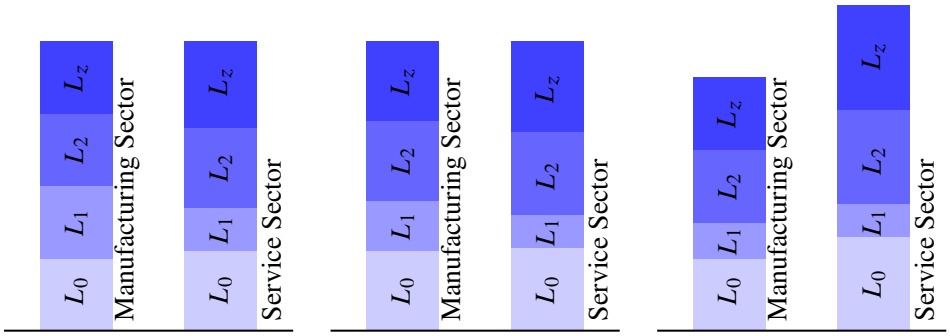
Corollary 3 (The reverse relative-productivity change) *A rise in $M(z)/M_h$, or a uniform rise in managerial skill $z = s_z z^0$ holding productivity fixed, reverses the signs in Proposition 5. Managers lose relative to every fixed worker type; the managerial wage-bill share falls, and managerial employment weakly falls, with a strict first-order fall under the active-selection condition. Under Assumption 8, worker wage differences weakly compress and workers move weakly toward less skill-intensive tasks.*

The proposition identifies a force consistent with faster managerial earnings growth and, under the sorting restriction, faster earnings growth for higher-skill workers. It does not require this force alone to account for all wage changes. The conclusions concern relative earnings of fixed types; absolute earnings and occupational mean earnings require separate accounting. The inequality restriction in part 2 is deliberately weak: the constant-elasticity benchmark changes managerial premia without changing worker wage gaps.

Partial effects and quantitative attribution. The preceding propositions are comparative statics of separate forces. They need not each reproduce the full observed wage ranking, and no analytical dominance condition is imposed on their joint change. The numerical exercise assesses their magnitudes, interaction over finite changes, and contributions to occupational mean earnings.

Occupational mean earnings. Let L_z denote managerial employment and τ_z its share of the labor wage bill. For each experiment,

$$\frac{\bar{w}_z}{\bar{w}_h} = \frac{\tau_z}{1-\tau_z}\frac{1-L_z}{L_z}, \qquad \partial_a \log \frac{\bar{w}_z}{\bar{w}_h} = \frac{\partial_a \tau_z}{\tau_z(1-\tau_z)} - \frac{\partial_a L_z}{L_z(1-L_z)}. \tag{36}$$



(a) Low $M(j)$, $j \in \mathcal{J}^1$ (b) High $M(j)$, $j \in \mathcal{J}^1$, κ fixed (c) High $M(j)$, $j \in \mathcal{J}^1$

Fig. 8: Comparative Statics, Across Sectors.

Consequently the observed managerial mean premium rises only when the wage-bill term in (36) exceeds the employment-composition term. Analogous within-group composition terms enter the mean wage of high-skill worker occupations. These must be evaluated when connecting the propositions to the empirical targets; fixed-type wage rankings alone do not prove the occupational mean rankings. Appendix D.3 gives an illustrative bounded-power mean comparison, but it is not imposed as an explanation of the observed managerial premium.

2.9 Sectoral Reallocation

Routinization can generate structural change through differential exposure to a common task-productivity shock. The sector using routine tasks more intensively experiences a larger productivity gain. Its relative output price falls, and complementarity between sectoral outputs reduces its expenditure share. Employment also depends on the occupational composition of the sector's wage bill. Proposition 6 gives alternative local conditions under which the more exposed sector loses employment share. They concern headcount composition as well as relative sectoral revenue.

For this proposition, initially $M(j) = M_0$ for all worker tasks and

$$M_a(j) = M_0 \exp\{a\mathbf{1}_{\mathcal{R}}(j)\}, \qquad r_i = \int_{\mathcal{R}} \nu_i(j)dj, \qquad r_m > r_s,$$

where $\mathcal{R} \subset (0, J)$ is the routine interval. Sectoral productivities and all other primitives are fixed. To isolate sectoral exposure, first use the homogeneous-worker benchmark $b(h, j) = 1$. Managerial skill has finite mean and a smooth density near the interior selection cutoff; the density may be zero, and atoms away from the cutoff are allowed. For these local headcount comparisons, sectoral managerial means and their first-order responses are equal, so managerial employment equals effective services divided by the common mean $\bar{z}$ to the required order. Sectoral distributions can otherwise differ. The constant-fraction allocation in (14) is one sufficient implementation. General equilibrium alone does not distinguish other assignments of managers that supply the same effective services.

Proposition 6 (Routinization and Structural Change) *Suppose $0 < \alpha < 1$ and $\omega, \sigma > 0$ are finite. If either set of conditions below holds, then a small increase in a raises Φ_m/Φ_s and Y_m/Y_s, and lowers p_m/p_s, K_m/K_s, and L_m/L_s. Hence manufacturing's employment share falls.*

1. ***Equal managerial weights.*** *The initial weights satisfy* $\eta_m = \eta_s$, $0 < \epsilon < 1$, *and*

$$(1-\epsilon)\theta + (1-\omega)\frac{(1-\phi)\theta(1-\theta)}{\phi + (1-\phi)\theta} > 0, \qquad \phi = t/\bar{z},$$

where θ *is the common baseline worker wage-bill share. This condition is automatic when* $0 < \omega \leq 1$*; it also permits* $\omega > 1$ *when relative-revenue contraction dominates headcount composition.*

2. ***Unequal managerial weights.*** *The initial weights need not be ordered, and* $0 < \epsilon < 1$. *The exact baseline price and headcount conditions* (77) *and* (76) *hold. No parametric managerial-skill distribution is required. A primitive sufficient example has* $\eta_m \leq \eta_s$, $0 < \omega < 1$, *a selected Pareto upper tail with exponent* $q > 1$, *and*

$$\epsilon \leq 1 - \frac{1-\omega}{q}.$$

The tail need hold only above some $z_0 < t$*; the distribution below* z_0 *is unrestricted.*

For each sufficiently small fixed positive shock, the strict comparisons persist under sufficiently small smooth perturbations of the respective benchmark with continuous equilibrium prices, aggregate effective allocations, and sectoral managerial means. In particular, part 1 permits a small positive gap $\eta_s - \eta_m$. *Both parts permit sufficiently small positive worker heterogeneity* $b_\delta(h,j) = \exp\{\delta\psi(h,j)\}$, *where* $\psi(0,0) = 0$ *and* $\psi_{hj} > 0$, *with* ψ *smooth, bounded, and uniformly continuous in the task index uniformly over worker skill. Extending the signs of the local derivatives themselves additionally requires differentiable continuations of prices, capital and headcount allocations, and sectoral managerial means whose derivatives are continuous at the benchmark.*

These are alternative sufficient regimes. The homogeneous-worker benchmark is used only for this sectoral result; positive δ restores strict log-supermodularity. Controlling all worker productivity dispersion is stronger than requiring a small cross-partial alone. The Pareto tail is only one sufficient example for part 2, not a general restriction on skills. The power-body example for upskilling likewise serves a different proof. Neither regime is asserted to hold in the empirical calibration.

The proof in Appendix D.5 separates the expenditure response from the change in headcounts per unit of wage bill. The exact headcount condition allows the relative-revenue contraction to dominate an opposing composition response. The simpler complementary-manager configuration reinforces the decline in part 1; part 2 permits larger managerial-intensity differences. Relative real output grows even as employment share falls. This is a simultaneous equilibrium comparison, not an initial employment expansion followed by a contraction.

Structural change also contributes to aggregate occupational change through composition. If manufacturing has the higher routine employment share, its contraction reduces the aggregate routine share; if services has the higher managerial employment share, the same reallocation raises the aggregate managerial share. Lemma 3 states this accounting decomposition. It does not require ranking the sectors' within-sector routine-share declines, or comparing the full equilibrium with a separately solved economy holding sector shares fixed. Figure 8 illustrates the within-sector and between-sector components of this adjustment.

3 Calibration

We first calibrate the model to 1980, then match 2019 by changing task-specific productivities, the Pareto skill-distribution parameters, and the skill requirements χ_j. The productivity changes are governed by the

managerial growth rate m_z and the worker-task growth rates $\{m_j\}_{j=0}^{9}$. The distribution parameters allowed to change are a and γ_h, with $\gamma_z = 1$ throughout. Sectoral task weights, sector-specific productivities, and substitution elasticities are common to both dates. Capital per worker is taken from the data at each date.

We estimate the aggregate technology (4) separately from sectoral prices and output. Within the model, the 1980 calibration matches occupational employment shares and relative wages, the manufacturing employment share, and output per worker. The 2019 calibration targets relative wages by occupation, employment shares within sectors, and aggregate output per worker. The elasticities σ and ω are disciplined jointly with task-productivity growth by the change between the two dates, but do not themselves change over time. The intervening decades (1990, 2000 and 2010) are computed by interpolating the changing parameters between the calibrated endpoints.

3.1 Parametrization

Discrete log-supermodularity In the quantitative analysis, we collapse the continuum of horizontally differentiated worker tasks into 10 groups, corresponding to the one-digit occupation groups in the census in Section 1 and Appendix A Table F.1. (There is still only one management task.) The 10 worker occupation groups are further categorized into low/medium/high skill tasks, or manual/routine/abstract jobs, according to the mean wages of each occupation group. These three broad skill groups are only used for exposition; the calibration uses the underlying occupation groups.

At the 1980 baseline, we index worker occupations by $j = 0, \ldots, 9$ and write the task technology as:

$$b(h, j) = \begin{cases} \bar{h} = 1 & \text{for } j = 0, \\ h - \chi_j & \text{for } j \in \{1, \ldots, 9\}, \quad 0 = \chi_1 < \chi_2 < \ldots < \chi_9. \end{cases}$$

The discrete specification replaces the continuous assignment function with task thresholds. This technology implies that, for the lowest-skill task 0, the worker's skill does not matter and everyone performs the task with equal efficiency. All skills (h) are used in task 1, but for tasks $j \in \{2, \ldots, 9\}$ there is a "skill loss," which increases with higher-order tasks.[22] The manual restriction has an economic role. It represents work that does not use additional worker skill, so the task has no direct upskilling channel even as its employment expands. Changes in requirements and skill supply can raise relative earnings in skill-sensitive tasks without giving manual workers the same gain. Setting the level $\bar{h} = 1$ only chooses units; assuming that $b(h, 0) = \bar{h}$ for every h is the quantitative restriction. It is motivated by the low-end wage–employment pattern, not implied by that pattern. Neither general equilibrium nor the general analytical assignment model requires it. The analytical requirement comparison instead uses the explicit additive benchmark $b(h, j) = c + h - \chi(j)$, which remains skill-dependent at the bottom; other special benchmarks are confined to their respective propositions.

For any two worker tasks with $\chi_{j'} > \chi_j$ and positive effective skills,

$$\frac{\partial}{\partial h} \log \frac{h - \chi_{j'}}{h - \chi_j} = \frac{\chi_{j'} - \chi_j}{(h - \chi_{j'})(h - \chi_j)} > 0.$$

Thus a simple additive requirement implements strict log-supermodularity: more-skilled workers have a comparative advantage in tasks with higher requirements. This is a property of a given requirement profile. Upskilling concerns how that profile changes and how the change affects relative returns to skill. Proposition 4 provides one sufficient class of such changes; neither a rise in a single χ_j nor a change in

[22] This can be interpreted as low-order skills not being used in high-order tasks, or high-order tasks requiring a fixed cost of preparation to perform the task, resulting in less skills utilized. Both the requirement parameters and the skill distribution are recalibrated for 2019. Their joint effect on effective skills depends on occupation and selection.

	(1)	(2)	(3)
γ	0.362**	0.339**	0.249**
	(0.010)	(0.005)	(0.004)
ϵ	0.004**	0.002**	0.003
	(0.000)	(0.000)	(0.001)
AIC	-573.298	-573.805	-573.584
$RMSE_1$	0.117	0.117	0.117
$RMSE_2$	0.041	0.041	0.041

Standard errors in parentheses

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Table 4: Aggregate Production Function

The manufacturing share parameter γ_m and the manufacturing-services elasticity parameter ϵ are estimated from the time series of output and price ratios from 1947 to 2019, from the National Industry Accounts. Standard errors in parentheses. For details of the estimation, we closely follow Herrendorf et al. (2013).

the observed effective-skill measure is by itself that analytical experiment. The quantitative requirement profile and the skill distribution are allowed to change jointly.

With 10 worker tasks, we solve for nine worker-task thresholds $\{\hat{h}_j\}_{j=1}^{9}$ and the managerial cutoff parameter $\hat{z}$. The discrete equilibrium conditions are summarized in Appendix E.

Bidimensional skill distribution For the quantitative analysis, we assume a skill distribution that is type IV bivariate Pareto (Arnold, 2014), with counter-c.d.f.

$$\bar{\mu}(h,z) = \left(1 + h^{1/\gamma_h} + z^{1/\gamma_z}\right)^{-a}.$$

We normalize $\gamma_z = 1$, since we cannot separately identify both skills from the skill-specific TFP's. This distribution is consistent with an establishment size distribution that is Pareto, and a wage distribution that is hump-shaped and has a thinner tail. Appendix Figure F.9 shows the marginal distributions of h and z for both 1980 and 2019.

3.2 Aggregate Production Function

The aggregate production function (4) is estimated outside of the model. For the estimation, we only look at manufacturing (inclusive of mining and construction) and services (inclusive of government). We estimate the parameters γ_m and ϵ from:

$$\log\left(\frac{p_m Y_m}{PY}\right) = \log\gamma_m + (1-\epsilon)\log p_m - \log\left[\gamma_m p_m^{1-\epsilon} + \gamma_s p_s^{1-\epsilon}\right] + u_1$$

$$\log(Y) \qquad = c + \frac{\epsilon}{\epsilon-1}\log\left[\gamma_m^{\frac{1}{\epsilon}} Y_m^{\frac{\epsilon-1}{\epsilon}} + \gamma_s^{\frac{1}{\epsilon}} Y_s^{\frac{\epsilon-1}{\epsilon}}\right] + u_2$$

where $\gamma_s \equiv 1 - \gamma_m$, using non-linear seemingly unrelated regression on all years of real and nominal sectoral output observed in the BEA Industry Accounts.[23] Real production by sector is computed by a cyclical expansion procedure as in Herrendorf et al. (2013) using production value-added to merge lower-digit industries (as opposed to consumption value-added in their analysis).

[23] The constant c is included since it is not levels but relative changes that identify ϵ.

Sectoral prices are implied from nominal versus real sectoral quantities, which may depend on the choice of base year. For robustness, we check results using three different base years, corresponding to columns (1)-(3) in Table 4. For each column, respectively, 1947 is the first year the required data is available, 1980 is the first year in our model, and 2005 is chosen as a year close to the present but before the Great Recession. The values are in a similar range as in Herrendorf et al. (2013). For the calibration, we use $(\gamma_m, \epsilon) = (0.339, 0.002)$ from column (2), as reported in Table 5.

The capital income share α is computed as the average of 1-(labor income/total income), and fixed at 0.363, and assumed to be equal across sectors.[24] Total income is GDI net of Mixed Income and Value-Added Tax from NIPA and Industry Accounts, and labor income is from NIPA. For the calibration we also need total capital stock (for manufacturing and services) for each decade, which we take from the Fixed Assets Account Table 3 and directly plug into the model.[25] Since we do not model population growth, in practice we normalize output per worker in 1980, y_{1980}, to one, and plug in $K_t = k_t / y_{1980}$ for $t \in \{1980, 1990, 2000, 2010, 2019\}$, where k_t is capital per worker in year t.

3.3 Setting Parameters

All parameters are in Table 5, except for the skill loss parameters χ_j, task intensity parameters (η_i, ν_{ij}), and task-TFP growth rates m_j, which are in Table 6. Below, we explain how these parameters are recovered. Appendix E has more detail.

Calibrating the distribution For given γ_h, a, and $\{\chi_j\}_{j=2}^9$, we numerically compute the thresholds $\{\hat{h}_j\}_{j=1}^9$ and $\hat{z}$ to match observed employment shares by occupation in 1980, by integrating over the skill distribution. With these thresholds, we can compute model-implied relative wages using the discrete version of (28), which is (81) in Appendix E:

$$\frac{w_1 \bar{h}_1}{w_0} = \frac{\bar{h}_1}{\hat{h}_1}, \quad \frac{w_z \bar{z}}{w_0} = \frac{\bar{z}}{\hat{z}}, \quad \text{and} \quad \frac{w_{j+1}(\bar{h}_{j+1} - \chi_{j+1})}{w_j(\bar{h}_j - \chi_j)} = \frac{\hat{h}_{j+1} - \chi_j}{\bar{h}_j - \chi_j} \cdot \frac{\bar{h}_{j+1} - \chi_{j+1}}{\hat{h}_{j+1} - \chi_{j+1}}.$$

Here, $\bar{h}_j$ and $\bar{z}$ denote the mean skills in each task. The left-hand side is the ratio of mean wages by occupation, which we observe in the data. The right-hand side depends on the thresholds and selected mean skills, which are determined by (γ_h, a, χ_j) and the employment shares. Hence, we iterate over γ_h, a, and $\{\chi_j\}_{j=2}^9$ to fit observed mean wage ratios, while computing the implied thresholds $\{\hat{h}_j\}_{j=1}^9$ and $\hat{z}$ to match 1980 employment shares by occupation. Relative wages are calibration targets; the retained benchmark does not match every detailed occupational wage ratio exactly.

We repeat this step using 2019 employment shares and relative wages, allowing (a, γ_h) and the requirement parameters to change. This produces separate endpoint distributions, requirements, and threshold vectors $\mathbf{x}_{1980}$ and $\mathbf{x}_{2019}$. Conditional on these endpoint values, we recover the production parameters and task-productivity growth rates so that both threshold vectors are consistent with equilibrium. Relative wages are therefore calibration targets at both dates.

Calibrated within the model We normalize $(\gamma_z, \bar{h}) = 1$, $\chi_1 = 0$ at the 1980 baseline, and $A_s = 1$. We also set $M_j = M$ for all worker and managerial tasks in 1980, since initial task-productivity levels are not separately identified from task weights in a static equilibrium. The common initial productivity M matches output per worker, while A_m matches the manufacturing employment share.

[24] The difference between the α's when we let them differ between the two sectors was negligible. Herrendorf et al. (2015) compares this assumption against sectoral production functions that are CES in capital and labor, and finds that both specifications capture the effect of differential productivity growth across sectors equally well.

[25] Real capital stock is aggregated using the same cyclical expansion procedure used for value-added.

	Parameter	Value	Target
Fixed from data	K_{1980}	2.900	Computed from BEA/NIPA
	K_{2019}	4.104	
	α	0.363	
	γ	0.339	Estimated in section 3.2
	ϵ	0.002	
Fit to 1980	ν_{ij} (18)	Table 6	Within-sector employment shares by occupation
	η_i (2)		Within-sector manager share
	A_m	1.038	Manufacturing employment share
	$M_j \equiv M_0$	0.974	Output per worker, normalization
	a	9.174	
	γ_h	0.210	Relative wages by occupation
	χ_j (8)	Table 6	
Common elasticities	σ	0.610	1980–2019 output growth and within-sector employment shares
	ω	0.250	
Fit to 2019	m_j (11)	Table 6	Output per worker growth and employment shares within sectors
	a	2.706	
	γ_h	0.258	Relative wages by occupation
	χ_j (8)	Table 6	
Fixed	$\bar{h}$	1.000	Not separately identified from M_0
	γ_z	1.000	Monotonic firm-size distribution
	A_s	1.000	Normalization

Table 5: Parameters

The population size is normalized, so K_t is capital per capita and is supplied from the data at each date. The elasticities σ and ω are common to 1980 and 2019. Only task productivities, (a, γ_h), and the χ_j's change between the two calibrated equilibria. Employment-share and relative-wage targets are computed from the Census and ACS using the definitions in Appendix A. Table F.1 reports the relative-wage targets at both endpoints.

Conditional on the common elasticities and the 1980 skill distribution and requirements, we recover η_i and $\{\nu_{ij}\}_{j=0}^{9}$ from 1980 employment shares within each sector. There are nine independent worker-task weights per sector because $\sum_{j=0}^{9} \nu_{ij} = 1$. Together with (M, A_m), these parameters fit the 1980 production and employment targets, with the employment shares shown in Figure 9.

For the 2019 equilibrium, we retain these sectoral task weights and sector-specific productivities. Task productivities evolve according to

$$M_j(t) = M(1 + m_j)^{t-1980}, \qquad j \in \{z, 0, \ldots, 9\}.$$

Thus managerial productivity in 2019 is $M(1 + m_z)^{39}$, with the analogous expression for each worker task. The benchmark has no exogenous sector-specific productivity growth: A_m and A_s remain at their 1980 values.

The 11 task-productivity growth rates, together with the common elasticities (σ, ω), are chosen to match aggregate output per worker growth and 2019 employment shares within sectors, conditional on the endpoint skill distributions and requirement parameters. The Pareto parameters (a, γ_h) and the χ_j's account for the changing relative-wage targets described above. Capital per worker is supplied from the data. All other parameters remain fixed across dates. The calibrated values are reported in Tables 5 and 6.

Ranked by mean wage (except management)	Emp Wgts (ν_{ij}, η_i) Manu.	Serv.	m_j(%)	χ_j 1980	2019
Low Skill Services	**0.017**	**0.143**	**0.566**	**-**	**-**
Middle Skill	**0.816**	**0.520**			
Administrative Support	0.089	0.174	3.573	0.000	0.069
Machine Operators	0.258	0.016	5.968	0.010	0.012
Transportation	0.112	0.080	3.078	0.026	0.000
Sales	0.026	0.121	1.386	0.047	0.168
Technicians	0.034	0.040	-1.104	0.047	0.205
Mechanics & Construction	0.165	0.065	2.740	0.047	0.116
Miners & Precision Workers	0.132	0.027	6.452	0.055	0.101
High Skill	**0.167**	**0.337**			
Professionals	0.069	0.193	-1.926	0.059	0.205
Management Support	0.098	0.144	-1.754	0.090	0.267
Management	0.077	0.131	-2.658	-	-

Table 6: Calibrated Employment Weights, Growth Rates and Skill Requirements

Discussion As implied by the employment data in Figure F.6(a), the manufacturing sector has higher intensity parameters among middle skill jobs and a lower intensity in managers.[26] Since the estimated elasticity between manufacturing and services is less than one, for structural change to occur, productivity needs to rise by relatively more in those occupations used more intensively by manufacturing, which are the middle-skill jobs. This is evident from the productivity-growth column in Table 6.

The calibrated elasticities are $\omega = 0.250 < \sigma = 0.610 < 1$, with both values held fixed across dates. Managerial and worker composites are therefore more complementary than worker tasks are to one another. A useful comparison for the elasticity across tasks is Goos et al. (2014). Their point estimate for σ is around 0.9, which is much closer to 1 than ours.[27] However, theirs is an empirical framework that does not take into account general equilibrium or aggregate effects. In our model, $(1 - \sigma)m_j$ is the direct task-demand component; equilibrium prices, worker reassignment, managerial selection and sectoral reallocation also affect occupational employment (Appendix D.1). If we were to set $\sigma = \omega = 0.9$, we would recover much higher values for m_j to explain the employment share changes in the data. This would result in sectoral and aggregate TFP growth rates that are unrealistically high.

4 Quantitative Analysis

The quantitative analysis asks which combinations of productivity, requirements, and skill supply explain the joint wage–employment pattern between 1980 and 2019, especially wage changes at the upper end. We first report the calibrated fit, then compare the mechanisms using the existing counterfactuals, and finally connect the recovered changes to occupational task content. Endpoint wages and employment discipline the calibration. The counterfactual comparisons and task-content associations give the fitted changes their economic interpretation.

4.1 Model Fit

Figure 9 plots the model implied trends in employment shares across occupations by sector, as well as sectoral shares and aggregate occupational employment shares, against the data. When computing

[26] Since we normalize $M_j \equiv M$ in 1980, the parameters correspond to skill-adjusted employment weights.

[27] Since manager is not a special occupation in that paper, it lacks a counter part to our ω.

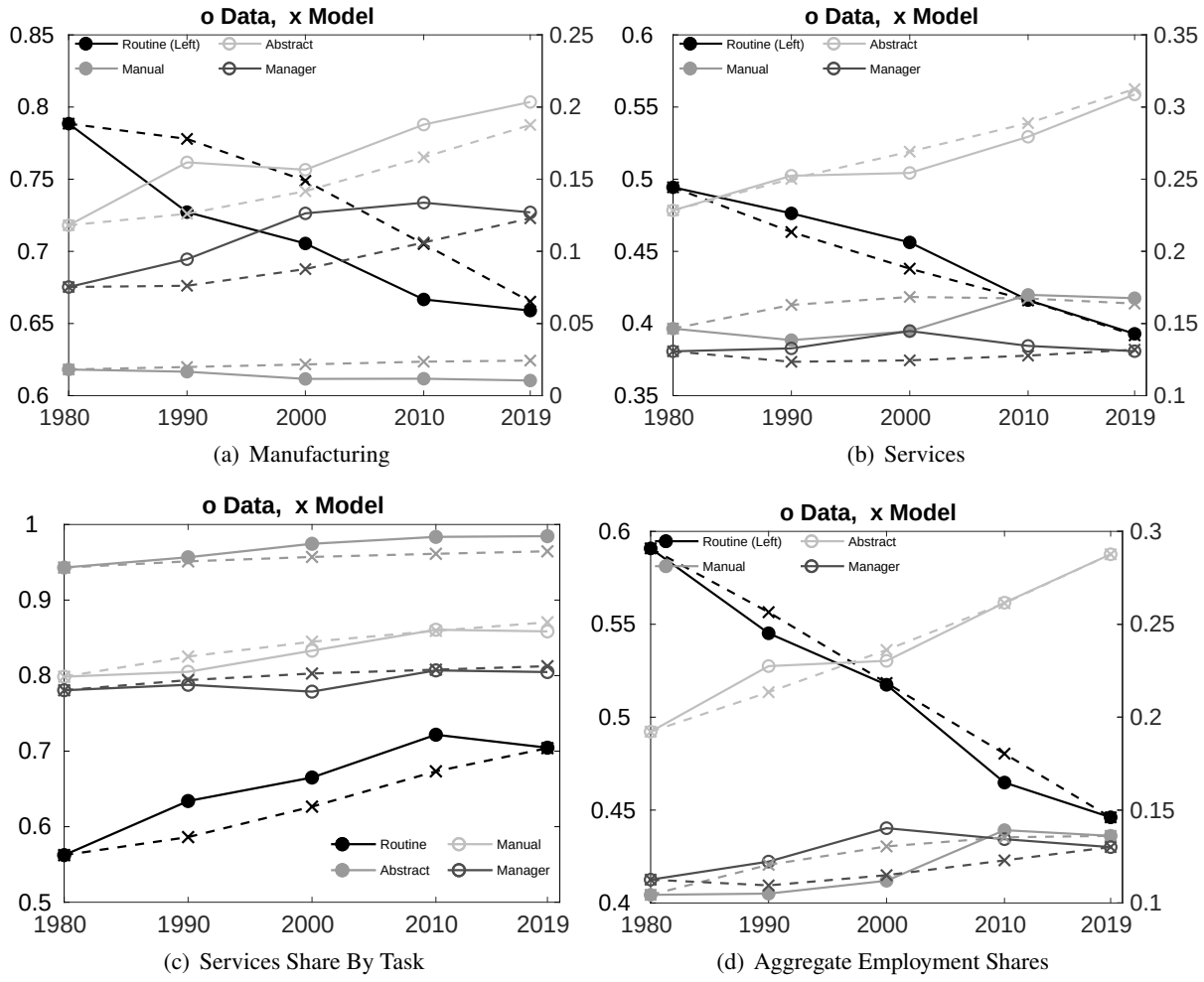


Fig. 9: Data vs. Model, Employment Shares by Task.

The quantitative model has 11 one-digit occupation groups. For graphical representation only, we re-group the 10 worker occupations into the 3 broader categories of manual, routine, and abstract as in Table 6. Panels (a) and (b) show within-sector occupational employment shares, with routine jobs on the left axis and the other groups on the right. Panel (c) shows the fraction working in services for each occupation group; Panel (d) shows aggregate occupational employment shares.

the simulated paths for 1990, 2000, and 2010 we plug in the empirical values of $K_t = k_t/y_{1980}$ and the task-specific TFP's implied by the calibrated growth rates, and solve for the respective equilibrium allocations. For ease of graphical representation, the figure groups the 10 worker occupations ranked by their 1980 mean wage into 3 broader categories summarized in Table 6: manual, routine, and abstract.

The fit in Figure 9 is assessed conditional on the endpoint skill distributions and requirements. The 11 task-productivity growth rates and two common elasticities are disciplined by within-sector occupational employment shares and aggregate output growth; the skill parameters are disciplined by the relative-wage targets. The reported paths exhibit larger routine-share declines and managerial-share increases in manufacturing than in services. These relative magnitudes are quantitative results, separate from the sufficient conditions for the signs in Proposition 6.

The manufacturing employment share is matched in 1980, but its subsequent change is not a separate calibration target. The fact that aggregate occupation shares and structural change by occupation (i.e., the rise in service share for each occupation group) are almost exactly replicated, as seen in Figure 9(c)-(d), is also a success of the quantitative model. This is consistent with the exposure mechanism in Proposition 6

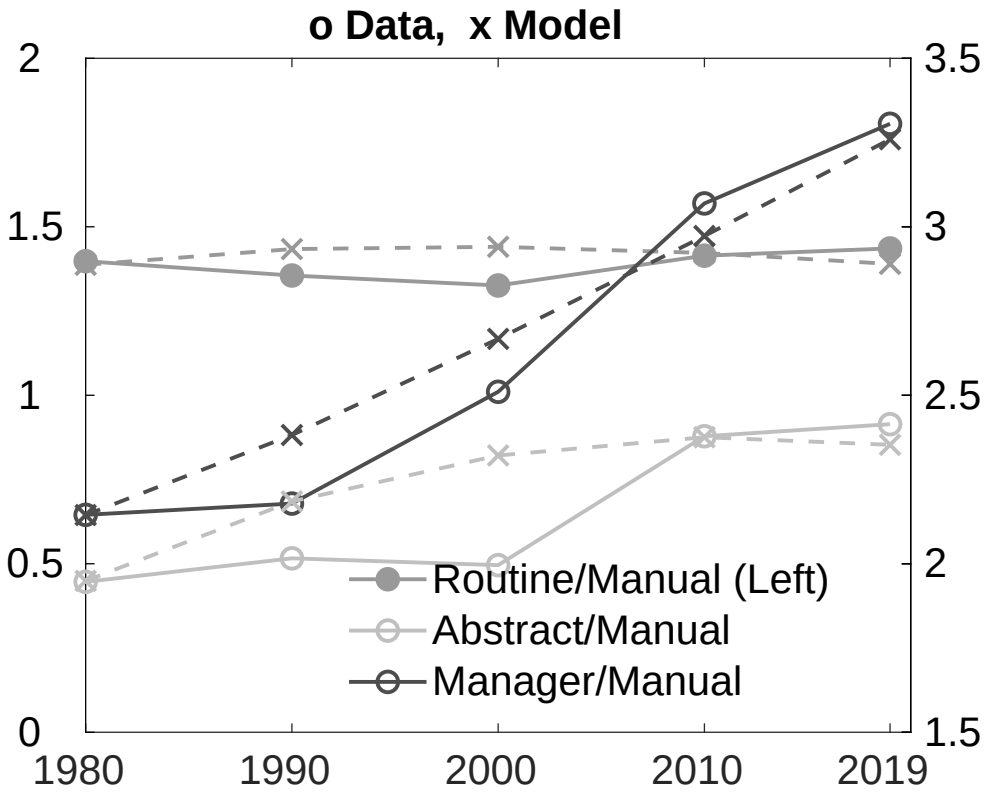


Fig. 10: Data vs. Model, Relative Occupational Wages.
Routine, abstract, and manager mean wages are each divided by the manual mean wage. Circles denote data and crosses denote the model. The routine ratio uses the left axis; the abstract and manager ratios use the right axis.

and the composition effect in Lemma 3; it does not establish that the calibration satisfies the proposition's sufficient restrictions.

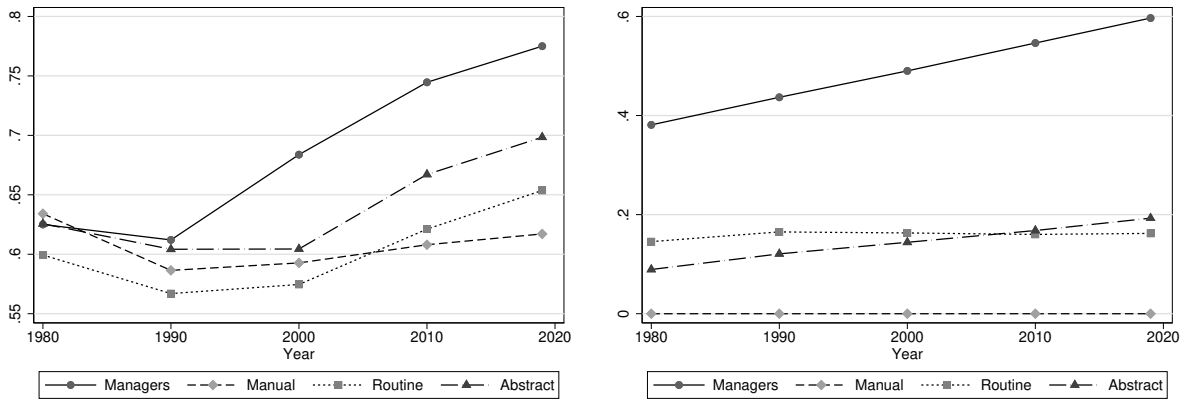
Relative wages Relative occupational wages in 1980 and 2019 discipline the skill distribution and requirements. Two features of the model matter for reading the fit. First, earnings at a fixed skill and occupation are common across sectors; broad-group sectoral means differ because manufacturing and services use detailed occupations in different proportions. The data also exhibit a persistent within-occupation manufacturing wage premium, which the model abstracts from.[28]

Second, occupational mean wages depend on selection as well as skill prices. An expanding occupation can draw workers with lower average skill, attenuating mean wage growth even when the wage paid per unit of skill rises. This composition effect applies where effective skills vary within the occupation; it need not accompany every expansion. In particular, the manual task has constant effective skill $\bar{h} = 1$, so changing selection does not directly change its mean wage. Complementarity and the equilibrium price w_0 govern that wage.

Appendix Figure F.1 reports the corresponding within-sector wage ratios using each sector's manual wage as the denominator. These broad-group comparisons also reflect the sectoral composition of detailed occupations.

Figure 10 compares routine, abstract and managerial mean wages with the manual mean wage. The model reproduces the broad rise in abstract and managerial relative wages, with the largest gains for

[28]The sectoral wage difference is relatively stable over the observation period; see Appendix Figure F.8(b). In the discrete model, mean skills within each occupation are equal across sectors.



(a) Data (b) Model

Fig. 11: Within-Occupation Wage Inequality

Standard deviation of log wages within the four occupation groups in the data and the model. Left scale for managers and right scale for the rest.

managers, though gaps remain in the abstract ratio and intermediate paths. The manual technology fixes effective skill at $\bar{h} = 1$, so manual wage changes come entirely through w_0. For the other occupations, requirements and selected skills also matter. The counterfactuals below distinguish their contributions.

Within-group wage inequality Because our quantitative model has a continuum of skills and discrete occupations, it has implications for wage inequality *within* occupation groups as well. As shown in Figure 11(a), the standard deviation of log wages rose substantially among managers, slightly among abstract workers, dropped among routine workers and remained more or less constant among manual workers.

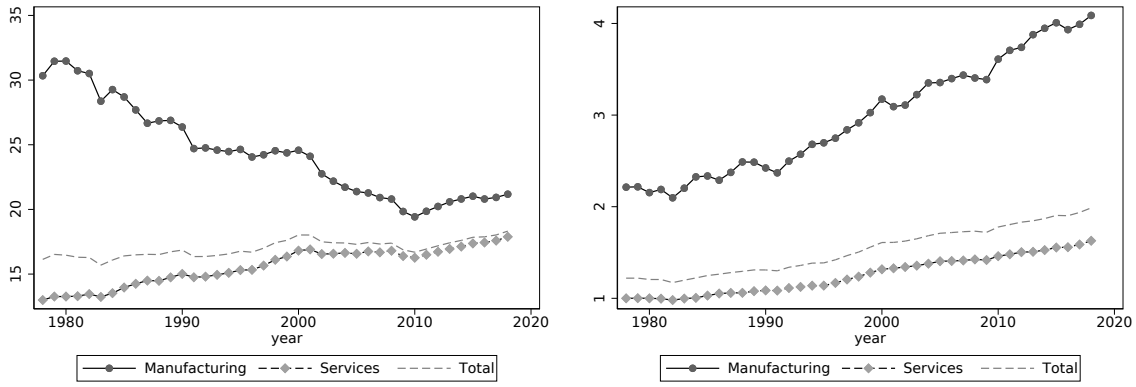
The reported model reproduces these qualitative within-occupation patterns, but understates their magnitudes. Since occupational mean wages do not determine the full within-group distribution, dispersion is a further dimension on which to assess the model. A richer account of skill accumulation or managerial heterogeneity could affect this comparison; the present exercise does not identify those additional channels.

Average size of establishments The managerial margin also has implications for establishments, conditional on a stable relationship between managers and establishments. With a constant number of managers per establishment, the faster rise in manufacturing's manager-to-worker ratio implies a faster decline in workers per establishment. Figure 12(a) shows this pattern in the BDS. Panel (b) combines BEA value added with establishment counts and shows the accompanying growth in output per establishment. The model has constant returns to scale and does not determine an establishment-size distribution; these comparisons connect the managerial mechanism to organization rather than constitute establishment-level predictions.

Appendix Figures F.3 and F.4 report the related cross-industry comparisons. Aggregate employees per establishment remain roughly constant despite the rise in managerial employment, indicating that managers cannot be identified one-for-one with establishments throughout the period. This motivates distinguishing managerial functions and hierarchies when interpreting the organizational margin.

4.2 Employment and Wage Dynamics

The main counterfactual result is the contrast between employment reallocation and wage growth. Productivity changes move jobs across occupations, but their wage and managerial-employment implications



(a) Average No. of Workers

(b) Average Value-added Output

Fig. 12: Establishment Size by Sector

Source: BDS and BEA Industry Accounts, 1978–2018. Left: Average number of workers per establishment, in aggregate and by sector. The BDS manufacturing group includes mining, utilities, construction and manufacturing; services comprise the remaining reported industries. Right: Real value-added output per establishment, normalized by service output per establishment in 1980. BEA services output includes government; BDS counts cover private establishments.

differ substantially from the data. Requirements and skill supply reconcile these margins. We examine that comparison first, then show how the occupational mechanism carries through to sectoral output and employment.

Wage growth, skill requirements, and skill supply Figure 13 makes the wage discrepancy clear. Task-productivity changes alone produce too much manual wage growth and too little abstract and managerial wage growth. Adding changes in requirements and the skill distribution brings the wage pattern closer to the data. Conversely, skill changes alone produce substantial abstract and managerial wage gains, but fail to reproduce the contraction of routine employment and manufacturing in Figure 14. The joint explanation requires both sets of forces.

The managerial comparison is especially informative. Managers organize complementary worker tasks, so improvements in those tasks can raise demand for managerial services as well as managerial earnings. In the productivity-only counterfactual, this benefit brings a much larger managerial employment increase than observed, even while producing too little managerial wage growth. In the data, managers remain the largest relative-wage gainers, but their employment share peaks and then falls; other high-paying worker groups continue to expand. The full model combines a smaller employment increase with larger earnings gains. Explaining management as simply another expanding high-skill occupation would miss this contrast. The model's separate managerial skill and occupational selection allow wages and employment to respond differently.

In the discrete model, mean managerial earnings relative to manual earnings satisfy $\bar{w}_z/w_0 = \bar{z}/\hat{z}$ under $\bar{h} = 1$ (Appendix E). The selected mean $\bar{z}$ and the relative skill price $1/\hat{z}$ can change even when managerial employment does not expand. Selection is based on managerial comparative advantage over worker earnings, rather than on h alone. This distinguishes management from both manual work, where effective skill per worker is fixed, and high-skill worker tasks, where requirements govern the return to h. The identity does not by itself assign the observed wage gain to selection or prices.

Upskilling does not automatically make managers the largest beneficiaries: Proposition 4 identifies requirement changes that favor high-h workers relative to fixed managerial types, while Proposition 5 isolates the scarcity of managerial services. These comparisons distinguish the mechanisms; their

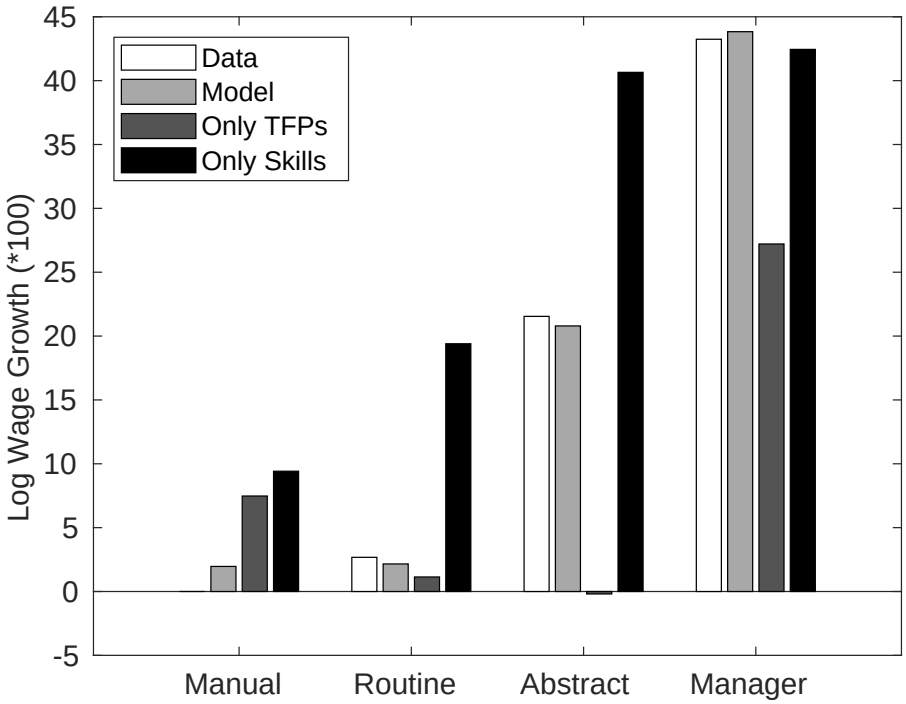


Fig. 13: Benchmark vs. Counterfactuals, Occupational Wages.
Changes in log mean occupational wages, less the observed manual-wage change, multiplied by 100; the same observed manual-wage change is subtracted from every series. The bars compare the data, full model, and the indicated combinations of productivity and skill changes.

assumptions do not identify the historical shocks.

These experiments are equilibrium comparisons, not additive contributions. Requirements and skill supply change both selected skills and skill prices; their effects depend on the productivity changes with which they interact. The analytical results establish these distinct partial effects under explicit sufficient conditions. The counterfactuals show their magnitudes and interaction over the calibrated period. Together, they explain why an employment-based account of technological change leaves essential wage information unused.

From occupational change to sectoral change. We next analyze the role of TSTC in structural change using two counterfactuals.

(1) We restrict all task-specific TFP growth to be equal, $m_j = m$ for all j, and instead let the exogenous sector-specific TFPs (A_m, A_s) grow at rates a_m and a_s, respectively. We jointly recalibrate m, a_m, and a_s to match the empirical growth rate of the aggregate and sectoral TFP's from 1980 to 2019. This version only has exogenous sector-specific TFP growth but no TSTC.[29]

(2) We allow both exogenous task- and sector-specific TFP growth, and recalibrate $\{m_j\}_{j=z,0}^{9}$, a_m, and a_s to match the change in employment shares *and* the empirical growth rates of the aggregate and sectoral TFP's from 1980 to 2019. Recall that our benchmark calibration of Section 3 restricted a_m and a_s to be 0 but did not target sectoral TFPs.

[29] Sectoral TFP is constructed from the NIPA accounts. Real value-added and capital are computed via cyclical expansion from the industry accounts, labor is computed from full-time equivalent persons in production in NIPA Table 6, and TFP is the Solow residual by sector.

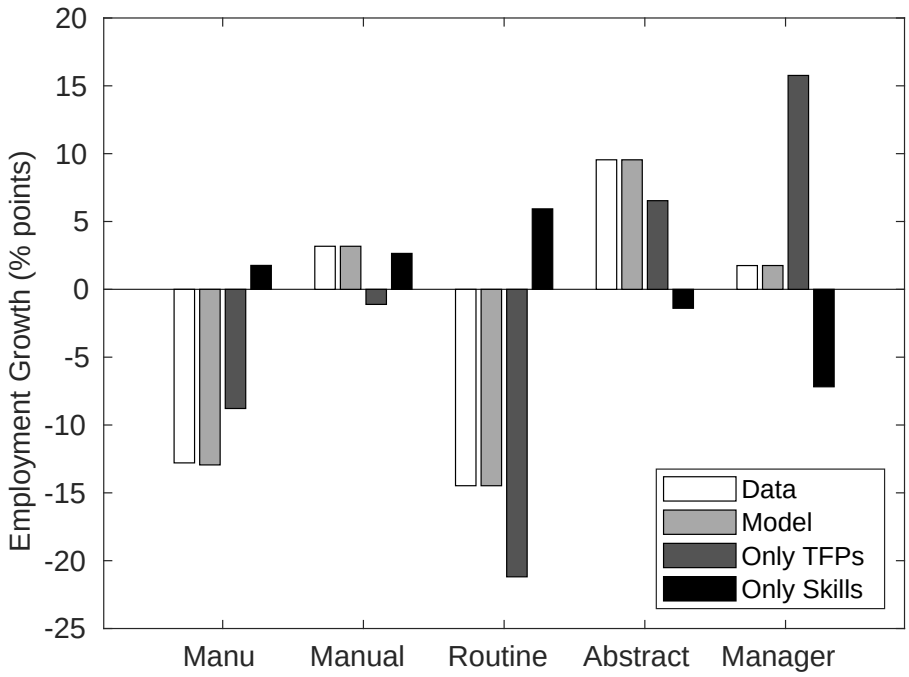


Fig. 14: Benchmark vs. Counterfactuals, Occupational Employment Shares.
Percentage-point changes in employment shares; the bars compare the data, full model, and the indicated combinations of productivity and skill changes.

In both cases, we keep all other parameters at their benchmark values of Tables 5-6, and only recalibrate the TFP growth rates.

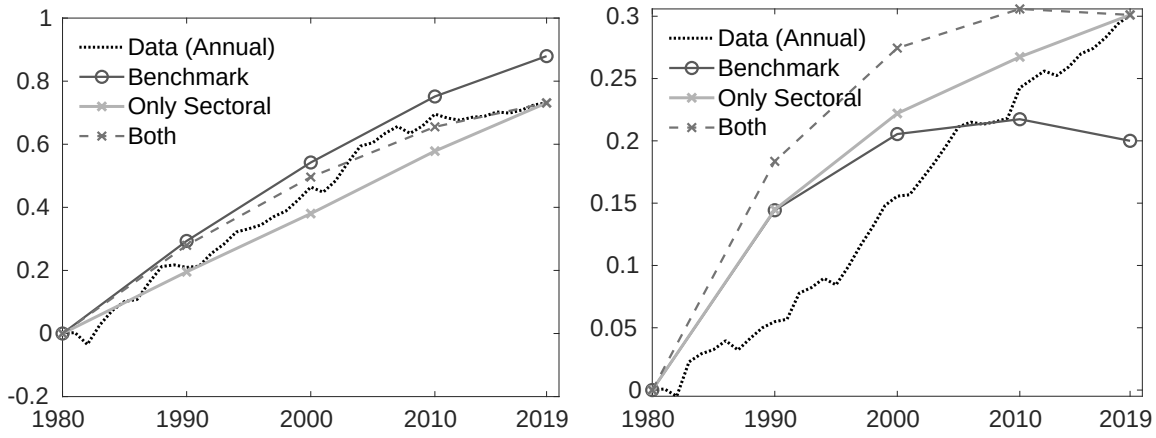
We focus on sectoral TFP's since in our model structural change results from the differential TFP growth between sectors—expressed in closed form in (21)—whether it is exogenous (caused by a_m and a_s) or endogenous (as in Section 2.9). The counterfactual paths are compared over the same 1980–2019 interval as the benchmark.

TFP and output growth Figure 15 shows the paths of sectoral log TFP's in the data, in our benchmark calibration, and in the two counterfactuals. By construction, all four match aggregate TFP and GDP growth between the two calibrated endpoints (Appendix Figure F.10).[30]

Instead, we focus on the evolution of sectoral TFP's. Note that the calibrated TSTC rates are higher among routine jobs, on which manufacturing is more reliant. Proposition 6 gives sufficient conditions under which greater routine exposure produces both higher relative manufacturing TFP and a declining employment share. However, our benchmark calibration did not target sectoral TFP's, so the question is whether their growth can be explained by our benchmark with only TSTC.

Figure 15 compares the sectoral TFP paths, while Appendix Figure F.11 reports sectoral output. Sectoral TFP and employment are distinct outcomes: capital allocation, occupational composition, and average earnings also enter the relationship between output and headcounts in (23). The sectoral comparisons therefore complement the wage–employment decomposition rather than identify the skill channels on their own.

[30]Denoting aggregate TFP as Z_t, since $Y_t = Z_t K_t^{\alpha}$ (labor is normalized to one) and we plug in the empirical values of K_t for all calibrations, it is the same whether we match aggregate TFP or GDP.



(a) Log TFP, Manufacturing (b) Log TFP, Services

Fig. 15: Benchmark vs. Counterfactuals, Sectoral TFP.
Data: NIPA. Log 1980 levels are normalized to 0, so the slopes of the lines are the growth rates.

Structural change and polarization Figure 16 compares the service employment share under the benchmark and the two counterfactuals. Appendix Figures F.12 and F.16 report the corresponding occupational outcomes. Matching sectoral TFP does not by itself guarantee a match to these employment shares, because relative average earnings also enter (23).

The comparison separates occupation-specific from sector-specific productivity changes. Lemma 3 further isolates the pure between-sector composition contribution; it does not restrict the endogenous within-sector responses to a sectoral shock. Section F.3 illustrates this accounting distinction using the 1980 and 2019 employment shares. The main quantitative question remains how productivity, upskilling, and skill supply jointly account for wages and employment.

4.3 Task Content, Productivity, and Wage Growth

The analytical and counterfactual results distinguish task productivity from changes in the use and supply of skills. We now ask whether the recovered changes correspond to different observable characteristics of work. Figure 18 relates them to employment and wages; Figure 19 relates them to task content and educational attainment. This is external evidence for interpreting the calibrated channels.

Figure 18(a) shows a correlation of -0.913 between task-productivity growth and employment-share changes. The relevant question is what kind of technological change those fitted productivity gains represent. We compare them with routine-task measures and then contrast the result with the effective-skill channel.

The aggregate routine task intensity (RTI) measures provide an initial comparison. Figure 17 uses the DOT-based measure in Autor and Dorn (2013), constructed from the task indices of Autor et al. (2003). We also consider the O*NET-based measure in Acemoglu and Autor (2011). Productivity growth is positively correlated with both, but their explanatory power is limited (Appendix Table F.3). The disaggregated task measures below are more informative.[31]

The college-share comparisons help distinguish the two channels. In the 1980–2019 regressions in Appendix Table F.3, task-productivity growth is negatively associated with the change in the college share in a univariate specification; that coefficient is not significant after routine-manual content is included.

[31] Appendix Figure F.6(b) reports the concentration of high-RTI occupations in the middle of the wage ranking, following Autor and Dorn (2013).

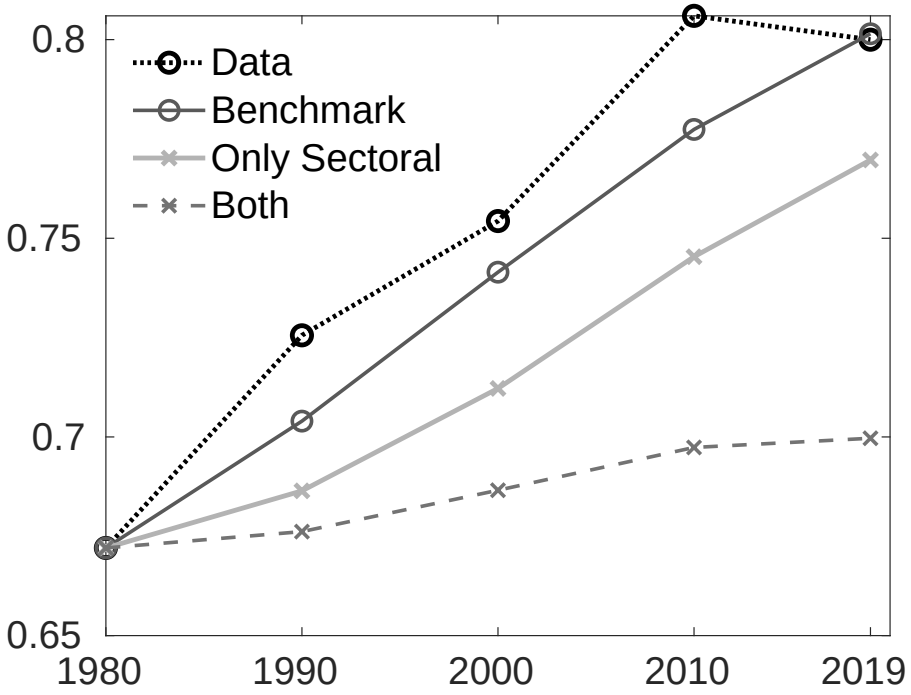


Fig. 16: Benchmark vs. Counterfactuals, Service Employment Share.
Vertical axis is the services share of overall employment.

By contrast, Appendix Table F.2 shows a positive association between the effective-skill change and the college-share change, alongside its association with nonroutine analytical content. These patterns motivate separate productivity and skill channels without making educational attainment a direct measure of either primitive. The distinction is between channels, not necessarily between their underlying causes. A common technological advance could change both task productivity and occupational skill requirements, with the two sets of task measures capturing different dimensions of that change. These associations help discipline the channels but do not identify whether their underlying source is common.

Routine-manual content is positively associated with task-productivity growth, while interpersonal content is negatively associated. The respective univariate regressions have R^2 values of 0.658 and 0.525 (Appendix Table F.3); Figure 19(a) shows the routine-manual comparison. The recovered productivity changes thus line up more closely with particular activities than with an aggregate routine index. They support a routine-automation interpretation of the employment channel.[32]

Task productivity can also reflect forces omitted from the model, including offshoring and differences in capital–skill substitutability.[33] The task-content comparison gives the fitted employment channel an economic interpretation without attributing every productivity gain to automation.

Task-productivity growth chiefly captures the reallocation associated with firms' demand for task output. The skill distribution and requirements jointly determine how workers' skills translate into effective input and earnings. Define the signed log change in effective skill relative to mean skill as

$$\Delta_\chi \equiv \log\left[1 - \frac{\chi_{1980}}{\bar{h}_{1980}}\right] - \log\left[1 - \frac{\chi_{2019}}{\bar{h}_{2019}}\right].$$

[32]Task productivity here is factor-neutral. A capital-augmenting improvement, such as a fall in task-specific capital costs, can generate a similar reallocation channel; see Section 2.6 and Goos et al. (2014).

[33]Task-specific capital–labor substitution differs from an elasticity that varies directly with worker skill, as in Krusell et al. (2000). Cheaper foreign labor can have a task-productivity-like effect; the calibrated rates have only a weak association with the offshorability measure of Firpo et al. (2011).

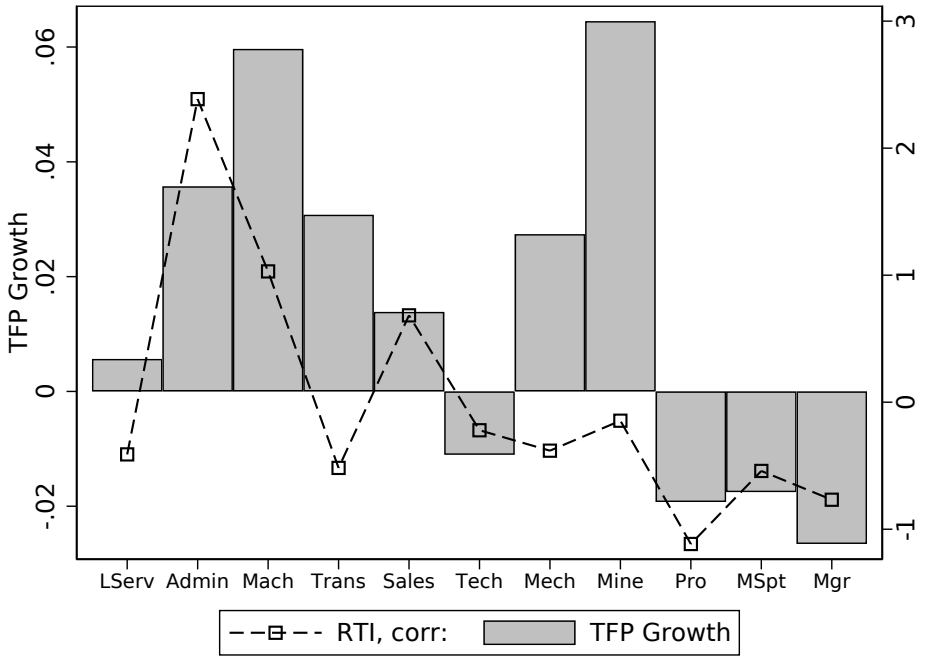


Fig. 17: Task-Specific Productivity Growth and Routine Task Intensity.
Calibrated task-specific productivity growth and the DOT-based routine task intensity (RTI) index across the 11 occupation groups. The two series use separate vertical axes. Here RTI is computed from broad-group mean task scores; Appendix Table F.3 instead uses the group average of detailed-occupation RTI.

The sign convention makes the index positive when effective skill relative to mean skill falls. It summarizes changes in requirements and skill composition; it is not an independently observed requirement shock. The figures and regressions use eleven occupation groups. For the managerial group, the plotted input is a stored managerial-skill statistic; it is not a worker requirement or a verified conditional mean.

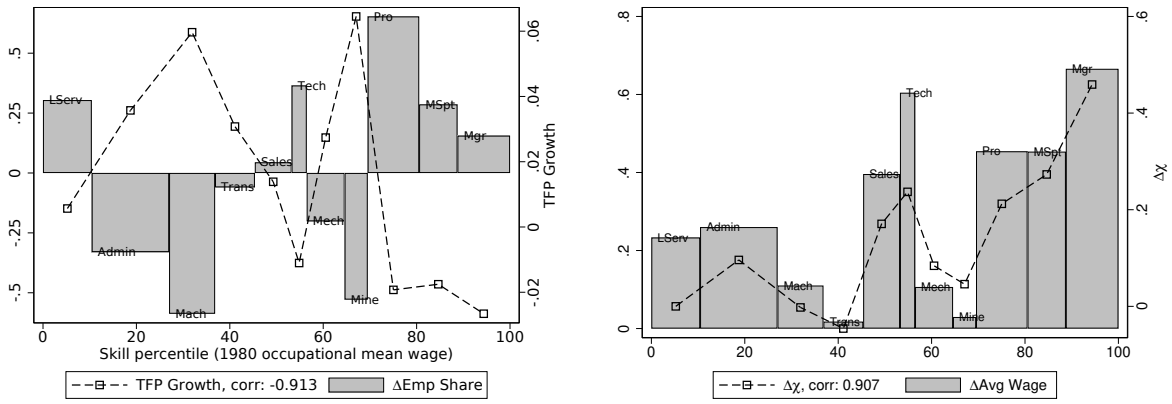
The measure combines requirements and skill supply. Its correlation with occupational wage growth is 0.907 in Figure 18(b). It is also associated with nonroutine analytical content and the college-share change, with correlations of 0.754 and 0.639 in Figure 19(b). These are correlations, not variance shares. The contrast with routine-manual productivity growth is the important result: the task characteristics associated with the movement of jobs differ from those associated with the changing rewards to skills. This evidence complements the counterfactuals' separation of the two channels.

5 AI and Other Applications

5.1 Artificial Intelligence, Wages, and Employment

AI raises the same joint question as the historical analysis: which activities become more productive, and how does this change the return to the skills used in them? Its exposure pattern need not resemble routine-manual automation. We first examine the existing occupational evidence for 2010–2019, then use an illustrative scenario to distinguish wage changes from employment-share changes.

Recent occupational evidence Table 7 reports the wage-growth regressions. Conditional on the preceding wage change, the AIOI measure of Felten et al. (2019) is positively associated with wage growth:



(a) Task TFP Growth and Employment Shares

(b) Skill Requirements and Wage Growth

Fig. 18: Calibrated Parameters vs Data

Panel (a) compares calibrated task-productivity growth with employment-share changes. Panel (b) compares occupational mean wage growth with Δ_χ, the change in effective skills relative to mean skill. The series use separate axes.

its coefficients are 0.015 and 0.018 in columns (3) and (4). The association is not statistically significant in the specifications without that control, and the Webb AI measure is not significantly associated with wage growth in either specification. The earlier nonroutine-analytical association is also weaker over this shorter period than in Table 3.

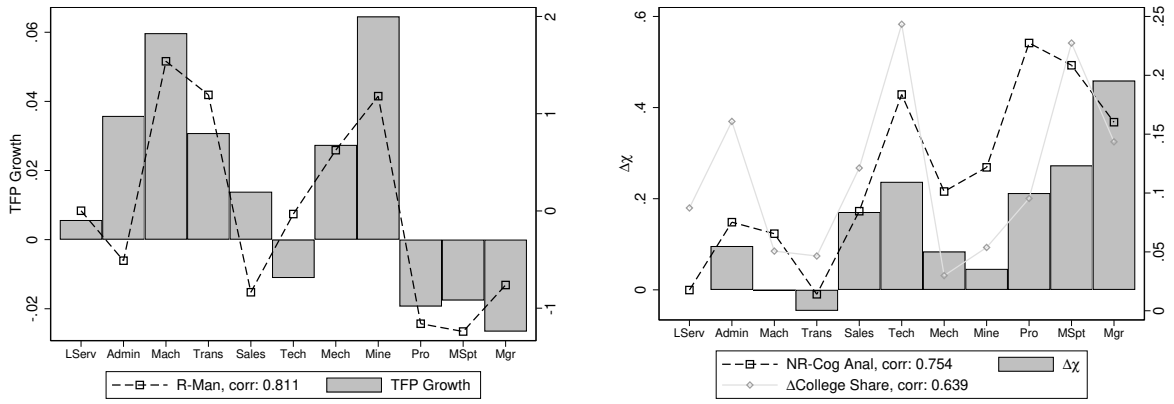
These results do not establish that AI caused the observed wage changes. They show that the relationship depends on both the exposure measure and the specification. In particular, an index measuring exposure need not measure displacement, and a positive wage association cannot by itself distinguish productivity gains, changing skill requirements, and selection.

Table 8 reports employment-share changes. AIOI has no statistically significant association in the reported specifications, whereas Webb's AI measure has a positive association. Thus the wage association of AIOI is not accompanied by a corresponding employment-share result, and the measures should not be collapsed into a single claim that AI uniformly expands or displaces exposed occupations.

An illustrative AI scenario The historical calibration and the O*NET comparisons in Section 4 motivate the productivity–skill distinction used in this application. The model permits AI to affect both task productivity and effective skills, but the retained AI illustration introduces exposure-linked changes in log effective skill while task productivities follow their historical continuation paths. Neither the exposure regressions nor the analytical comparative statics identify this particular mapping or determine the full equilibrium response.

Figure 20 illustrates the distinction using the existing AI scenario. Abstract employment shares rise in both paths, but rise less under AI than under the continuation benchmark. The AI path also gives higher abstract relative-wage growth than the projected 2010–2050 benchmark, while managerial relative-wage growth is lower than in that benchmark. These comparisons are within-occupation differences between the two projected paths. The plotted wage bars use a common historical normalization, so their position relative to zero should not be interpreted as an absolute wage gain or loss.

The preserved matching implementation constructs the scenario through exposure-linked changes in $q = \log(1 - \chi/\bar{h})$, with the manual exposure loading set to zero. An intercept matches continuation-benchmark GDP in 2030. Its managerial entry instead changes a managerial-skill statistic used to recover distribution parameters; it does not change a managerial task requirement. An isolated replay of the



(a) Task-specific TFP Growth and O*NET

(b) $\Delta\chi$, O*NET and College

Fig. 19: Parameters vs Measures

Panel (a) compares task-productivity growth with routine-manual task content. Panel (b) compares Δ_χ with nonroutine-cognitive analytical content and the change in the college-educated share. Legend entries report correlations; the series use separate axes.

October implementation matches both retained AI panels at plotted precision, although the original future workspace and generating invocation were not recovered. We therefore interpret the displayed comparisons as illustrative and conditional on the selected exposure-to-effective-skill mapping, not as forecasts or causal AI effects.

Exposure measures and distributional incidence We retain this broad-AI illustration and add one scenario using the human-rated β measure of Eloundou et al. (2024). The AIOI measure links progress in AI applications, including image recognition, translation and strategic games, to workplace abilities and then occupations. The LLM measure instead assesses whether assistance can halve the time needed for a task at comparable quality. Its $\beta = E1 + 0.5E2$ combines direct LLM assistance with assistance requiring additional software. These measures differ in technology scope, unit of measurement and assessment method, rather than being successive vintages of the same index.

We aggregate both measures with the existing 2010 labor-supply weights and occupational definitions, including self-employed people in the manager group, and standardize with equal weights across the eleven groups. The new measure covers 99.60% of labor supply; unmatched ratings remain missing. The group-level Pearson and Spearman correlations are 0.938 and 0.845. Agreement is weaker across 471 matched Census occupations (Pearson 0.840), and restricting both group measures to common support changes their correlation very little. Appendix F.1 reports the vectors, coverage, detailed correlations and exposure scatterplot. Administrative support becomes the most exposed group, ahead of management support. The standardized exposures of professionals, management-support workers and model managers fall by 0.234, 0.241 and 0.141, respectively, while sales exposure rises by 0.615. These are changes on a relative index scale, not changes in the percentage of workers displaced.

The additional scenario uses the same effective-skill mapping, historical dispersion loading and continuation rules as the retained illustration, with its intercept re-solved to match continuation GDP in 2030. The intercept falls from 0.471 to 0.391. Figure 21 compares the resulting paths. In 2050, abstract employment is 2.93 percentage points below continuation under LLM exposure, compared with 4.74 points under broad AI; abstract mean wages relative to the economy-wide mean are higher by 5.82 and 12.53 log points, respectively. Managerial relative mean wages are lower in both scenarios. Routine incidence differs: LLM exposure gives a smaller routine employment share and higher relative mean

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ_{2010} Wage	-0.201*	-0.201*	-0.233*	-0.208*				
	(0.086)	(0.087)	(0.090)	(0.082)				
NR-Cog Anal	0.015	0.015	0.008		0.002	0.003	-0.003	
	(0.008)	(0.008)	(0.008)		(0.007)	(0.007)	(0.008)	
AI (Webb)		0.000				-0.000		
		(0.006)				(0.007)		
AIOI (Felten et al.)			0.015**	0.018**			0.008	0.006
			(0.006)	(0.006)			(0.006)	(0.005)
Constant	0.048**	0.049**	0.051**	0.049**	0.034**	0.034**	0.034**	0.034**
	(0.008)	(0.008)	(0.008)	(0.008)	(0.006)	(0.006)	(0.006)	(0.006)
R^2	0.030	0.030	0.042	0.039	0.001	0.001	0.004	0.004

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$

Table 7: AI Exposure and Occupational Wage Growth, 2010–2019.
The dependent variable is the change in log occupational mean wages over 2010–2019. Columns (1)–(4) control for the corresponding 2000–2010 log change, denoted Δ_{2010}; columns (5)–(8) omit this control. Exposure and task-content regressors are standardized. Regressions are unweighted across occupations and use heteroskedasticity-robust standard errors. Exposure measures are from Webb (2020) and Felten et al. (2019). Entries reproduce the existing regressions; coefficients are descriptive associations.

wages than continuation, whereas broad AI gives a larger share and lower relative mean wages. Table F.5 reports the 2030 and 2050 levels. Matching GDP in 2030 isolates distributional incidence at that date; GDP need not agree thereafter.

The reduced abstract exposure under the LLM measure accompanies smaller abstract wage and employment effects under the imposed mapping. But standardized magnitudes and the GDP-matching intercept both change, and the equilibrium adjusts occupational prices, sorting and worker composition. The comparison therefore does not assign each outcome to a single ranking change. Its differing routine outcomes show that the detailed incidence is sensitive to how exposure is measured.

These remain conditional projections from the 2019 economy, not forecasts of adoption. The LLM ratings reflect 2023/2024-era capabilities, not measured 2026 capabilities, and technical time-saving potential need not translate into adoption or displacement. Applying the same exposure-to-effective-skill mapping to both measures is an assumption; neither measure identifies a requirement shock or the balance between task-productivity and requirements channels. Crosswalks, equal weighting within SOC and broad occupational aggregation lose within-group detail, while equal-group standardization fixes a relative intensity convention.

The comparison highlights why employment shares alone are an incomplete measure of the consequences of AI. A smaller employment share relative to a counterfactual can coexist with higher growth in an occupation's wage relative to the economy-wide mean, while both prices and the composition of its workers change. The scenarios make that possibility concrete; their quantitative paths remain conditional on the assumed evolution of technology and skills.

5.2 Robot Taxes

The AI exercise changes the direction of technological exposure. A related policy question is whether taxing capital used in routine tasks can preserve routine employment. Table 9 reports this experiment separately at the calibrated 1980 and 2019 equilibria. Here "robots" refer to capital assigned to routine

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ_{2010} Share	0.204*	0.226**	0.198*	0.198*				
	(0.079)	(0.084)	(0.084)	(0.084)				
R-Manual	-0.019	-0.023	-0.028	-0.028	-0.042	-0.048*	-0.060*	-0.060*
	(0.023)	(0.023)	(0.030)	(0.030)	(0.024)	(0.024)	(0.029)	(0.029)
AI (Webb)		0.083**				0.077**		
		(0.027)				(0.027)		
AIOI (Felten et al.)			-0.013	-0.013			-0.027	-0.027
			(0.045)	(0.045)			(0.044)	(0.044)
Constant	0.030	0.030	0.030	0.030	0.027	0.027	0.027	0.027
	(0.023)	(0.023)	(0.023)	(0.023)	(0.023)	(0.023)	(0.023)	(0.023)
R^2	0.035	0.077	0.035	0.035	0.011	0.048	0.014	0.014

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$

Table 8: AI Exposure and Occupational Employment-Share Changes, 2010–2019.
The dependent variable is the proportional change in the occupational employment share, $s_{2019}/s_{2010} - 1$, rather than a percentage-point change. Columns (1)–(4) control for $s_{2010}/s_{2000} - 1$, denoted Δ_{2010}; columns (5)–(8) omit this control. Exposure and task-content regressors are standardized. Regressions are unweighted across occupations and use heteroskedasticity-robust standard errors. Exposure measures are from Webb (2020) and Felten et al. (2019). Entries reproduce the existing regressions; coefficients are descriptive associations.

tasks, rather than a separately modelled technology or adoption decision.

In the reported 2019 experiment, a 50 percent tax raises the routine employment share from 44.61 to 46.12 percent and lowers output by 7.65 percent. The corresponding output loss in the 1980 economy is 11.82 percent, with the routine employment share rising from 59.08 to 60.56 percent. Thus protecting an occupational employment share can be costly in output, and the trade-off differs between the two calibrated economies. Both productivities and skills differ across those economies; this comparison alone does not isolate the contribution of upskilling. The exercise characterizes output and allocation effects, not an optimal tax or a welfare ranking with endogenous training and retirement.

5.3 Manufacturing Subsidies

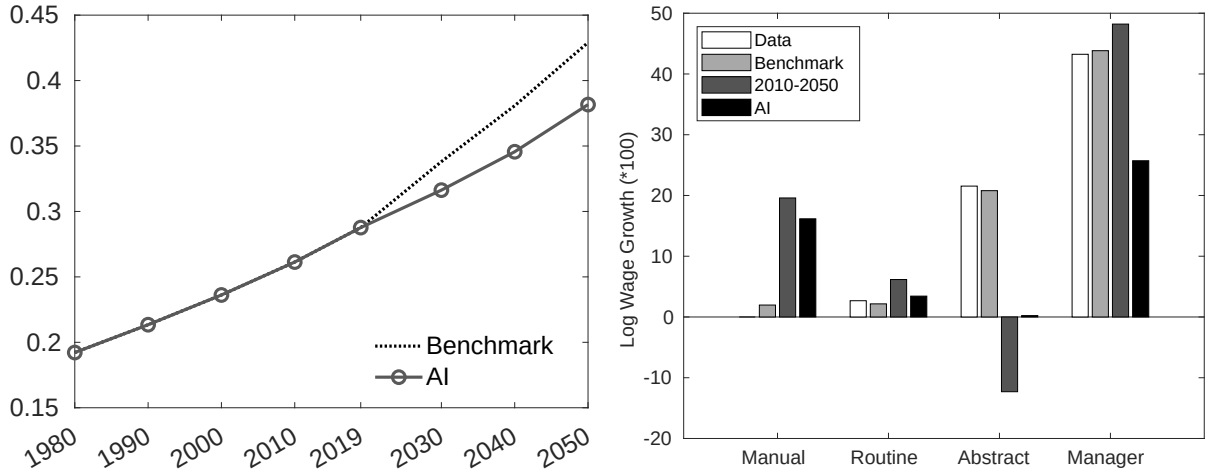
We consider an illustrative subsidy financed by a tax on services. For this calculation, impose equal sectoral average earnings before and after the policy, so the capital ratio $\kappa = K_s/K_m$ equals the employment ratio L_s/L_m. With total employment normalized to one,

$$L_m = 1/(1+\kappa),\ L_s = \kappa/(1+\kappa). \tag{37}$$

From (37), if the manufacturing employment share increases by x (measured as a fraction of total employment), κ must change to

$$\kappa' = \frac{\kappa - x(1+\kappa)}{1 + x(1+\kappa)}.$$

If service expenditures are taxed to subsidize manufacturing (change prices to $(1+\tau_s)p_s, (1-\tau_m)p_m$),



(a) Abstract Employment Share

(b) Occupational Wage Growth

Fig. 20: Wages and Employment in an Illustrative AI Scenario.

Panel (a) compares abstract employment shares under the continuation benchmark and AI scenario. Panel (b) compares log growth in occupational mean wages relative to the economy-wide mean wage, multiplied by 100, after subtracting historical manual relative-wage growth from every bar. The Data and Benchmark bars describe the historical comparison; the 2010–2050 and AI bars compare the projected continuation and AI scenarios. Interpret AI effects as differences between the last two bars within each occupation, rather than relative to zero. These are the existing illustrative simulations, not estimates of causal AI effects or forecasts of adoption.

(23) then changes to

$$\kappa' \equiv \frac{K_s}{K_m} = \frac{L_s}{L_m} = \left(\frac{1-\tau_m}{1+\tau_s}\right)\left(\frac{\gamma_s}{\gamma_m}\right)^{\frac{1}{\epsilon}}\left(\frac{Y_s}{Y_m}\right)^{\frac{\epsilon-1}{\epsilon}} = \frac{\gamma_s}{\gamma_m}\cdot\left(\frac{\Phi_s}{\Phi_m}\right)^{\epsilon-1}\left(\frac{1-\tau_m}{1+\tau_s}\right)^{\epsilon}$$

while budget balance requires

$$\tau_s p_s Y_s = \tau_m p_m Y_m \quad \Rightarrow \quad \tau_s = \tau_m \cdot \frac{p_m Y_m}{p_s Y_s} = \tau_m/\kappa'.$$

Combining these equations gives the tax and subsidy rates for a specified employment-share increase x. Table 10 reports $x = 0.05$ and $x = 0.10$ for the 1980 and 2019 economies. In 2019, a 5-percentage-point increase in manufacturing employment raises the routine employment share by only 1.22 percentage points. The reason is occupational composition: manufacturing also employs nonroutine workers, while routine jobs are used in services as well. Targeting a sector and targeting an occupation are different policies.

5.4 Work from Home

As a final illustration, the WFH and infectious-exposure scenarios redirect task-productivity growth toward occupations that are harder to perform remotely or more exposed to infection. They start from the calibrated 2019 economy, extrapolate the historical skill-distribution and requirement changes, and choose the productivity shift to match continuation-benchmark GDP in 2030. Appendix F.2 reports the measures and existing projections. These scenarios illustrate how the direction of task-level change affects occupational and sectoral employment; they are not estimates of realized pandemic effects.

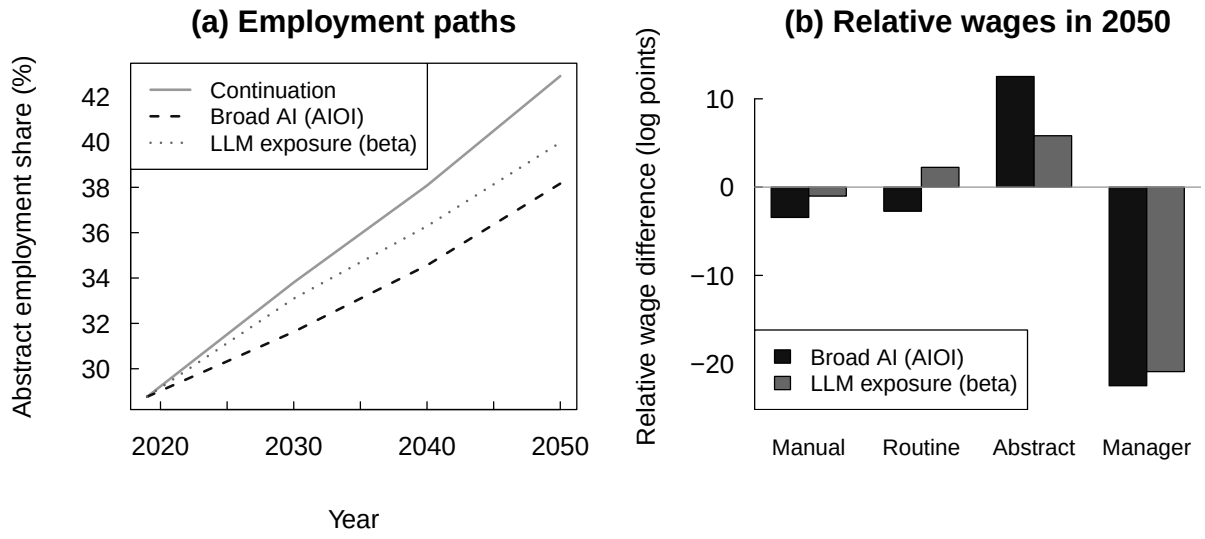


Fig. 21: AI Exposure Measures, Employment and Relative Wages.
Panel (a) compares abstract employment shares under continuation, the retained broad-AI scenario (AIOI) and the additional human-rated LLM-exposure scenario (β). Panel (b) reports 2050 differences from continuation in log occupational mean wages relative to the economy-wide mean, multiplied by 100. Within-occupation subtraction removes the common historical normalization used in Figure 20. Both exposure scenarios start from the existing 2019 baseline and match continuation GDP in 2030.

6 Conclusion

We have developed a framework that connects changes in occupational employment and wages to task productivity, skill requirements, and the distribution of worker and managerial skills. The model remains tractable despite its different layers: an additive requirement generates comparative advantage across worker tasks, while selection between workers and managers links the rewards to skills with occupational allocation. The analytical results distinguish the effects of each force under explicit sufficient conditions. Separating managerial skill from worker skill is motivated by the contrast within high-paying occupations: abstract workers gained employment share and relative wages, whereas managerial relative wages continued to rise after employment share began to fall. The model therefore accounts for the return to worker skill and the scarcity of managerial services jointly.

Our quantitative analysis of the United States from 1980 to 2019 shows that task-productivity changes account for much of employment reallocation, but produce too little abstract and managerial relative-wage growth and too much managerial employment growth. The two outcomes alone do not establish that two distinct causes are necessary. In our model, however, changes in requirements and skill supply help reconcile the wage pattern that task productivity alone misses. The occupational task evidence supports this channel: routine-manual content is associated with the recovered productivity changes, while analytical content and educational attainment are associated with changes in effective skill. These associations provide descriptive empirical support for upskilling, but do not establish it as the unique cause.

The applications illustrate why this distinction matters for technologies and policies with different occupational incidence. In both the broad-AI and LLM-exposure illustrations, abstract wages relative to the economy-wide mean grow faster but abstract employment grows more slowly than in the continuation benchmark. This finding survives the change in measure, while its magnitude depends on the exposure pattern. Relative to the same continuation benchmark, routine employment is higher and relative wages lower under broad AI, but employment is lower and relative wages higher under LLM exposure. These

τ_K (%)	Y Loss (%)	T/Y (%)	K_R/K (%)	L_R (%)
		1980		
0	-	-	53.25	59.08
10	2.86	1.72	51.46	59.44
20	5.43	3.11	49.82	59.75
30	7.75	4.24	48.32	60.04
40	9.87	5.17	46.93	60.31
50	**11.82**	**5.94**	**45.64**	**60.56**
		2019		
0	-	-	33.58	44.61
10	1.82	1.10	31.90	44.96
20	3.47	2.01	30.41	45.29
30	4.98	2.77	29.08	45.59
40	6.37	3.41	27.88	45.86
50	**7.65**	**3.96**	**26.79**	**46.12**

Table 9: Taxing Capital Used in Routine Tasks

τ_K is the tax rate on routine-task capital, T/Y is tax revenue as a share of output, and K_R/K and L_R are routine-task capital and employment shares. Output losses are relative to the no-tax equilibrium in the same year. All entries are percentages.

are conditional model comparisons, not forecasts. Robot taxes, manufacturing subsidies, and WFH shift task demand in other ways. In each case, changes in employment alone give an incomplete account of the distributional consequences, which also depend on relative wages and occupational selection.

L_m (ppt)		τ_s (%)		L_R (%)	
1980	2019	1980	2019	1980	2019
32.79	19.86	-	-	59.08	44.61
+5		60.76	33.09	60.60	45.83
+10		**74.81**	**42.58**	**62.11**	**47.06**

Table 10: Manufacturing Subsidies in the 1980 and 2019 Economies

Columns compare the calibrated 1980 and 2019 economies. The first row gives no-policy employment shares. Subsequent rows target increases of 5 and 10 percentage points in manufacturing's employment share; τ_s is the required service tax and L_R is the aggregate routine employment share. Employment shares and tax rates are in percent. The calculation retains the equal-sectoral-average-earnings condition stated in the text.

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Occupation Group		occ1990dd
Managers	self-employment+	4– 19
Management Support		22– 37
Professionals		43–199
Technicians		203–235
Sales		243–283
Administrative Support		303–389
Low Skill Services		405–472
Mechanics and Construction		503–599
Miners and Precision		614–699
Machine Operators		703–799
Transportation Workers		803–899

Table A.1: Census Occupation Groups

322 non-farm occupations according occ1990dd (Dorn, 2009), itself harmonized from occ1990 (Meyer and Osborne, 2005), are grouped into 11 occupation groups in order of their occ1990dd code. All other occupation groups correspond to their 1-digit census occupation group except for management support, technicians and sales. Groups are presented in their (contiguous), ascending order of their codes, excluding agricultural occupations 473–498 which are dropped. In the main text, occupation groups are presented in ascending order of skill (mean hourly wage).

Online Appendix

A Census Employment/Wages/Occupations

We use the 5% census samples from IPUMS USA. We drop military, unpaid family workers, and individuals who were in correctional or mental facilities. We also drop workers who work either in an agricultural occupation or industry.

For each individual, (annual) employment is defined as the product of weeks worked times usual weekly hours, weighted by census sampling weights. Missing usual weekly hours are imputed by hours worked last week when possible. Missing observations are imputed from workers in the same year-occupation-education cell with 322 occupations×6 hierarchical education categories: less than high school, some high school, high school, some college, college, and more than college.

Hourly wages are computed as annual labor income divided by annual employment at the individual level. Hence while employment shares include the self-employed, hourly wages do not include self-employment income.[34] We correct for top-coded incomes by multiplying them by 1.5, and hourly wages are set to not exceed this value divided by 50 weeks × 35 hours (full-time, full-year work). Low incomes are bottom-coded to first percentile of each year's wage distribution.

For the line graphs in Figures 1–2(b), we ranked occupations by their hourly wages defined as above, and smoothed across skill percentiles using a bandwidth of 0.75 for employment and 0.4 for wages; these are the same values used in Autor and Dorn (2013). For the bar graphs in Figures 1–2(b), 19 and F.6, we grouped the 322 occupations vaguely up to their 1-digit Census Occupation Codes, resulting with the 11 categories summarized in Table A.1 and used for our quantitative analysis. In the figures and in Table F.1, these groups are then ranked by the mean wage of the entire group. In particular, in Figures 1–2(b), and F.6, the horizontal length of a bar is set to equal the corresponding group's 1980 employment share, which does not necessarily coincide with the 3-digit occupations used to generate the smooth graphs by

[34]While we have only considered labor income in the paper, we have conducted robustness checks by including business income as well. Hourly business income is defined similarly as hourly wages. We also separately corrected for top-coding (the top-codes for labor and business income differ) and bottom-coding in a similar fashion.

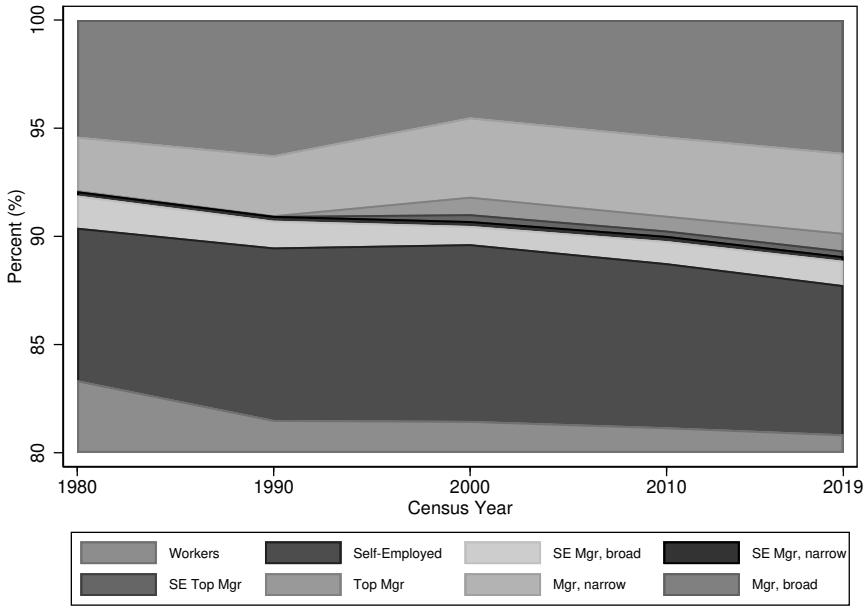


Fig. A.1: Managers in the Census

Source: U.S. Census (5%). Top managers are coded 4 in `occ1990dd` while broad managers include code 22 which are not-elsewhere-classified managers, or manager occupations that do not exist across all 4 censuses.

percentile.

Our benchmark employment classification includes the self-employed in the manager group. Figure A.1 separates the managerial and self-employed categories and shows that the employment share of non-managerial self-employed workers is comparatively stable. This is relevant to interpreting the managerial employment trend, but does not by itself establish invariance of wage moments or calibrated results to the classification. The benchmark includes the three self-employed categories, top managers and narrow managers, but excludes broad managers. Top managers are coded 4 in `occ1990dd` and include CEOs, public administrators and legislators. Broad managers are coded 22 and are either not elsewhere classified or unavailable in some censuses.

B Definition of Competitive Equilibrium

All markets are perfectly competitive. Individuals only care about which task they perform and are indifferent across sectors, and choose the task that yields the highest wage. Let $w_h(h)$ denote the wage of a worker with human capital h, and let w_z denote the efficiency wage per unit of managerial skill z for managers—the latter is possible since all managers perform the same task and we assume a constant returns to scale technology. Then an individual with skill $s = (h, z)$ can choose management if

$$w_z z \geq w_h(h). \tag{38}$$

At equality either occupation is optimal, and a positive mass of tied types can be split across occupations. As in the planner's problem, the density notation below describes the regular case; general type allocations are measures adding up to μ.

The final good is produced by a representative firm that takes sectoral prices as given, whose profits are given by

$$PY - p_m Y_m - p_s Y_s,$$

where Y is given in (4), P is the final good price and p_i the price of the sector $i \in \{m, s\}$ good. Normalizing $P = 1$, profit maximization yields (27) in the main text.

A manager with skill z in sector i maximizes profits taking R, the rental rate of capital R, and $w_h(h)$, the wage of a worker with human capital h, as given. For expositional purposes, it will be easier to exploit our constant returns assumptions to aggregate over managers in (1) for all quantities.[35] As in the planner's problem, let $l_{iz}(s)$ denote the number of individuals with skill $s = (h, z)$ working as managers in sector i, then $Z_i \equiv \int z l_{iz}(s) ds$ is the total amount of manager skill in sector i. To this end, define

$$[l_{iz}(s), l_{ih}(s, j), K_{iz}, K_{ih}(j), Z_i, H_i(j), X_{iz}, X_{ih}, T_i(j)]$$

identically as in the main text, except now they are equilibrium allocations to solve for rather than the solution to the planner's problem. In what follows, we solve the problem of a representative sector i firm who uses technology (7).

Let c_z and $c_h(j)$ denote the unit cost of producing 1 unit of task z and task j output, respectively. Then the sector i firm's problem is

$$\begin{aligned}
&\max \left\{ p_i Y_i - c_z X_{iz} - \int_j^J c_h(j) T_i(j) dj \right\}, \quad \text{where} \\
&c_z = \min \left\{ R K_{iz} + \int w_z Z_i \right\} \quad \text{s.t.} \quad X_{iz} = 1, \\
&c_h(j) = \min \left\{ R K_{ih}(j) + \int w_h(h) l_{ih}(s, j) ds \right\} \quad \text{s.t.} \quad T_i(j) = 1.
\end{aligned}$$

In equilibrium, all firms maximize profits and capital and labor markets clear. Profit maximization implies

$$c_z = p_i A_i^{1-1/\omega} \left[\frac{\eta_i Y_i}{X_{iz}} \right]^{\frac{1}{\omega}}, \quad c_h(j) = p_i A_i^{1-1/\omega} \left[\frac{(1-\eta_i) Y_i}{X_{ih}} \right]^{\frac{1}{\omega}} \left[\frac{\nu_i(j) X_{ih}}{T_i(j)} \right]^{\frac{1}{\sigma}} \tag{39a}$$

and cost minimization implies that the amount of capital chosen for each task, and total units of managerial skill chosen for the managerial task, satisfy

$$\alpha c_z X_{iz} = R K_{iz}, \quad \alpha c_h(j) T_i(j) = R K_{ih}(j), \quad (1-\alpha) c_z X_{iz} = w_z Z_i. \tag{40a}$$

Similarly, the optimal choice of labor satisfies

$$(1-\alpha) c_h(j) T_i(j) \cdot b(h, j) \leq w_h(h) H_i(j) \quad \text{with equality if } l_{ih}(s, j) > 0. \tag{40b}$$

We already imposed market clearing of the sectoral goods in (7) and (27), that is, we used Y_i to denote both the supply and demand for sectoral goods. The equilibrium capital and labor market clearing conditions

[35] While straightforward, solving for the aggregation yields several additional implicit prices at the indivdual manager level without adding any insights. Details are available upon request.

are identical to the resource constraints (5) and (6), respectively, where K is the exogenously given supply of capital and K_i are the sectoral supply and demand.

Definition 1 *A competitive equilibrium is a set of capital allocations* $\{K_{iz}, K_{ih}(j)\}$*, labor allocations* $\{l_{iz}(s), l_{ih}(s,j)\}$*, and prices* $\{p_i, R, w_z, w_h(h)\}$ *satisfying* (5)-(6)*,* (27)*, and* (40) *for* $i \in \{m, s\}$*, with each individual choosing an earnings-maximizing activity. Management is optimal under* (38)*, and tied types may be split. Labor allocations and clearing are understood as measures for general skill distributions and as the displayed densities in the regular case.*

Most equations from (9) can be replicated from (40), with the additional step of plugging in the values for c_z and $c_h(j)$ at every step. Of course, the implicit cost of tasks can be of interest of themselves, but since we did not pursue this route in our main analysis, we focus on the planner's problem for ease of notation.

C Proofs

C.1 Proof of Lemma 1 and Corollary 1

The feasibility constraint (6) and existence of $\hat{h}(j)$ and $\hat{z}(j)$ imply that the number of people with skill s assigned to task j is

$$l_h(s,j)ds = \delta(j - \hat{j}(h)) \cdot \mathcal{I}\left[z \leq \tilde{z}(h)\right] d\mu$$

where $\delta(\cdot)$ is the Dirac delta function and $\mathcal{I}$ the indicator function. Hence the allocation rule is completely determined by the assignment functions $\hat{h}(j)$ and $\hat{z}(j)$, and the productivity of all workers assigned to task $j = \hat{j}(h)$ is

$$H(j) = \int b(h', \hat{j}(h')) \cdot \delta(j - \hat{j}(h')) \cdot F(\tilde{z}(h')|h') dG(h').$$

With the change of variables $j' = \hat{j}(h')$, we can instead integrate over j':

$$\begin{aligned} H(j) &= \int b(\hat{h}(j'), j') \cdot \delta(j - j') \cdot F(\hat{z}(j')|\hat{h}(j')) g(\hat{h}(j')) \cdot \hat{h}'(j') dj' \\ &= b(\hat{h}(j), j) \cdot F(\hat{z}(j)|\hat{h}(j)) g(\hat{h}(j)) \cdot \hat{h}'(j), \end{aligned}$$

which is (17).

For the optimal allocation, there can be no marginal gain from switching any worker's assignment. So for any $j' = j + dj$,

$$\begin{aligned} \frac{MPT_i(j) \cdot T_i(j)}{H_i(j)} \cdot b(\hat{h}(j), j) &\geq \frac{MPT_i(j') \cdot T_i(j')}{H_i(j')} \cdot b(\hat{h}(j), j'), \\ \frac{MPT_i(j') \cdot T_i(j')}{H_i(j')} \cdot b(\hat{h}(j'), j') &\geq \frac{MPT_i(j) \cdot T_i(j)}{H_i(j)} \cdot b(\hat{h}(j'), j), \end{aligned}$$

with equality if $|dj| = 0$, and where $MPT_i(j)$ is the marginal product of X_{ih} w.r.t. $T_i(j)$. Using (9) to substitute out $MPT_i(j) \cdot T_i(j)$ ratios, and substituting for $H_i(j) = q_{ih}(j)H(j)$ using (17), we obtain

$$\frac{b(\hat{h}(j'), j')}{b(\hat{h}(j), j')} \geq \frac{\pi_{ih}(j')}{\pi_{ih}(j)} \cdot \frac{q_{ih}(j)F(\hat{z}(j)|\hat{h}(j))g(\hat{h}(j))\hat{h}'(j)}{q_{ih}(j')F(\hat{z}(j')|\hat{h}(j'))g(\hat{h}(j'))\hat{h}'(j')} \geq \frac{b(\hat{h}(j'), j)}{b(\hat{h}(j), j)},$$

and as $|dj| \to 0$,

$$\left[\partial \log b(\hat{h}(j), j) \big/ \partial h\right] \cdot \hat{h}'(j) = d \log \left\{ \pi_{ih}(j) \Big/ \left[q_{ih}(j) F(\hat{z}(j)|\hat{h}(j)) g(\hat{h}(j)) \hat{h}'(j) \right] \right\} \Big/ dj.$$

Now using the total derivative of $b(\hat{h}(j), j)$:

$$d \log b(\hat{h}(j), j) \big/ dj = \left[\partial \log b(\hat{h}(j), j) \big/ \partial h\right] \cdot \hat{h}'(j) + \partial \log b(\hat{h}(j), j) \big/ \partial j, \tag{41}$$

and applying $\pi_{ih}(0) = 1$, we obtain (18):

$$H_i(j) \big/ \pi_{ih}(j) H_i(0) = \exp\left[\int_0^j \frac{\partial \log b(\hat{h}(j'), j')}{\partial j'} dj' \right] \equiv B_j(j; \hat{h}). \tag{42}$$

Plugging (13) and (18) into (12) yields the first equality in (20) in the corollary, and note that (41) implies that $b(h, \hat{j}(h)) = B_h(h; \hat{j}) \cdot B_j(\hat{j}(h); \hat{h})$, which yields the second equality.

C.2 Proof of Proposition 1 and Corollary 2

First, we re-express all capital input ratios only in terms of the thresholds $[\hat{h}(j), \hat{z}]$. Plugging (18) into (9), and applying the task production function (8) we obtain

$$\pi_{ih}(j) = v_{ih}(j) \Big/ \left[\tilde{M}(j) B_j(j; \hat{h})^{1-\alpha} \right]^{1-\sigma}, \quad \text{where } v_{ih}(j) \equiv \frac{\nu_i(j)}{\nu_i(0)} \tag{43}$$

and $\tilde{M}(j) \equiv M(j)/M(0)$. Similarly, plugging (10) and (13) in (11) we obtain

$$\pi_{iz} = v_{iz} \cdot \left(\tilde{M}(z) \cdot \hat{z}^{1-\alpha} \right)^{\omega - 1} \cdot \Pi_{ih}^{\frac{\sigma-\omega}{\sigma-1}}, \quad \text{where } v_{iz} \equiv \frac{\eta_i \nu_i(0)^{\frac{1-\omega}{\sigma-1}}}{1-\eta_i} \tag{44}$$

and $\tilde{M}(z) \equiv M(z)/M(0)$.

Now given a between-sector allocation rule $[q_{ih}(j), q_{iz}]$, the optimal within-sector allocation is described by $[\hat{h}(j)]_{j=0}^{J}$ that solves a fixed point defined by (17)-(18) in Lemma 1 and (43), and $\hat{z}$ that solves the fixed point defined by (13) and (44):

$$q_{ih}(j) \hat{h}'(j) = \frac{v_{ih}(j)}{\tilde{M}(j)^{1-\sigma}} \cdot \frac{B_j(j)^{\alpha + \sigma(1-\alpha)}}{b(\hat{h}(j), j)} \cdot \frac{H_i(0)}{F(\hat{z}(j)|\hat{h}(j)) g(\hat{h}(j))} \tag{45a}$$

$$\hat{z}^{\alpha + \omega(1-\alpha)} = \frac{\tilde{M}(z)^{1-\omega}}{v_{iz}} \cdot \Pi_{ih}^{\frac{\sigma-\omega}{1-\sigma}} \cdot \frac{q_{iz} Z}{H_i(0)} \tag{45b}$$

where the boundary conditions for the ODE in (45a) are $\hat{h}(0) = 0$ and $\hat{h}(J) = h_M$, which determines $H_i(0)$:

$$H_i(0) \cdot \int_0^J \left[\frac{v_{ih}(j) \cdot B_j(j)^{\alpha + \sigma(1-\alpha)}}{q_{ih}(j) \tilde{M}(j)^{1-\sigma} \cdot b(\hat{h}(j), j) \cdot F(\hat{z}(j)|\hat{h}(j)) g(\hat{h}(j))} \right] dj = h_M. \tag{45c}$$

The functions $\left[B_j(j), B_h(h), \hat{z}(j), \tilde{z}(h)\right]$, which represent relative wages in equilibrium, are defined in (16), (19) and (20) and are functions only of $[\hat{h}(j), \hat{z}]$. That is, system (45) is a fixed point only in terms of the thresholds, and unaffected by the total amount of physical capital and labor in either sector. All that matters is relative masses *across tasks*.

These equations characterize any regular equilibrium. Existence with endogenous sectoral weights follows from Proposition 2; it is not inferred from separate solutions of the boundary-value equation and managerial cutoff equation.

Sectoral production function The corollary expresses sectoral output only in terms of sectoral capital and labor, and the optimal assignment rules. To derive this, first note that using (9)-(11), sectoral capital can be written as $K_i = K_i(0)\Pi_{K_i}$, which is the first equation in (22). Next, from (13), we know that Z_i is linear in $H_i(0)$:

$$Z_i = \underbrace{q_{iz}L_z}_{\equiv L_{iz}}\bar{z} = L_{iz}\bar{z} = H_i(0)\cdot\hat{z}\pi_{iz}, \quad \text{where} \quad L_z \equiv \int_{z>\tilde{z}(h)} d\mu, \quad \bar{z} = Z/L_z,$$

and using Lemma 1 and (43), so is total worker productivity:

$$\begin{aligned}\int \left[H_i(j)\Big/ b(\hat{h}(j),j)\right] dj &= \int q_{ih}(\hat{j}(h))F(\tilde{z}(h)|h)g(h)dh \\ &= H_i(0)\cdot\int \left[\pi_{ih}(j)/B_h(\hat{h}(j))\right] dj \quad = L_i - L_{iz}.\end{aligned}$$

So rearranging, we can represent sectoral labor input as $L_i = H_i(0)\Pi_{L_i}$, which is the second equation in (22). Finally, use (8)-(11) to rewrite (7) as

$$Y_i = \psi_i\cdot\Pi_{K_i}^{\frac{\omega}{\omega-1}}\Pi_{ih}^{\frac{\sigma-\omega}{(\sigma-1)(1-\omega)}}M(0)K_i(0)^{\alpha}H_i(0)^{1-\alpha},$$

and replacing $[K_i(0), H_i(0)]$ with the expressions in (22) yields (21).

C.3 Proof of Lemma 2

The lemma is the one-sector case of Proposition 2, proved below. It concerns an equilibrium with endogenous occupation choices and prices, rather than a solution for arbitrary externally imposed sectoral allocation weights. No limiting argument in the strength of log-supermodularity is needed.

C.4 Proof of Proposition 2

Unit costs. The unit cost of $MK^{\alpha}H^{1-\alpha}$ is $\alpha^{-\alpha}(1-\alpha)^{-(1-\alpha)}M^{-1}R^{\alpha}p^{1-\alpha}$. Applying the CES unit-cost formula successively to worker tasks, worker and managerial composites, and sectors gives (25). Each transformed elasticity is finite and positive. A positive-weight CES cost is concave, homogeneous of degree one, and strictly concave except along proportional input-price vectors. Applying this property through the nests establishes the corresponding property of C. Each sector has a positive managerial weight, and aggregate task weights are positive almost everywhere. Equality in cost concavity therefore requires proportional worker prices on every sector's used tasks, with the proportionality factor fixed by the common managerial price. Aggregate task coverage forces the same factor throughout the task interval. Distinct continuous worker-price profiles cannot differ only on a cost-irrelevant open set.

Attainment with a continuum of tasks. Let $b_- = \inf_{h,j} r_b(h,j) > 0$, $b_+ = \sup_{h,j} r_b(h,j) < \infty$, and $B_0 = \mathbb{E}b(h,0) \in (0,\infty)$. Uniform continuity gives the task modulus

$$m_b(r) = \sup_{h,\,|j-j'|\le r} |\log r_b(h,j) - \log r_b(h,j')| \longrightarrow 0 \quad \text{as } r \downarrow 0.$$

For any continuous nonnegative worker-price profile let $w_p(h) = \max_j p_h(j)b(h,j)$, and define its maximal supporting profile

$$\bar{p}_h(j) = \inf_{h\in\mathcal{H}} \frac{w_p(h)}{b(h,j)}.$$

Then $\bar{p}_h \ge p_h$ and $w_{\bar{p}}(h) = w_p(h)$ for every h. Replacing p_h by $\bar{p}_h$ leaves $\mathcal{G}$ unchanged and weakly raises C. Whenever the worker profile is nonzero, $\log \bar{p}_h$ has modulus of continuity bounded by m_b, and

$$\frac{b_-}{b_+}\|p_h\|_\infty \le \bar{p}_h(j) \le \frac{b_+}{b_-}\|p_h\|_\infty.$$

On the feasible set $\mathcal{G} \le 1$, worker prices are bounded by $1/(b_-B_0)$ and the managerial price by $1/\mathbb{E}z$. Supporting worker profiles are uniformly bounded and equicontinuous, including sequences tending uniformly to zero. The supporting-profile transformation is continuous in the uniform norm, so its fixed-point class is closed. Arzela–Ascoli and the bounded managerial-price interval give a compact search set. The bound

$$|\mathcal{G}(p) - \mathcal{G}(\tilde{p})| \le b_+B_0\|p_h - \tilde{p}_h\|_\infty + \mathbb{E}z\,|p_z - \tilde{p}_z|$$

establishes continuity of $\mathcal{G}$. The CES cost index is continuous on this compact class: nonzero supporting worker profiles are uniformly comparable, and profiles tending to zero have worker cost indices tending to zero. The outer finite-dimensional CES limits are continuous. Hence (26) attains its maximum.

Positive prices and uniqueness. An all-positive feasible price vector has positive cost, and homogeneity makes $\mathcal{G} = 1$ at the optimum. If $\Omega \ge 1$, a zero managerial price or an identically zero worker-price profile makes every sectoral cost, hence C, zero. Such a vector cannot maximize. If $0 < \Omega < 1$, introducing a missing price component at size $d > 0$ raises the sectoral cost by order $d^{1-\Omega}$; the required rescaling to preserve $\mathcal{G} \le 1$ costs at most order d. The former dominates for small d. Thus both components are positive. The supporting-profile bound then makes every worker-task price positive.

Suppose two distinct maximizers satisfy $\mathcal{G} = 1$. Convexity of $\mathcal{G}$ makes their midpoint feasible. Strict concavity of C makes its value strictly larger unless the price vectors are proportional. Proportional maximizers with the same earnings normalization coincide. Thus normalized equilibrium prices are unique.

Market clearing and decentralization. The convex constraint has strictly feasible points. At a positive maximizer the first-order condition is

$$\nabla C = \lambda S, \qquad S \in \partial\mathcal{G}, \qquad \lambda = C,$$

where the last equality uses homogeneity and $\mathcal{G} = 1$. The subgradient S is an occupational service supply induced by earnings-maximizing choices, with tied types split when necessary; the cost gradient is

conditional service demand. Consequently $S = \nabla C/C$ clears the labor-service markets. Writing C_* for the maximizing cost, restore the final-good numeraire by multiplying all service prices by

$$s_* = (1-\alpha)K^{\alpha}C_*^{-(1-\alpha)}, \qquad R = \frac{\alpha s_*}{(1-\alpha)K}, \qquad Y = K^{\alpha}C_*^{-(1-\alpha)}.$$

These values make output unit cost one and satisfy $RK = \alpha Y$ and total labor earnings $(1-\alpha)Y = s_*$. CES conditional demands uniquely determine aggregate sectoral output, capital, and effective service allocations.

Sectoral allocation identities. The following identities describe the regular interior case with positive sectoral task weights; they are not the existence or uniqueness proof. At zero task weights, use the conditional cost demands directly. Unit elasticities are interpreted by limits. Here $q_h(j)$ and q_z are service-sector shares.

$$Q_h(j) \equiv \frac{q_h(j)}{1-q_h(j)} = Q_h(0) \cdot \frac{v_s(j)}{v_m(j)}, \tag{46}$$

so the dependence of the between-sector allocation rule on the within-sector rule can be suppressed into $Q_h(0)$. Likewise, the rule for splitting individuals between managers and workers, (45b), implies

$$Q_z \equiv \frac{q_z}{1-q_z} = Q_h(0) \cdot \frac{v_{sz}}{v_{mz}} \cdot \left(\frac{\Pi_{mh}}{\Pi_{sh}}\right)^{\frac{\sigma-\omega}{1-\sigma}} \tag{47}$$

where Π_{ih} is defined by (10) and (43), and thus depends on the thresholds $[\hat{h}(j), \hat{z}]$ through B_j. So in equilibrium, $Q_h(0)$ alone must solve the implied sectoral shares in (23) given (21):

$$Q_h(0) = \frac{\gamma_s}{\gamma_m}\left(\frac{\psi_s}{\psi_m}\right)^{\epsilon-1}\left(\frac{\Pi_{sh}}{\Pi_{mh}}\right)^{\frac{(\sigma-\omega)(1-\epsilon)}{(1-\sigma)(1-\omega)}}\left(\frac{\Pi_{K_s}}{\Pi_{K_m}}\right)^{\frac{\omega-\epsilon}{1-\omega}}. \tag{48}$$

Here $L_s/L_m = Q_h(0)\Pi_{L_s}/\Pi_{L_m}$ and $\bar{w}_s/\bar{w}_m = (\Pi_{K_s}/\Pi_{K_m})/(\Pi_{L_s}/\Pi_{L_m})$; substituting these ratios into (23) cancels the Π_L terms.

Uniqueness of occupational assignment. Strict log-supermodularity implies that maximizers of $p_h(j)b(h,j)$ are ordered in h. If one h has two distinct maximizing tasks, no smaller h can choose above the lower one and no larger h can choose below the upper one. The open task intervals associated with such ties are disjoint; choosing a rational point in each shows that there are at most countably many tied worker-skill types. The atomless selected worker-skill measure assigns them zero worker mass; atoms among types who only manage do not affect this conclusion. The stated zero-probability condition on $z = w_p(h)/p_z$ removes worker–manager ties; conditional managerial atomlessness suffices for it. Occupational choices are therefore unique almost everywhere.

Under the regular interior characterization, positive demand densities and positive worker participation at each h make the inverse assignment continuous and strictly increasing. Under the additional regularity imposed for differential characterization it satisfies the density equation (17). The service-price uniqueness does not distinguish observationally equivalent individual assignments across sectors or managers. Such assignments leave the same aggregate effective services and output.

D Comparative Statics Proofs

This appendix establishes the local task-automation result, the skill and requirement comparative statics, and sectoral reallocation.

D.1 Proof of Proposition 3

This proof retains endogenous occupational selection. The managerial share restriction is sufficient, not necessary, and is not imposed on the quantitative calibration.

Baseline coefficients and regularity. Normalize $p_z = 1$, write $\ell = \log b_\delta$, $t(h) = w(h)/p_z$, and let $\mathcal{G}$ be total labor earnings. Define

$$\tau(j) = \frac{p(j)H(j)}{\mathcal{G}}, \quad \tau_z = \frac{Z}{\mathcal{G}}, \quad T = \int_0^J \tau = 1 - \tau_z, \quad e(j) = \frac{t(\hat{h}(j))f(t(\hat{h}(j)) \mid \hat{h}(j))}{F(t(\hat{h}(j)) \mid \hat{h}(j))},$$

$$D(j) = \frac{w(\hat{h}(j))F(t(\hat{h}(j)) \mid \hat{h}(j))g(\hat{h}(j))}{\mathcal{G}\ell_{hj}(\hat{h}(j), j)}, \quad c = 1 - \sigma, \quad d = 1 - \omega, \quad \zeta = \Omega + \frac{\Sigma - \Omega}{T}.$$

Here $c, d > 0$, $\Sigma = \alpha + (1-\alpha)\sigma$, and $\Omega = \alpha + (1-\alpha)\omega$. A subscript 0 denotes the limiting baseline as $\delta \downarrow 0$, not the shocked equilibrium. Positivity and smoothness in Assumption 5 apply on the compact worker/task domains and a neighborhood of the selected cutoffs. The total managerial first moment is finite; derivatives of selected moments follow from the density at the moving cutoffs, without a smooth global tail.

At $\delta = 0$, optimality gives $p_0(j)k_0(j) = q_0$ and $w_0(h) = q_0 b_0(h)$. The selected worker wage bill and managerial services are

$$W(q) = \int qb_0(h)F(qb_0(h) \mid h)g(h)dh, \quad Z(q) = \int \int_{(qb_0(h),\infty)} z\, dF(z \mid h)dG(h).$$

The worker cost index is proportional to q, so demand requires $W(q)/Z(q) = Cq^{1-\Omega}$ for a fixed $C > 0$. The logarithmic elasticity of W is at least one, Z is nonincreasing, and the demand elasticity $1 - \Omega$ is below one. This gives a unique interior q_0; existence and positive aggregate supplies follow from the price argument, which does not require strict log-supermodularity. The limiting inverse assignment is the unique cumulative wage-bill match

$$\frac{1}{\mathcal{G}_0}\int_0^{\hat{h}_0(j)} w_0(h)F(w_0(h) \mid h)g(h)dh = \int_0^j \tau_0(s)ds. \tag{49}$$

It is a canonical limit of positive sorting, not a uniqueness claim for individual assignments at the exactly separable technology.

For small positive δ, task optimality implies

$$p_\delta(j)k_0(j) = p_\delta(0)k_0(0)\exp\left\{-\delta\int_0^j u_j(\hat{h}_\delta(s), s)ds\right\}.$$

The bounded integrand makes the relative price spread $O(\delta)$. The strictly monotone scalar clearing equation above bounds its level and gives $q_\delta \to q_0$. Hence wages, selected supplies, and demand densities converge uniformly. Their positive density bounds and cumulative clearing give $\hat{h}_\delta \to \hat{h}_0$ in C^1. In

particular, on each smooth task interval,

$$\tau_\delta \to \tau_0, \quad e_\delta \to e_0, \quad \delta D_\delta \to \bar{D}_0(j) = \frac{w_0(\hat{h}_0(j))F(w_0(\hat{h}_0(j)) \mid \hat{h}_0(j))g(\hat{h}_0(j))}{\mathcal{G}_0 u_{hj}(\hat{h}_0(j), j)} > 0. \tag{50}$$

The smoothness and uniform positive bounds make these limits uniform, with the corresponding first derivatives. Assumption 6 requires only that e_0 be nondecreasing along the reference selection curve. No sign restriction on e'_δ away from that limit, constant selection elasticity, or exact power conditional c.d.f. is required.

Exact local response. Let $R = (\underline{j}, \overline{j})$, $r = T^{-1}\int_R \tau$, and $x(j) = \partial_m \log[p(j)/p_z]$, evaluated at $m = 0$ for fixed $\delta > 0$. The envelope theorem and task optimality give

$$\partial_m \log[w(h)/p_z] = x(\hat{j}(h)), \qquad \partial_m \hat{h}(j) = -\frac{x'(j)}{\ell_{hj}(\hat{h}(j), j)}. \tag{51}$$

Set $V = Dx'$ and $I = \int \tau x$. Linearized supply and CES demand clearing give

$$-V' + \tau(e + \Sigma)x = \tau\{B - c\mathbf{1}_R\}, \qquad B = \zeta I + (c - d\tau_z)r, \qquad V(0) = V(J) = 0. \tag{52}$$

Both x and V are continuous at the shock boundaries. The one-sided derivatives obey $V'(\underline{j}+) - V'(\underline{j}-) = c\tau(\underline{j})$ and $V'(\overline{j}+) - V'(\overline{j}-) = -c\tau(\overline{j})$. For completeness, supply-share growth is $\tau\{(1 + e)x - I\} - V'$. Within-worker demand-share growth is $(1 - \Sigma)(x - I/T) - c(\mathbf{1}_R - r)$; total worker-share growth is $-(1 - \Omega)\tau_z\{I/T - cr/(1 - \Sigma)\}$. Equating supply and demand and using $(1 - \Omega)c/(1 - \Sigma) = d$ gives (52). This includes the managerial clearing row through adding up.

Equivalently the response operator is

$$\begin{aligned}\mathcal{A}x = &-(Dx')' + \tau(e + \Omega\tau_z)x + \zeta\tau(j)\int \tau(s)[x(j) - x(s)]ds \\ &= \tau\{(c - d\tau_z)r - c\mathbf{1}_R\}.\end{aligned} \tag{53}$$

Its quadratic form, with $\bar{x} = T^{-1}\int \tau x$, can be written

$$\int D(x')^2 + \int \tau e x^2 + \Sigma \int \tau(x - \bar{x})^2 + \Omega\tau_z T\bar{x}^2 > 0$$

for every nonzero x. This identity uses $\Sigma - \zeta T = \Omega\tau_z$ and remains coercive when $\zeta < 0$; the ordering $\omega < \sigma$ is unnecessary. The positive bounds on τ_z, Σ and Ω give a uniform weighted L^2 bound on x. Since D is bounded below by a positive constant times $1/\delta$, the energy identity also bounds $\|x'\|_2$ by $O(\sqrt{\delta})$. The one-dimensional bound on oscillations gives a uniform $\|x\|_\infty$ bound. Equation (52) then bounds V' and V, while (50) implies $x' = O(\delta)$.

The local differentiability used here can be formulated in piecewise smooth spaces with fixed breakpoints $\underline{j}, \overline{j}$. Write $y = \log p$ and recover $h_y(j)$ from $y'(j) = -\ell_j(h_y(j), j)$, using $\ell_{hj} > 0$. Set $w_y(h_y(j)) = e^{y(j)}b(h_y(j), j)$, and compute $W[y]$, $Z[y]$ and $\mathcal{G}[y] = W[y] + Z[y]$ from the selected supplies. Let $\tau_d[y, m]$ be the CES demand wage-bill density. The nonlinear clearing residual is

$$\mathcal{R}[y, m](j) = \frac{w_y(h_y(j))F(w_y(h_y(j)) \mid h_y(j))g(h_y(j))}{\mathcal{G}[y]} h'_y(j) - \tau_d[y, m](j).$$

The boundary conditions are $y'(0) = -\ell_j(0,0)$ and $y'(J) = -\ell_j(h_M, J)$, with y, y' continuous at the two breakpoints. A zero of this residual clears workers task by task; managerial clearing follows by adding up. For fixed $\delta > 0$, this is a smooth map from piecewise C^2 prices with these boundary and matching conditions to piecewise C^0 residuals, near the interior equilibrium with $h'_y > 0$. Its derivative, up to an overall nonzero sign, is the operator in (53). Coercivity, one-dimensional elliptic regularity and the finite-rank integral term give its bounded inverse. The implicit function theorem supplies a locally differentiable piecewise C^2 price branch and piecewise C^1 assignment branch. Finite difference quotients converge in those spaces. This argument is local around the positive interior equilibrium; it makes no statement about a nonregular boundary allocation.

Limiting wages and flux. Integrating (52), using $V(0) = V(J) = 0$ and $\Sigma - \zeta T = \Omega\tau_z$, gives

$$\int \tau e x + \Omega\tau_z I = -d\tau_z T r.$$

Consequently x_δ converges uniformly to the negative constant

$$x_0 = -\frac{d\tau_{z0} r_0}{\bar{e}_0 + \Omega\tau_{z0}} < 0, \qquad \bar{e}_0 = T_0^{-1} \int \tau_0 e_0. \tag{54}$$

Substituting this into (52) gives a continuous limiting flux, with piecewise C^1 convergence,

$$V_0'(j) = \tau_0(j)\{x_0[e_0(j) - \bar{e}_0] + c[\mathbf{1}_R(j) - r_0]\}, \qquad V_0(0) = V_0(J) = 0. \tag{55}$$

Define $H(j) = x_0[e_0(j) - \bar{e}_0] - cr_0$. This function is nonincreasing. Moreover,

$$H(0) < 0 \quad \Longleftrightarrow \quad d\tau_{z0}[\bar{e}_0 - e_0(0)] < c[\bar{e}_0 + \Omega\tau_{z0}]. \tag{56}$$

The proposition imposes this sharper inequality directly. The simpler check $\tau_{z0} \le c/d$ implies it, including equality in that share bound, since $e_0(0) \ge 0$ and $\Omega\tau_{z0} > 0$. Higher managerial shares are allowed when participation elasticities vary little across worker skills.

The single crossing. Since $H < 0$, integrating (55) from zero gives $V_0 < 0$ on $(0, \underline{j}]$; integrating backward from J gives $V_0 > 0$ on $[\overline{j}, J)$. Within R, in the wage-bill coordinate $s = \int_0^j \tau_0$, the derivative $dV_0/ds = H(j) + c$ is nonincreasing. Thus V_0 is concave there. Its negative left value and positive right value imply exactly one zero $j_0^* \in R$, with strictly positive derivative at that zero. The endpoint derivatives are strictly negative. Uniform convergence and piecewise C^1 convergence preserve these endpoint signs, the transverse interior crossing, and strict signs away from those three zeros. Hence, for all sufficiently small positive δ, V_δ has exactly one interior zero $j_\delta^* \in R$, is negative below it and positive above it.

By (51), the inverse-assignment response has the opposite sign because $D\ell_{hj} > 0$. It is positive below j_δ^* and negative above it. It vanishes at the fixed endpoints with nonzero inward slopes. Piecewise C^1 differentiability of the equilibrium branch preserves these signs and the unique transverse crossing for each fixed sufficiently small δ and every sufficiently small positive m. This proves part 1; the allowed shock size may depend on δ.

Wages and occupational entry. Uniform convergence to the strictly negative constant (54) gives $x_\delta(j) < 0$ everywhere for small positive δ. The cutoff $t(h)$ therefore falls at every fixed h, and every fixed

managerial type gains relative to every fixed worker type. In particular $\hat{z} = t(0)$ falls. Locally,

$$\partial_m L_z = -\int t(h) f(t(h) \mid h) g(h) x(\hat{j}(h)) dh \geq 0, \qquad \partial_m \tau_z = -\int \tau e x - \tau_z \int \tau x > 0.$$

For $h^* = \hat{h}(j^*_\delta)$,

$$\frac{d}{dh}\, \partial_m \log[w(h)/p_z] = \frac{x'(\hat{j}(h))}{\hat{h}'(\hat{j}(h))}$$

is negative below h^* and positive above it. Worker wage growth is therefore locally U-shaped. The finite-shock envelope identity in the text gives the corresponding relative wage comparisons. Managerial entry is strict when the selection density is positive on a set of positive worker-skill measure. These fixed-type results do not sign average managerial earnings: the wage-bill and headcount effects in (36) must still be compared. Nor does the inverse-assignment crossing alone establish headcount or capital-share changes for every occupation, given endogenous entry into management.

D.2 Proof of Proposition 5

We work with a one-sector equilibrium. Let $p(j) = p_h(j)$, $p_z = w_z$, $w(h) = \max_j p(j) b(h, j)$, and $t(h) = w(h)/p_z$. Write $\mathcal{G}$ for total labor earnings and define the wage-bill density and managerial share by

$$\tau(j) = \frac{p(j) H(j)}{\mathcal{G}}, \qquad \tau_z = \frac{p_z Z}{\mathcal{G}}, \qquad \tau_W = \int \tau(j) dj = 1 - \tau_z.$$

Use Σ, Ω from (25), and put

$$e(j) = \frac{t(\hat{h}(j)) f(t(\hat{h}(j)) \mid \hat{h}(j))}{F(t(\hat{h}(j)) \mid \hat{h}(j))}, \quad D(j) = \frac{w(\hat{h}(j)) F(t(\hat{h}(j)) \mid \hat{h}(j)) g(\hat{h}(j))}{\mathcal{G}\, \partial_{hj} \log b(\hat{h}(j), j)}, \quad \zeta = \Omega + \frac{\Sigma - \Omega}{\tau_W}.$$

All derivatives below are local and evaluated at the baseline. Set $a = \log[M_h/M(z)]$ and hold M_h fixed, so $\partial_a \log M(z) = -1$. Normalize $\partial_a \log p_z = 0$ and define the managerial-premium response

$$x(j) = -\partial_a \log p(j).$$

The worker envelope gives $\partial_a \log[w(h)/p_z] = -x(\hat{j}(h))$ for fixed h. This normalization removes a common price-growth component and does not affect relative earnings.

Continuous price response. Differentiating task optimality, $(\log p)'(j) + \partial_j \log b(\hat{h}(j), j) = 0$, gives

$$\partial_a \hat{h}(j) = \frac{x'(j)}{\partial_{hj} \log b(\hat{h}(j), j)}. \tag{57}$$

The fixed-skill task response is $-x'/(\partial_{hj} \log b\, \hat{h}')$ and its earnings-weighted flux is $-Dx'$. Expressing linearized market clearing in terms of the premium x gives

$$-(Dx')' + \tau(e + \Sigma)x = \tau B_M, \quad B_M = (1 - \omega)\tau_z + \zeta \int \tau x, \quad x'(0) = x'(J) = 0. \tag{58}$$

The source $(1-\omega)\tau\tau_z$ reflects the fall in managerial productivity relative to workers. The Neumann conditions follow from fixed endpoint inverse assignments. Equivalently, using $\Omega\tau_z + \zeta\tau_W = \Sigma$,

$$-(Dx')' + \tau(e + \Omega\tau_z)x + \zeta\tau \int \tau(k)[x(j) - x(k)]dk = (1-\omega)\tau\tau_z > 0.$$

This includes worker reassignment, occupational switching, and CES demand responses. The managerial clearing row is redundant by adding up wage-bill shares.

Managers relative to workers. Let $\mathcal{L}q = -(Dq')' + \tau(e + \Sigma)q = \tau$ with Neumann boundary conditions. Its positive zero-order coefficient gives a unique $q > 0$. Put $Q = \int \tau q$. Integrating gives $\int \tau(e + \Sigma)q = \tau_W$, so $Q \leq \tau_W/\Sigma$. Equation (58) therefore has the unique solution

$$x = B_M q, \qquad B_M = \frac{(1-\omega)\tau_z}{1 - \zeta Q} > 0.$$

If $\zeta \leq 0$ the denominator is at least one; if $\zeta > 0$, it is at least $1 - \zeta\tau_W/\Sigma = \Omega\tau_z/\Sigma > 0$. Thus no ordering of ω and σ is needed. Every fixed managerial type gains relative to every fixed worker type, and $\partial_a \log t(h) = -x(\hat{j}(h)) < 0$. Consequently

$$\partial_a L_z = \int t(h) f(t(h) \mid h) g(h) x(\hat{j}(h)) dh \geq 0, \qquad \partial_a \tau_z = \int \tau e x + \tau_z \int \tau x > 0.$$

The employment inequality is strict exactly when the selection density is positive on a set of positive worker-skill measure.

Wage differences among workers. Assumption 8 makes e nondecreasing across tasks along the increasing baseline assignment. A rising segment of x would require a local minimum at j_0 followed by a strictly larger local maximum at $j_1 > j_0$, allowing endpoints with their Neumann conditions. Equation (58) gives

$$[e(j_0) + \Sigma]x(j_0) \geq B_M \geq [e(j_1) + \Sigma]x(j_1),$$

a contradiction. Thus $x' \leq 0$, $\partial_a \hat{h} \leq 0$, and $\partial_a \hat{j} \geq 0$. For $h_2 > h_1$,

$$\partial_a \log \frac{w(h_2)}{w(h_1)} = x(\hat{j}(h_1)) - x(\hat{j}(h_2)) \geq 0.$$

For $V = Dx'$, differentiating (58) after division by τ gives

$$-\left(\frac{V'}{\tau}\right)' + \frac{e + \Sigma}{D} V = -e'x, \qquad V(0) = V(J) = 0.$$

If e is nonconstant and nondecreasing, the strong maximum principle gives $V < 0$ in the interior, establishing the strict version.

If $e = k$ is constant, the unique response is $x = (1-\omega)\tau_z/(k + \Omega\tau_z)$, so worker wage growth is constant and task assignments do not change. Linearity of the local response gives the reversed signs in Corollary 3.

Exact abundance equivalence. For any allocation of the same people and capital, multiplying managerial skills by s replaces Z by sZ and

$$M(z)K_z^{\alpha}(sZ)^{1-\alpha} = M(z)s^{1-\alpha}K_z^{\alpha}Z^{1-\alpha}.$$

The production possibilities and individual earnings therefore coincide with those of the productivity experiment. If p_z^A is the price per unit of scaled skill and p_z^P the equivalent productivity-economy price, $sp_z^A = p_z^P$. In particular, the cutoff in original z units is $w^A/(sp_z^A) = w^P/p_z^P$; the cutoff in scaled units is s times this value. The equivalence in (35) uses original units. A common multiplier on all worker-task productivities can likewise be factored out of every production technology, leaving the relative parameter $M(z)/M_h$. This establishes the relative-productivity parameterization. A uniform multiplication of b itself by s_h is equivalent to $M_h \mapsto M_h s_h^{1-\alpha}$; scaling the argument h is a different experiment unless b is homogeneous in h.

D.3 Proof of Proposition 4 and Extensions

Requirement-shock equilibrium. Retain the baseline coefficients above and put $b = c + h - \chi(j)$. In this proof, a denotes the requirement perturbation ε. For $\chi_a = \chi + ad$, now let $x = \partial_a \log p$ with $\partial_a \log p_z = 0$, and define

$$v(j) = \frac{d(j)}{b(\hat{h}(j), j)}, \quad A(j) = \frac{d(j)}{\chi'(j)}, \quad y(j) = x(j) - v(j), \quad I_y = \int \tau y, \quad I_v = \int \tau v.$$

The task-indexed function $y(j)$ in this proof is $y_J(j)$ in the statement: it evaluates fixed-h wage growth on the baseline assignment. It does not follow a skill type changing with a. With $k_b(h, j) = \partial_a \log b_a(h, j)|_0 = -d(j)/b(h, j)$, differentiating task optimality gives the task flux

$$Q = D\{x' + \partial_j k_b(\hat{h}(j), j)\} = Dy' - \tau A, \qquad Q(0) = Q(J) = 0. \tag{59}$$

Worker supply shares change by $\tau(y - I_y + ey) - Q'$, while demand shares change by $\tau(x - \int \tau x) - \Omega\tau\tau_z x - \zeta\tau \int \tau(k)[x(j) - x(k)]dk$. The baseline demand identity

$$\frac{\tau'}{\tau} = A_0' + (1 - \Sigma)\frac{\chi'}{b}$$

therefore yields the full, assignment-inclusive response equation

$$\begin{aligned} -(Dy')' + \tau(e + \Sigma)y &= \tau\{B_R - A' - AA_0'\}, \\ B_R = \zeta I_y + (\zeta - 1)I_v, &\qquad Dy'|_{0,J} = \tau A|_{0,J}. \end{aligned} \tag{60}$$

Managerial selection is included through ey; no price or allocation response has been suppressed.

Increasing relative worker wage growth. First put $B_A = e^{A_0}A$ and $S_A = -[A' + AA_0']' \geq 0$. Then

$$-B_A'' + A_0'B_A' = e^{A_0}S_A \geq 0, \qquad B_A(0), B_A(J) \geq 0.$$

The one-dimensional maximum principle gives $A \geq 0$. Since $A \not\equiv 0$, either the forcing or a boundary value is nonzero, and the strong maximum principle gives $A > 0$ in the interior. In particular $I_v = \int \tau\chi' A/b > 0$.

Under Assumption 7, $e = k$ along the baseline selection curve. Set $V = Dy'$. Divide (60) by τ and differentiate to eliminate B_R:

$$-\left(\frac{V'}{\tau}\right)' + \frac{k+\Sigma}{D}V = -A'' - A'A_0' - AA_0'' \geq 0,$$
$$V(0) = \tau(0)A(0) \geq 0, \qquad V(J) = \tau(J)A(J) \geq 0. \tag{61}$$

The forcing and boundary values cannot all vanish, since that would imply $A \equiv 0$. The strong maximum principle for this second-order boundary-value problem therefore gives $V > 0$ on $(0, J)$. Hence

$$\partial_a \partial_h \log \frac{w_a(h)}{p_{z,a}} = \frac{y'(\hat{j}(h))}{\hat{h}'(\hat{j}(h))} > 0.$$

Integrating (60), the boundary flux cancels $\int (\tau A)'$. Using $\tau'/\tau = A_0' + (1-\Sigma)\chi'/b$ then gives

$$I_y = \frac{(1-\Omega)\tau_z}{k + \Omega\tau_z} I_v. \tag{62}$$

For $0 < \omega < 1$, $I_y > 0$. A strictly increasing y with a positive wage-weighted average satisfies $y(J) > 0$, so a nonempty upper skill interval gains relative to managers. If $y(0) < 0$, continuity and strict monotonicity give a unique zero; otherwise no worker type strictly loses. Moreover $\partial_a Z = -k\mathcal{G}I_y/p_z$, $\partial_a \mathcal{G}/\mathcal{G} = I_y$, and $\partial_a \tau_z = -(k + \tau_z)I_y$, so its sign is opposite to $1 - \omega$. The strict wage-gradient conclusion holds for every positive ω, σ. Finally, $Q/\tau = \partial_a \hat{j}_a(h)$ at $j = \hat{j}(h)$ establishes (34).

A sufficient primitive sorting restriction. For the index stretch $A = j$, suppose additionally that $0 < \sigma < 1$, $\chi(j) = \gamma j$, $\gamma > 0$, $c > \gamma J$, $M(j) = 1$, $\nu(j) = 1/J$, and $g(h) = CS(h)^k$, with C normalizing the density. If

$$\frac{h_M}{J} \leq \frac{(1-\Sigma)\gamma}{k+1}, \tag{63}$$

then the entire interior assignment shifts upward. Indeed $gF = C\rho t^k$ cancels the conditional scale, and baseline supply and demand imply

$$\frac{\hat{h}''}{\hat{h}'} = \frac{(1-\Sigma)\gamma - (k+1)\hat{h}'}{c + \hat{h} - \gamma j}.$$

By uniqueness of this smooth ODE, its slope cannot cross $q_* = (1-\Sigma)\gamma/(k+1)$; a slope reaching q_* remains constant there. Its mean is $h_M/J \leq q_*$, so $\hat{h}' \leq q_*$ throughout. For $\mathcal{P}V = -(V'/\tau)' + (k+\Sigma)V/D$,

$$\mathcal{P}(\tau j) = \frac{\gamma}{b^2}[(k+1)j\hat{h}' - (1-\Sigma)(c+\hat{h})] \leq -\frac{(1-\Sigma)\gamma}{b} < 0.$$

Since (61) has zero forcing here, $Q = V - \tau j$ satisfies $\mathcal{P}Q > 0$ with zero endpoints, so $Q > 0$ on the interior. Thus $\partial_a \hat{j}_a(h) > 0$ and $\partial_a \hat{h}_a(j) < 0$ away from endpoints. This restriction on skill dispersion relative to the requirement gradient is sufficient and strong; it is not part of the general model or the wage-gradient proposition.

An occupational-mean comparison under an additional restriction. For the further independent, bounded-power case $S(h) = S_0$ and $\rho = 1$, put $u = t/S_0$, $A = \mathbb{E}[u^k]$, and $B_1 = \mathbb{E}[u^{k+1}]$.

Then

$$\frac{\bar{w}_z}{\bar{w}_h} = \frac{k}{k+1}\frac{A(1-B_1)}{B_1(1-A)}.$$

Let m_L and m_W be worker-employment- and worker-wage-bill-weighted averages of y, respectively. Both w and y increase with h, so $m_L \le m_W$ by nonnegative covariance. Equation (62) gives $m_W > 0$, and

$$\partial_a \log(\bar{w}_z/\bar{w}_h) = \frac{k}{1-A}m_L - \frac{k+1}{1-B_1}m_W < 0.$$

The inequality follows because $(k+1)u^k - ku^{k+1} < 1$ for $0 < u < 1$, hence $k/(1-A) < (k+1)/(1-B_1)$. For the relative managerial-productivity decline in Proposition 5, this independent benchmark has constant negative worker-relative growth $y = -x < 0$. Thus $m_L = m_W < 0$ and the same mean-ratio formula gives a rising managerial mean premium. Corollary 3 reverses that sign. Neither conclusion is imposed on correlated skills or a different upper tail.

Smooth common-support skills and compatibility. Consider

$$f_{U,m}(u) = \frac{m}{\mathrm{B}(k/m, q/m)}u^{k-1}(1+u^m)^{-(k+q)/m}, \qquad u > 0, \quad k > 0, \quad q > 1. \tag{64}$$

It has a positive smooth density, common unbounded support, and finite mean. With $S(h) = S_0 e^{\beta h}$,

$$\partial_{hz} \log f(z \mid h) = \frac{\beta(k+q)mu^{m-1}}{S(h)(1+u^m)^2} > 0, \qquad u = z/S(h).$$

The log-density of $\log U$ is strictly concave:

$$L(v) = kv - \frac{k+q}{m}\log(1+e^{mv}) + \text{constant}, \qquad L''(v) < 0.$$

Its reverse hazard is strictly decreasing. To see this directly, write $R(v) = e^{L(v)}/\int_{-\infty}^{v} e^{L(t)}dt$, so $R' = R(L' - R)$. If $L' \le 0$, $R' < 0$. If $L' > 0$, strict concavity bounds the integral below $e^{L(v)}/L'(v)$, again giving $R' < 0$. For $\Phi(v) = \log F_U(e^v)$ and $G(x,h) = \Phi(x - \log S(h))$,

$$G_{xx} = \Phi'' < 0, \qquad G_{xh} = -\beta\Phi'' > 0, \qquad G_{xh}/|G_{xx}| = \beta.$$

Thus $\beta > \sup \partial_h \log b$ establishes both density MLR and the stronger global inequality $G_{xh} > |G_{xx}|\partial_h \log b$, hence the reference participation ordering. These global restrictions are properties of this example, not general equilibrium assumptions.

As $m \to \infty$ the density in (64) tends to $kq/(k+q)$ times u^{k-1} below one and u^{-q-1} above one. Its lower body is $[q/(k+q)]u^k$, while

$$\mathbb{E}U_m = \frac{\mathrm{B}((k+1)/m, (q-1)/m)}{\mathrm{B}(k/m, q/m)} \longrightarrow \frac{kq}{(k+1)(q-1)}.$$

The CDF and density converge in the required differentiable norms on compact subsets of $(0,1)$; truncated managerial first moments converge there as well. For the index stretch $A = j$, take a regular limiting equilibrium with selection arguments strictly below one. Its linearized price operator has the coercive quadratic form used in (53), with $e = k$, and therefore a bounded inverse on the fixed-endpoint function

spaces. Together with the stated coefficient and tail-moment convergence, the implicit-function theorem gives continuous differentiable dependence on the skill distribution. Assignments and shock responses consequently converge differentiably; differentiability in the shock alone would not suffice. No additional nonsingularity assumption is needed. At the limit, the solution V_0 of (61) satisfies $V_0(j) > 0$ for $j > 0$ and $V_0'(0) > 0$ by the boundary point lemma. The continuous extension of $V_0(j)/j$ has a positive minimum d_0. For sufficiently large finite m,

$$\|V_m' - V_0'\|_\infty < d_0/2, \qquad V_m(0) = 0,$$

so $V_m(j) \geq d_0 j/2 > 0$ for all $j > 0$. Positive upper-tail relative growth and the falling managerial wage-bill share persist by continuity. This supplies a uniform endpoint-aware robustness argument, not an uncontrolled vanishing-error approximation. No numerical threshold for m is asserted. The strictly decreasing lower-tail elasticity of this family and the scale dominance condition also give nonzero managerial sorting in Proposition 5.

A primitive sufficient bound for the selection region. For completeness, the power-body/Pareto-tail limiting family admits a conservative bound avoiding any assumption on the location of endogenous cutoffs. Suppose $M(j) = 1$, $0 < \omega < 1$, any finite $\sigma > 0$, and $\rho = q/(k+q)$. Put

$$b_0 = c - \chi(0), \quad b_- = c - \max \chi, \quad b_+ = c + h_M - \min \chi, \quad P = e^{[\chi(J)-\chi(0)]/b_-},$$

$$r_0 = b_0/S_{\max}, \quad r_+ = Pb_+/S_{\min}, \quad u_- = b_0/(r_+ S_{\max}), \quad E_U = \frac{kq}{(k+1)(q-1)},$$

$$B_a = \frac{(1-\rho)q}{(q-1)\rho} P^{1-\Omega}(S_{\max}/b_0)^{1-\Omega}, \quad B_b = \frac{r_+ S_{\max} E_U}{b_0 \rho u_-^k}(P/r_0)^{1-\Omega}.$$

A sufficient restriction on fundamentals is

$$\frac{\eta}{1-\eta} M(z)^{\omega-1} > \max\{B_a, B_b\}. \tag{65}$$

Indeed task optimality gives $p(0) \leq p(j) \leq Pp(0)$ and $C_h \leq Pp(0)$. If any type has $t(h)/S(h) \geq 1$, then $r = p_z/p(0) \leq r_+$. For $r \leq r_0$, all cutoffs lie in the Pareto region: the worker wage bill is at least $p(0)b_0\rho$ and the managerial wage bill is at most $p(0)(1-\rho)qb_0^{1-q}S_{\max}^q r^q/(q-1)$. Their ratio divided by $(r/P)^{1-\Omega}$ is at most B_a. For $r_0 < r \leq r_+$, worker participation is at least ρu_-^k and managerial services at most $S_{\max}E_U$, giving the upper bound B_b. CES demand requires the same ratio to equal $[\eta/(1-\eta)]M(z)^{\omega-1}(p_z/C_h)^{1-\Omega}$, contradicting (65). Hence all selection arguments are below one, with a uniform margin by compactness. The bound is deliberately conservative, sufficient rather than necessary, and is not prescribed as an empirical restriction.

D.4 Decomposing polarization

Let $V_i(j)$ for each task $j \in \mathcal{J}$ denote the "unnormalized" labor in job j in sector i, defined as

$$\begin{aligned} V_i(j) &\equiv (1-\eta_i)\nu_i(0)M(0)^{\sigma-1} \cdot \pi_{ih}(j)/B_h(\hat{h}(j)) \\ &= (1-\eta_i)\nu_i(j)\left[M(j)B_j(j)^{1-\alpha}\right]^{\sigma-1}/B_h(\hat{h}(j)) \end{aligned} \tag{66a}$$

where we have replaced for $\pi_{ih}(j)$ using (43). Similarly we can define

$$\begin{aligned} V_i(z) &\equiv (1-\eta_i)\nu_i(0)M(0)^{\sigma-1}\cdot(\hat{z}/\bar{z})\pi_{iz} \\ &= \eta_i M(z)^{\omega-1}\left[\nu_i(0)M(0)^{\sigma-1}\Pi_{ih}\right]^{\frac{\sigma-\omega}{\sigma-1}}\cdot\hat{z}^{\alpha+\omega(1-\alpha)}/\bar{z} \\ &= \eta_i M(z)^{\omega-1}V_{ih}^{\frac{\sigma-\omega}{\sigma-1}}\cdot\hat{z}^{\alpha+\omega(1-\alpha)}/\bar{z} \end{aligned} \tag{66b}$$

$$V_{ih} \equiv \nu_i(0)M(0)^{\sigma-1}\Pi_{ih} = \int\left\{\nu_i(j)\left[M(j)B_j(j)^{1-\alpha}\right]^{\sigma-1}\right\}dj \tag{66c}$$

where V_{ih} is the unnormalized counterpart of Π_{ih} in (10), and to derive $V_i(z)$, we replaced for π_{iz} using (44). These $[V_i(j)]_{j\in\mathcal{J}^z}$ are simply the amount of labor in task j unnormalized by $H_i(0)$, so we know that taking the ratio between any pair yields the labor input ratio between the two tasks. Using these expressions we can define the "unnormalized" labor in sector i:

$$V_i = (1-\eta_i)\nu_i(0)M(0)^{\sigma-1}\Pi_{L_i} = V_{il} + V_i(z), \quad \text{where } V_{il} = \int V_i(j)dj. \tag{67}$$

So the definition of Π_{L_i} in (22) implies that the amount of labor (or worker human capital) in each task across both sectors can be expressed as

$$L(j) = \sum_{i\in\{m,s\}}\frac{L_i(j)}{L_i}\cdot L_i = \sum_{i\in\{m,s\}}\frac{V_i(j)}{V_i}\cdot L_i.$$

Consider the same exercise as in Proof D.1 to Proposition 3, that $M(j)$ grows to $M^1(j) = M(j)e^m$, $m > 0$, for all $j \in \mathcal{J}^1 \equiv (\underline{j}, \bar{j})$. Denoting the log-derivative of X w.r.t. $\tilde{m}$ as Δ_X,

$$\begin{aligned} \Delta_{L(j)} &= \sum_{i\in\{m,s\}}\frac{L_i(j)}{L(j)}\cdot\left[\Delta_{V_i(j)} - \Delta_{V_i} + \Delta_{L_i}\right] \\ &= \sum_{i\in\{m,s\}}\frac{L_i(j)}{L(j)}\cdot\Bigg\{\underbrace{\Delta_{V_i(j)} - \int_{\mathcal{J}^z}\left[\frac{V_i(j')}{V_i}\cdot\Delta_{V_i(j')}\right]dj'}_{W_{ij}} + \Delta_{L_i}\Bigg\}. \end{aligned} \tag{68}$$

Equation (68) separates within-sector occupational changes from sectoral reallocation. The sectoral comparison below concerns the latter and includes the response of sectoral average earnings.

D.5 Proof of Proposition 6

Accounting and part 1. Write W_i for the sectoral labor wage bill and $\bar{w}_i = W_i/L_i$. Common capital shares imply

$$\frac{K_m}{K_s} = \frac{W_m}{W_s} = \frac{p_m Y_m}{p_s Y_s} = \frac{\gamma_m}{\gamma_s}\left(\frac{p_m}{p_s}\right)^{1-\epsilon}, \qquad \frac{L_m}{L_s} = \frac{W_m}{W_s}\frac{\bar{w}_s}{\bar{w}_m}. \tag{69}$$

Normalize the managerial service price to one. All workers earn $w = t$, and managers are those with $z > t$. Let θ be the common initial worker share of sectoral labor wage bills. Put

$$\phi = \frac{t}{\bar{z}} \in (0,1), \qquad T = \phi + (1-\phi)\theta, \qquad f_\theta = \frac{(1-\phi)\theta(1-\theta)}{T}, \qquad d = r_m - r_s > 0.$$

Write a dot for the derivative with respect to a at zero, and $x = \dot{\log} t$. The worker-service cost index has derivative $\dot{\log} C_{ih} = x - r_i/(1-\alpha)$. The sectoral cost index has derivative $\dot{\log} C_i = \theta\{x - r_i/(1-\alpha)\}$. The common rental-rate term cancels from relative output prices, giving

$$\dot{\log}(p_m/p_s) = -\theta d < 0. \tag{70}$$

CES final demand and (69) imply

$$\dot{\log}(Y_m/Y_s) = \epsilon\theta d > 0, \qquad \dot{\log}(W_m/W_s) = -(1-\epsilon)\theta d < 0. \tag{71}$$

Worker wage-bill shares satisfy

$$\dot{\theta}_i = (1-\omega)\theta(1-\theta)\{(1-\alpha)x - r_i\}.$$

Since $L_i = (W_i/w)T_i$, with $T_i = \phi + (1-\phi)\theta_i$, common changes in w and ϕ cancel in the relative derivative at the equal-θ baseline. Hence

$$\boxed{\dot{\log}(L_m/L_s) = -\{(1-\epsilon)\theta + (1-\omega)f_\theta\}d < 0.} \tag{72}$$

The first term is the relative-revenue contraction under between-sector complementarity. When $\omega < 1$, the second is the change toward higher-paid managers, reducing headcounts per unit of wage bill more in manufacturing. If $\omega > 1$, this composition term opposes the decline; the stated baseline inequality ensures that the first term dominates. If $\lambda_m = L_m/(L_m + L_s)$, then $\dot{\lambda}_m = \lambda_m(1-\lambda_m)\dot{\log}(L_m/L_s) < 0$.

The productivity defined by $Y_i = \Phi_i K_i^{\alpha} L_i^{1-\alpha}$ obeys

$$\dot{\log}(\Phi_m/\Phi_s) = \{\theta + (1-\alpha)(1-\omega)f_\theta\}d > 0. \tag{73}$$

This follows from (69), (71), and (72). Writing $G_\theta = (1-\epsilon)\theta + (1-\omega)f_\theta$, the coefficient equals $\Xi\theta + (1-\alpha)G_\theta > 0$ under the stated headcount condition, including when $\omega > 1$. It includes the change in employment composition.

Part 2. First allow an arbitrary smooth managerial-skill distribution. Let $Z(t) = \int_t^\infty z\, dF(z)$ and $\bar{z} = Z(t)/[1-F(t)]$. Write

$$u_\phi = \frac{d\log(t/\bar{z})}{d\log t} = 1 + \frac{t^2 f(t)}{Z(t)} - \frac{t f(t)}{1-F(t)}.$$

Only this local selected-mean elasticity enters the additional headcount-composition term.

Let s_i be sectoral revenue shares and θ_i worker shares of sectoral wage bills. Write F and f here for the marginal managerial-skill CDF and density, integrating out h at the homogeneous-worker benchmark. Define $E_i = \theta_i r_i$. The formulas below need no ordering of θ_m and θ_s. In the sufficient ordered-weight example, $\eta_m \le \eta_s$ implies $\theta_m \ge \theta_s$ and hence $E_m > E_s$. Let

$$A = \sum_i s_i\theta_i(1-\theta_i), \quad B = s_m s_s(\theta_m - \theta_s)^2, \quad C = \sum_i s_i\theta_i(1-\theta_i)r_i,$$
$$\theta = \sum_i s_i\theta_i, \qquad e = tf(t)/F(t), \qquad \Xi = \alpha + (1-\alpha)\epsilon.$$

Differentiating aggregate worker wage-bill supply and demand gives

$$x = -\frac{(1-\omega)C + (1-\epsilon)s_m s_s(\theta_m - \theta_s)(E_m - E_s)}{\Omega A + \Xi B + \theta e}. \tag{74}$$

Indeed supply's derivative with respect to $\log t$ is $\theta(1-\theta) + \theta e$, demand's is $(1-\Omega)A + (1-\Xi)B$, and $\theta(1-\theta) = A + B$. The fixed-price shock to demand is the negative numerator in (74). This accounts for sectoral reallocation as well as occupational selection.

For any θ_i, put

$$T_i = \phi + (1-\phi)\theta_i, \quad f_i = \frac{(1-\phi)\theta_i(1-\theta_i)}{T_i}, \quad G_i = (1-\epsilon)\theta_i + (1-\omega)f_i.$$

Allowing ϕ to change with the selection cutoff, the exact relative-headcount response is

$$\begin{aligned}\log(\dot{L_m}/L_s) = (1-\alpha)x(G_m - G_s) - (r_m G_m - r_s G_s) \\ + \phi u_\phi x\left[\frac{1-\theta_m}{T_m} - \frac{1-\theta_s}{T_s}\right].\end{aligned} \tag{75}$$

All terms on the right are computed from the baseline. The exact condition for the relative employment decline is therefore

$$\begin{aligned}r_m G_m - r_s G_s > (1-\alpha)x(G_m - G_s) \\ + \phi u_\phi x\left[\frac{1-\theta_m}{T_m} - \frac{1-\theta_s}{T_s}\right],\end{aligned} \tag{76}$$

where x is the explicit baseline expression in (74). This condition permits either sign of u_ϕ and requires no tail shape away from the cutoff.

Relative prices obey

$$P = \log(\dot{p_m}/p_s) = (1-\alpha)(\theta_m - \theta_s)x - (E_m - E_s).$$

Thus the exact condition for the price decline is

$$(1-\alpha)(\theta_m - \theta_s)x < E_m - E_s. \tag{77}$$

Revenue and output signs follow from final CES demand. When the two baseline conditions give $P < 0$ and $L = \log(\dot{L_m}/L_s) < 0$, accounting gives

$$\log(\dot{\Phi_m}/\Phi_s) = -\Xi P - (1-\alpha)L > 0.$$

Thus the stated headcount condition also suffices for the TFP sign; no separate selected-mean restriction is needed.

A simple tail example. If $\Pr(z > u) = \pi(z_0/u)^q$ for $u \geq z_0$, with $q > 1$, $0 < \pi < 1$, and $t > z_0$, then $\bar{z} = qt/(q-1)$, $\phi = (q-1)/q$, and $u_\phi = 0$. For $f(\theta) = (1-\phi)\theta(1-\theta)/[\phi + (1-\phi)\theta]$,

$$f'(\theta) = \frac{\phi}{[\phi + (1-\phi)\theta]^2} - 1 \geq -(1-\phi) = -1/q.$$

Consequently $G'(\theta) \geq (1-\epsilon) - (1-\omega)/q \geq 0$ under the primitive bound in the statement, and it is strictly positive at interior worker shares. Under $\eta_m \leq \eta_s$ and $\omega < 1$, (74) gives $x < 0$ and $E_m > E_s$, hence (77). Also $G_m \geq G_s > 0$, so (76) holds. The earlier restriction $\epsilon \leq \omega$ is sufficient but unnecessarily tight. Even the new primitive bound can be relaxed whenever the exact baseline condition holds. More generally, under the same ordered-weight complementary-manager configuration, $u_\phi \leq 0$ reinforces the decline when $G_m \geq G_s$, because the final bracket in (75) is nonpositive.

A primitive sufficient condition for tail selection. The tail-selection condition can be ensured from primitives. Set $p_z = 1$ and evaluate worker cost $c_0 = z_0 M_0^{-1/(1-\alpha)}$ at $t = z_0$. If

$$\frac{(1-\eta_s)c_0^{1-\Omega}}{\eta_s M(z)^{\omega-1} + (1-\eta_s)c_0^{1-\Omega}} > \frac{1-\pi}{1-\pi+\pi q/(q-1)}, \qquad \Omega = \alpha + (1-\alpha)\omega,$$

even the more management-intensive sector demands a greater worker wage-bill share than supply at $t = z_0$. Equilibrium therefore has $t > z_0$: supply worker/manager wage-bill odds grow more than proportionally in t, while CES demand odds grow less than proportionally. The inequality is sufficient, not necessary.

Extension to positive worker heterogeneity. At each benchmark all reported signs are strict. For each sufficiently small fixed shock, continuity of equilibrium prices and aggregate allocations therefore preserves these comparisons under sufficiently small perturbations. At $b = 1$, effective worker services equal worker headcounts even though assignments across tasks need not be unique. Under $b_\delta \to 1$ uniformly, that distinction vanishes uniformly; a common managerial mean in the compared allocations likewise identifies managerial headcounts. For local derivative signs, equality of baseline means and their first-order responses is enough. A sufficiently small positive gap in managerial weights is permitted in part 1. Strict log-supermodularity holds for $\delta > 0$. To extend local derivative signs, rather than comparisons for a fixed shock, use the additional continuous-derivative condition in the proposition. Finite-shock continuity alone is insufficient for this stronger conclusion. No quantitative size for these neighborhoods is established.

D.6 Composition feedback

Let $n_{Ri} = L_i(\mathcal{R})/L_i$ be the within-sector routine employment share, $\lambda_m = L_m/(L_m + L_s)$, and n_R the aggregate routine share. Then

$$\dot{n}_R = \lambda_m \dot{n}_{Rm} + (1-\lambda_m)\dot{n}_{Rs} + (n_{Rm} - n_{Rs})\dot{\lambda}_m. \tag{78}$$

Lemma 3 (Composition effect) *If manufacturing's employment share falls and $n_{Rm} > n_{Rs}$, the between-sector term in* (78) *is negative. If services has a higher managerial employment share, the corresponding between-sector contribution to the aggregate managerial share is positive.*

Proof *The first statement follows directly from* (78)*; the second applies the same identity to managers. At the homogeneous-worker benchmarks, $n_{Ri} = \theta_i r_i/T_i$. Since θ_i/T_i increases in θ_i, $r_m > r_s$ and $\theta_m \geq \theta_s$ imply $n_{Rm} > n_{Rs}$. The managerial share is $\phi(1-\theta_i)/T_i$, which decreases in θ_i. It is therefore higher in services in the ordered-weight example with $\eta_s > \eta_m$. At the equal-weight benchmark the managerial composition term is zero. The strict routine-share ordering persists nearby for that example. In the broader*

exact-condition cases, the routine and managerial composition orderings must be checked separately. ∎

This decomposition is evaluated at the full equilibrium response. It does not establish a comparison with an alternative equilibrium holding sector shares fixed.

E Quantitative Model and Numerical Details

With discrete tasks, it must be that the marginal product of the threshold worker is equalized between tasks:

$$MPT_{i0} \cdot \frac{(1-\alpha)T_{i0}}{L_{i0}} = MPT_{i1} \cdot \frac{(1-\alpha)T_{i1}}{\bar{h}_1 L_{i1}} \cdot \hat{h}_1,$$
$$MPT_{ij} \cdot \frac{(1-\alpha)T_{ij}}{(\bar{h}_j - \chi_j)L_{ij}} \cdot (\hat{h}_{j+1} - \chi_j) = MPT_{ij+1} \cdot \frac{(1-\alpha)T_{i,j+1}}{(\bar{h}_{j+1} - \chi_{j+1})L_{i,j+1}} \cdot (\hat{h}_{j+1} - \chi_{j+1})$$

using Assumption 3, and L_{ij} is the measure of workers in sector i, task j and $\bar{h}_j$ is their mean raw worker skill. For $j \geq 1$, effective worker skill is $H_{ij} = (\bar{h}_j - \chi_j)L_{ij}$; the manual task instead has $H_{i0} = L_{i0}$. We retain the representative within-task allocation with equal mean raw skills across sectors $i \in \{m, s\}$. This equality is automatic when tasks are a continuum. Then

$$\hat{h}_1 = \frac{\bar{h}_1 L_{i1}}{\pi_{i1} L_{i0}}, \qquad \frac{\hat{h}_{j+1} - \chi_{j+1}}{\hat{h}_{j+1} - \chi_j} = \frac{\pi_{ij}(\bar{h}_{j+1} - \chi_{j+1})L_{i,j+1}}{\pi_{i,j+1}(\bar{h}_j - \chi_j)L_{ij}}, \tag{79}$$

where π_{ij} is the discrete version of (9), and can be expressed using (79) as

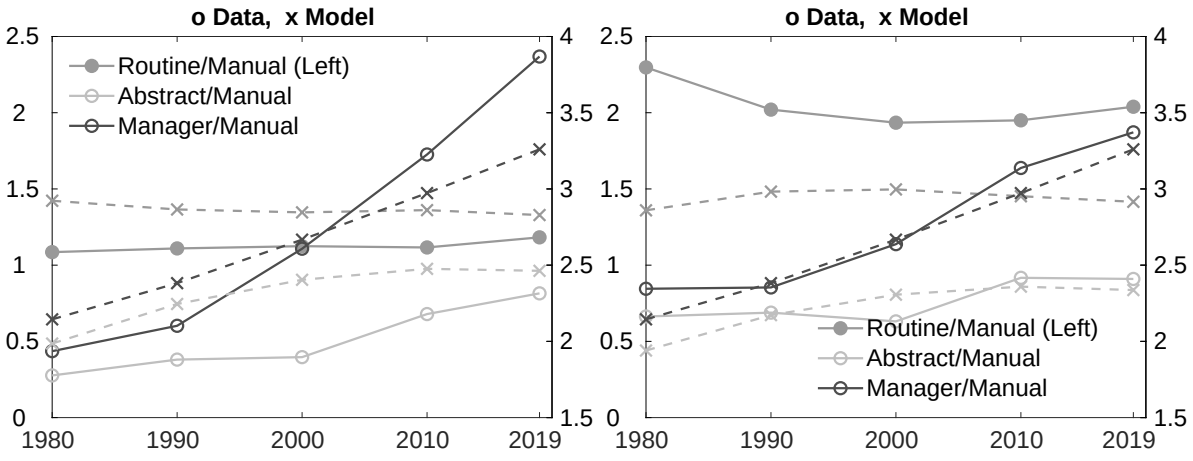
$$\pi_{i1} = \frac{\nu_{i1}}{\nu_{i0}} \cdot \left(\frac{M_1}{M_0} \cdot \hat{h}_1^{1-\alpha}\right)^{\sigma-1}, \quad \frac{\pi_{i,j+1}}{\pi_{ij}} = \frac{\nu_{i,j+1}}{\nu_{ij}} \cdot \left[\frac{M_{j+1}}{M_j}\left(\frac{\hat{h}_{j+1} - \chi_{j+1}}{\hat{h}_{j+1} - \chi_j}\right)^{1-\alpha}\right]^{\sigma-1}. \tag{80}$$

In equilibrium, indifference across tasks for threshold workers imply

$$w_0 = w_z \hat{z} = w_1 \hat{h}_1, \quad w_j(\hat{h}_{j+1} - \chi_j) = w_{j+1}(\hat{h}_{j+1} - \chi_{j+1})$$
$$\Rightarrow \quad w_z/w_0 = 1/\hat{z}, \quad w_1/w_0 = 1/\hat{h}_1, \quad w_{j+1}/w_j = \frac{\hat{h}_{j+1} - \chi_j}{\hat{h}_{j+1} - \chi_{j+1}}. \tag{81}$$

which is used to calibrate the skill distribution and requirement profile at each endpoint, as described in Section 3.3. The remaining calibration has the following blocks:

1. Choose candidate common elasticities (σ, ω). Their values apply to both the 1980 and 2019 equilibria.
2. Conditional on the 1980 distribution, requirements, and threshold vector $\mathbf{x}_{1980}$, recover the sectoral task weights from 1980 within-sector occupational employment shares. Normalize all initial task productivities to the common level M. Choose (M, A_m) to match output per worker and the manufacturing employment share, with $A_s = 1$.
3. Retain these task weights and sector-specific productivities in 2019. Use the separately calibrated 2019 distribution, requirements, and threshold vector $\mathbf{x}_{2019}$, together with observed capital per worker.



(a) Manufacturing (b) Services

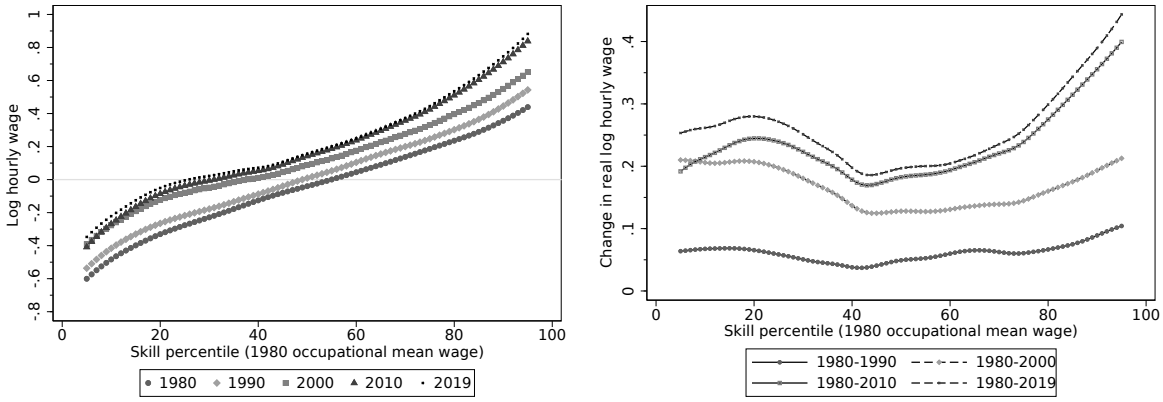
Fig. F.1: Data vs. Model, Relative Occupational Wages by Sector, 1980–2019.

Routine, abstract, and manager mean wages are divided by the manual mean wage within the same sector and year. Circles denote data and crosses denote the model. The routine ratio uses the left axis; the abstract and manager ratios use the right axis. The broad occupation groups are those in Table 6; these panels complement the aggregate wage fit in Figure 10.

Choose the 11 task-productivity growth rates $\{m_z, m_0, \ldots, m_9\}$ to fit output growth and within-sector occupational employment shares. The endpoint task productivities are $M_j(2019) = M(1 + m_j)^{39}$.

4. Update the common elasticities jointly with the growth rates to fit the change between the two endpoints. Relative-wage targets discipline (a, γ_h) and the requirement profile at each date. Tables 5 and 6 report the resulting 1980–2019 benchmark.

F Tables and Figures Not in Text



(a) Wage Profiles

(b) Cumulative Wage Growth

Fig. F.2: Occupational Wage Profiles and Growth by Initial Wage Rank, 1980–2019.
Occupations are ranked by their 1980 mean hourly wage, using employment-weighted skill percentiles. Panel (a) shows log mean hourly wages relative to the 1980 mean across percentile bins, for 1980, 1990, 2000, 2010, and 2019. Panel (b) shows cumulative changes from 1980 to each subsequent date, rather than separate decade changes. Occupational means use labor-supply weights; the profiles use the same fixed initial rank and LOWESS bandwidth of 0.4 as the endpoint wage comparison. These are changes across occupation ranks, not longitudinal wage changes for individual workers.

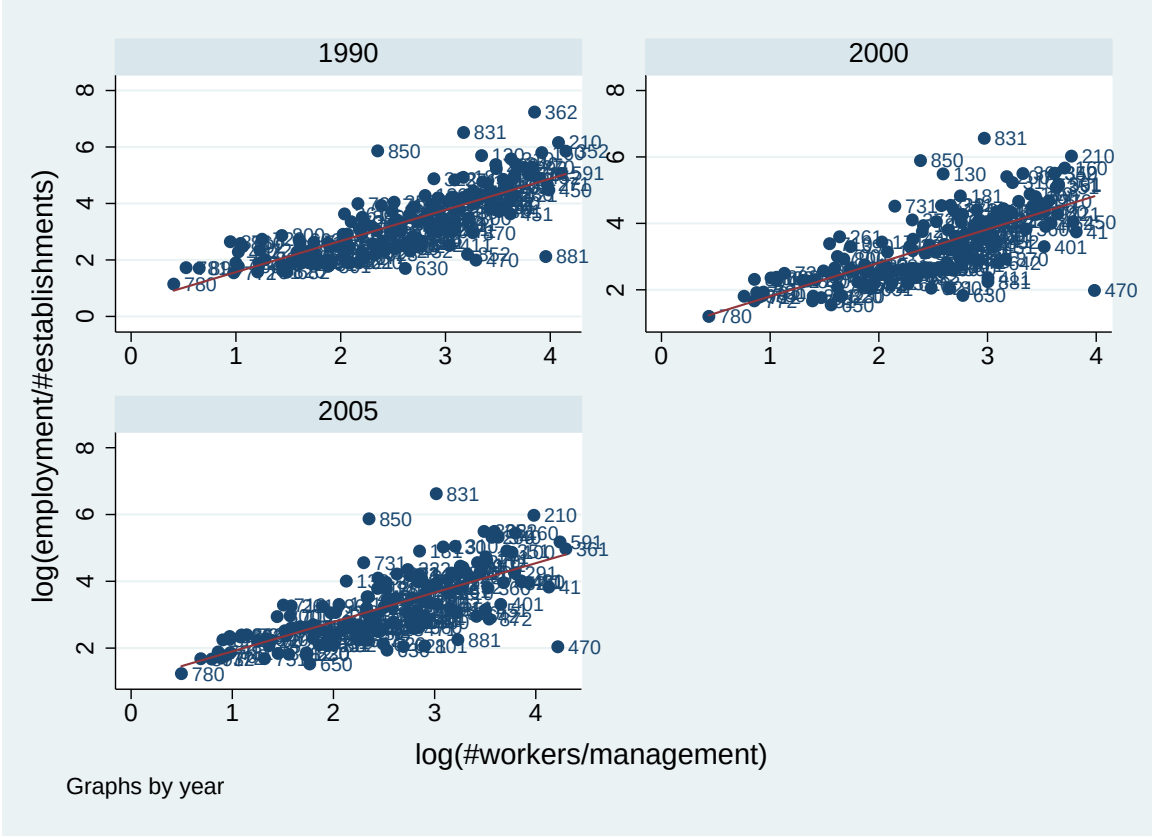


Fig. F.3: Establishment Size and Managerial Span of Control.
Cross-industry comparison of log employment per establishment (vertical axis, SUSB) and log workers per manager (horizontal axis, Census), by year. The narrative slides report 170 industries. These descriptive associations do not establish a one-to-one mapping between managers and establishments.

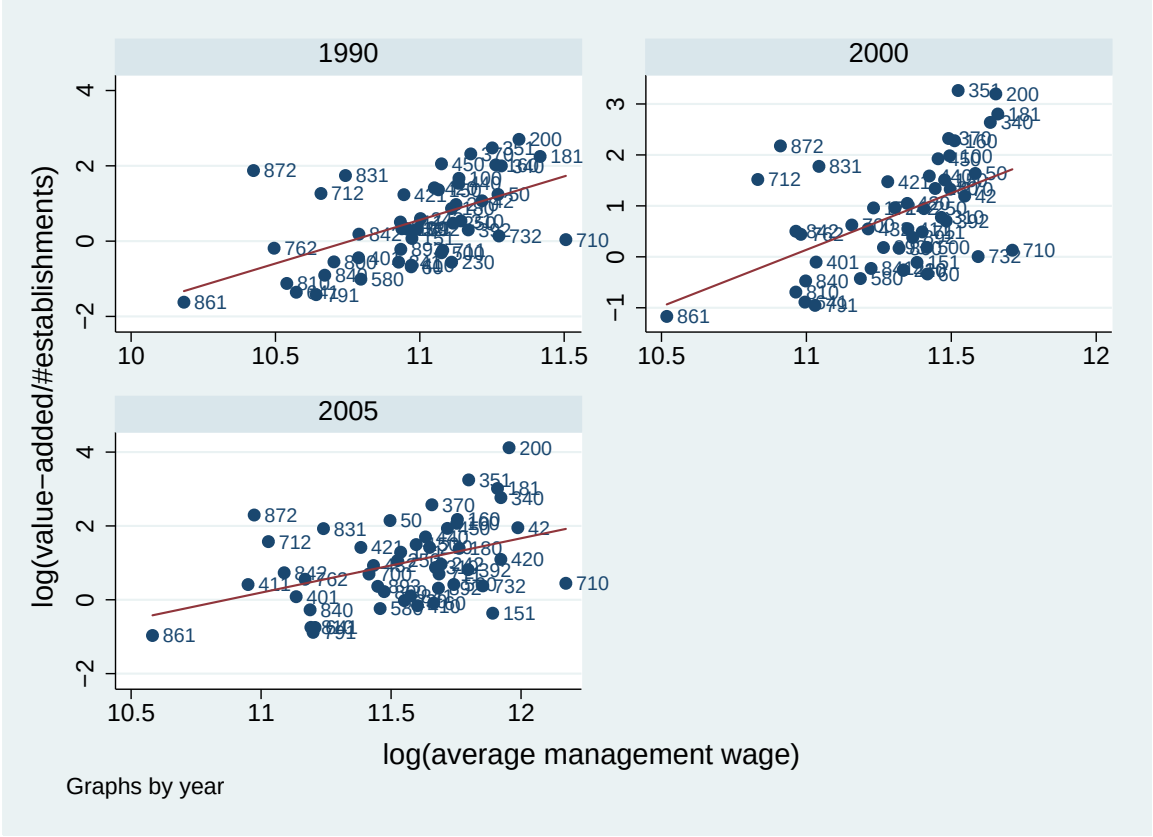


Fig. F.4: Establishment Output and Manager Compensation.
Cross-industry comparison of log value added per establishment (vertical axis, BEA-based measure) and log average managerial wages (horizontal axis, Census), by year. The narrative slides report 48 industries. The comparison is descriptive.

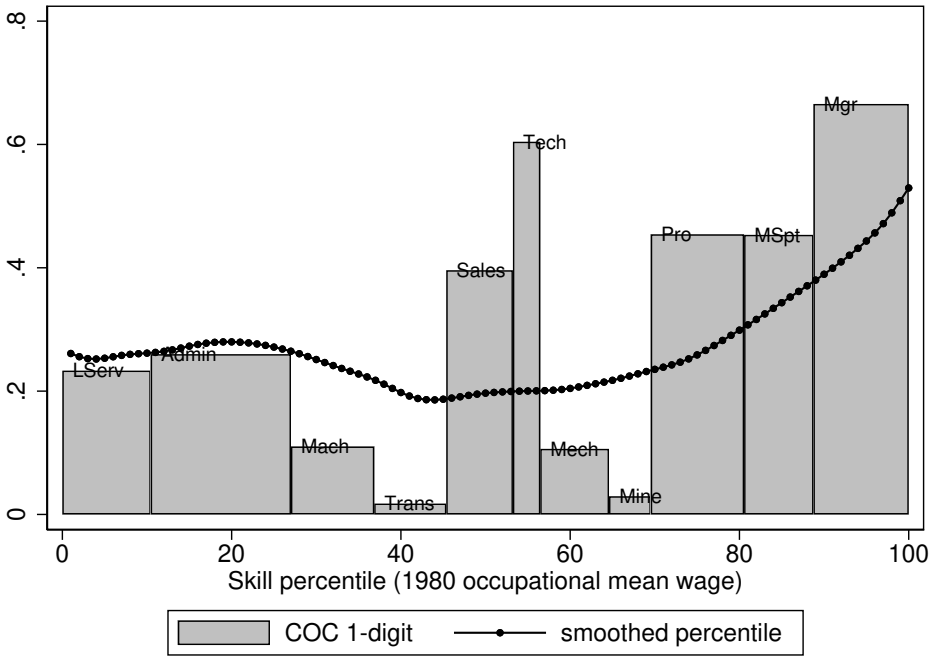


Fig. F.5: Occupational Wage Growth, 1980–2019.

Source: U.S. Census (5%) and the 2019 ACS. Occupations are ranked by their 1980 mean wage; bars represent the 11 broad occupation groups and the line smooths across the 310 detailed occupations. This growth panel complements the wage levels in Figure 1(b). Figure F.2 retains the intermediate cumulative comparisons along the initial wage ranking.

Ranked by mean wage (except management)	COC Code	Relative Wages 1980	2019
Low Skill Services	400	**0.65**	**0.53**
Middle Skill		**0.90**	**0.76**
Administrative Support	300	0.78	0.65
Machine Operators	700	0.84	0.61
Transportation	800	0.88	0.58
Sales	240	0.94	0.90
Technicians	200	1.04	1.22
Mechanics & Construction	500	1.06	0.76
Miners & Precision Workers	600	1.09	0.72
High Skill		**1.26**	**1.27**
Professionals	40	1.21	1.23
Management Support	20	1.32	1.34
Management	1	**1.39**	**1.74**

(average wage in each year normalized to 1)

Table F.1: Occupational Relative-Wage Targets, 1980–2019

Source: 1980 Census and 2019 ACS; occupation definitions follow Appendix A. Mean wages are normalized by the overall mean wage in each year. Broad middle- and high-skill rows summarize the underlying occupation groups. Employment targets are shown in Figures 9 and 4.

ΔAvg Wage	(1)	(2)	(3)
TFP Growth	-6.233**		-2.974*
	(0.945)		(0.921)
Δ_χ		1.377**	0.886**
		(0.190)	(0.184)
Constant	0.395**	0.111*	0.224**
	(0.038)	(0.040)	(0.044)
R^2	0.756	0.822	0.889

Δ_χ	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
RTI (DOT)	-0.053							
	(0.045)							
R-Manual		-0.109**				-0.064*		-0.053
		(0.025)				(0.020)		(0.029)
NR-Man Pers			0.093*					
			(0.030)					
NR-Cog Anal				0.113**		0.074*	0.086**	0.068*
				(0.026)		(0.024)	(0.025)	(0.028)
ΔCol Share					1.311**		0.642	0.300
					(0.340)		(0.303)	(0.411)
Constant	0.139*	0.139**	0.139**	0.139**	-0.011	0.139**	0.065	0.105
	(0.044)	(0.033)	(0.037)	(0.031)	(0.044)	(0.028)	(0.043)	(0.066)
R^2	0.126	0.527	0.383	0.568	0.408	0.687	0.635	0.698

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$

Table F.2: Effective Skills, Wage Growth, and Task Content, 1980–2019

The first panel relates log occupational mean-wage growth to task-productivity growth and Δ_χ; the second relates Δ_χ to task content and the change in the college share. Both panels use the benchmark calibration and the signed log effective-skill index defined in Section 4, matching the figures. There are eleven occupation groups, including the managerial-skill statistic rather than a worker requirement in the manager slot. Regressions are unweighted OLS with heteroskedasticity-robust standard errors (HC1). Task indices are standardized across the eleven groups; the college-share change is in levels. Wages are calibration targets, so their association with the inferred index is not independent validation. The task-content associations provide additional descriptive evidence.

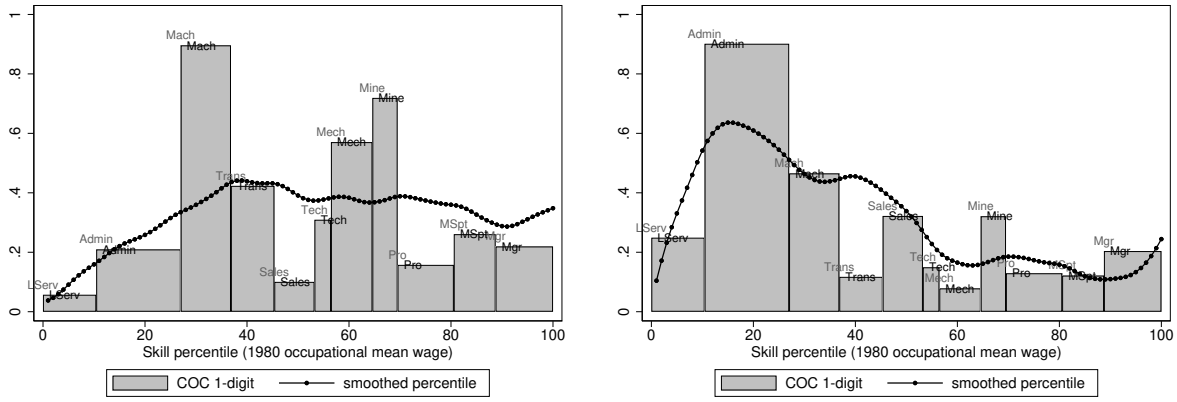
ΔEmp Share	(1)	(2)	(3)
TFP Growth	-11.304**		-16.260**
	(1.895)		(2.088)
Δ_χ		1.340	-1.347**
		(0.717)	(0.340)
Constant	0.177*	-0.177	0.438**
	(0.074)	(0.159)	(0.078)
R^2	0.834	0.261	0.937

TFP Growth	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
RTI (DOT)	0.019							
	(0.009)							
R-Manual		0.026**				0.028	0.022*	0.022*
		(0.005)				(0.016)	(0.009)	(0.007)
NR-Man Pers			-0.023**			0.003		
			(0.004)			(0.015)		
NR-Cog Anal				-0.019*			-0.006	
				(0.006)			(0.009)	
ΔCol Share					-0.271**			-0.070
					(0.077)			(0.096)
Constant	0.015	0.015*	0.015	0.015	0.046**	0.015*	0.015*	0.023
	(0.008)	(0.006)	(0.007)	(0.008)	(0.013)	(0.006)	(0.006)	(0.011)
R^2	0.352	0.658	0.525	0.374	0.388	0.659	0.683	0.673

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$

Table F.3: Task Productivity, Employment, and Task Content, 1980–2019

The first panel relates proportional occupational employment-share growth to task-productivity growth and Δ_χ. The second relates task-productivity growth to task content and changes in the college share. Both panels use the benchmark calibration and eleven occupation groups, with the same index, sample, standardization and unweighted OLS/HC1 conventions as Table F.2. Statistical associations do not identify causal effects.

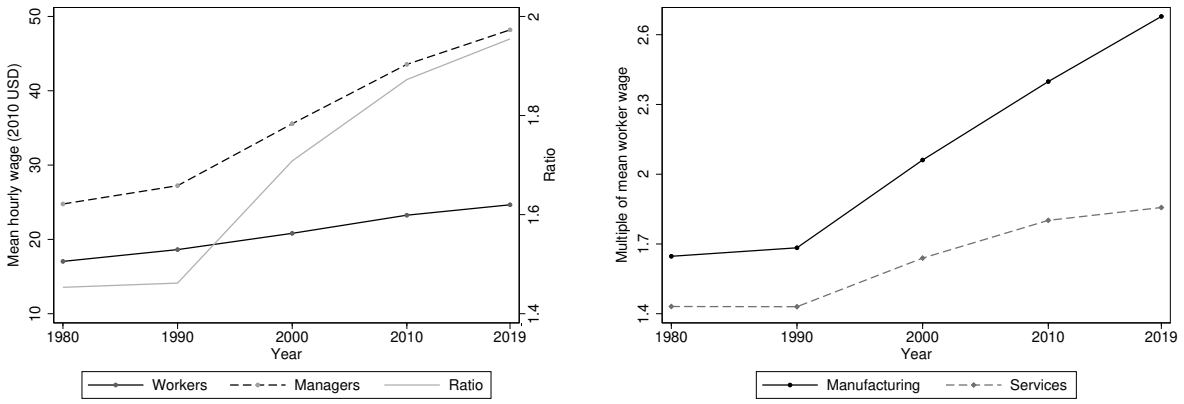


(a) Manufacturing

(b) Routinization

Fig. F.6: Manufacturing Employment Shares and Routine Job Shares

Source: U.S. Census (5%). Left: Manufacturing employment share by occupation-skill percentile in 1980. Right: Share of top employment-weighted third of occupations in terms of RTI by skill percentile, replicates Autor and Dorn (2013) who construct RTI from detailed task requirements by occupation in DOT. Occupations are ranked by their 1980 mean wage for 11 one-digit groups and smoothed across 322 three-digit groups, separately. The x-axis units are in percent share of employment. Further details in text and Appendix A.

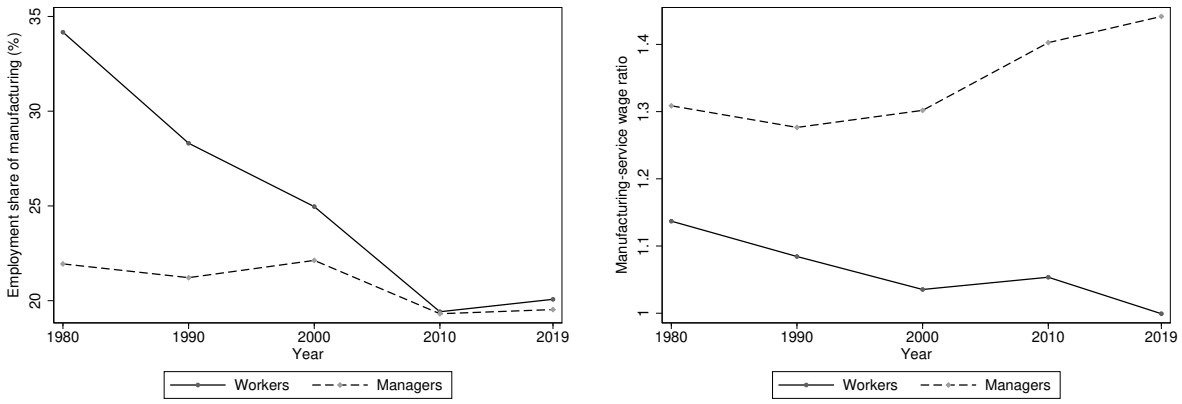


(a) Aggregate Wage Levels and Ratio

(b) Manager Wage by Sector

Fig. F.7: Relative Manager Wages

Source: U.S. Census (5%). Left: levels and ratio of mean wages or managers and all other workers in aggregate. Right: relative mean wage of managers over all other workers within manufacturing and services. "Manufacturing" combines manufacturing, mining and construction, and services subsumes service and government. See Appendix A for how we define management in the census and Figure A.1 for a detailed breakdown of the manager group.

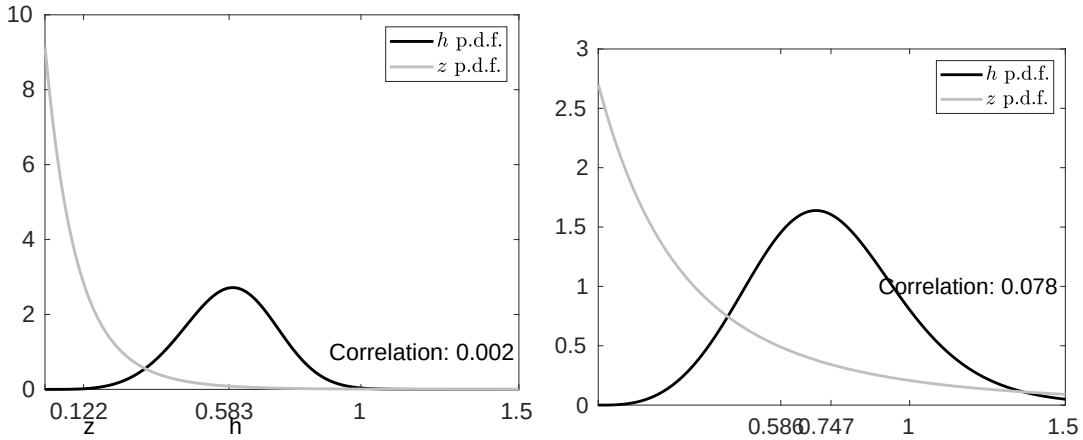


(a) Manufacturing Employment Share

(b) Manufacturing-Services Average Wage Ratio

Fig. F.8: Manufacturing vs. Services by Occuaption

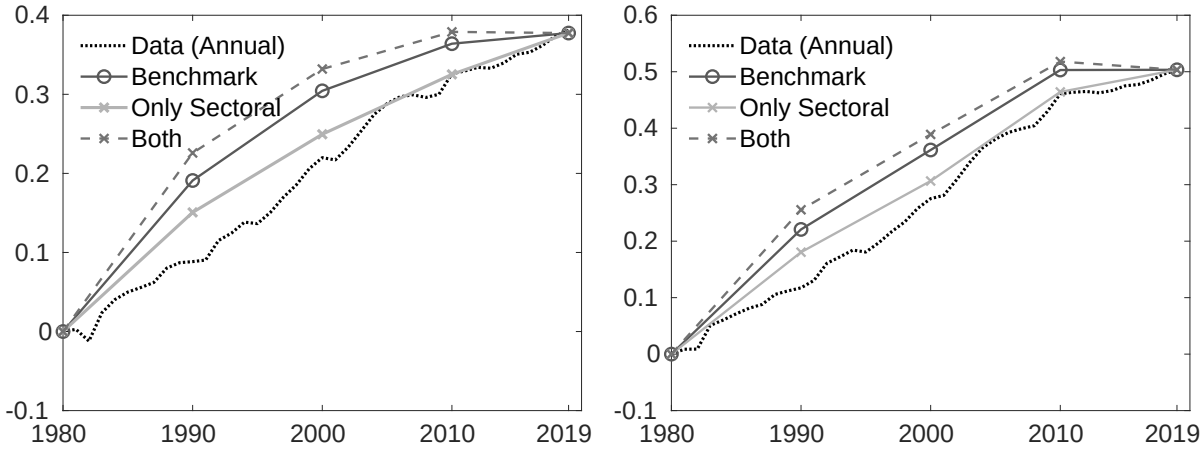
Source: U.S. Census (5%). Left: manufacturing employment share within the manager occupation group and all other workers. Right: mean wage of manufacturing employment relative to services employment within the manager occupation group and all other workers. "Manufacturing" combines manufacturing, mining and construction, and services subsumes service and government. See Appendix A for how we define management in the census and Figure A.1 for a detailed breakdown of the manager group.



(a) 1980 (b) 2019

Fig. F.9: Calibrated Skill Distribution

We use a type IV bivariate Pareto distribution to model the distribution over worker and manager skills (h, z). The figure depicts the marginal distributions of each skill, and also their mean below the x-axis.

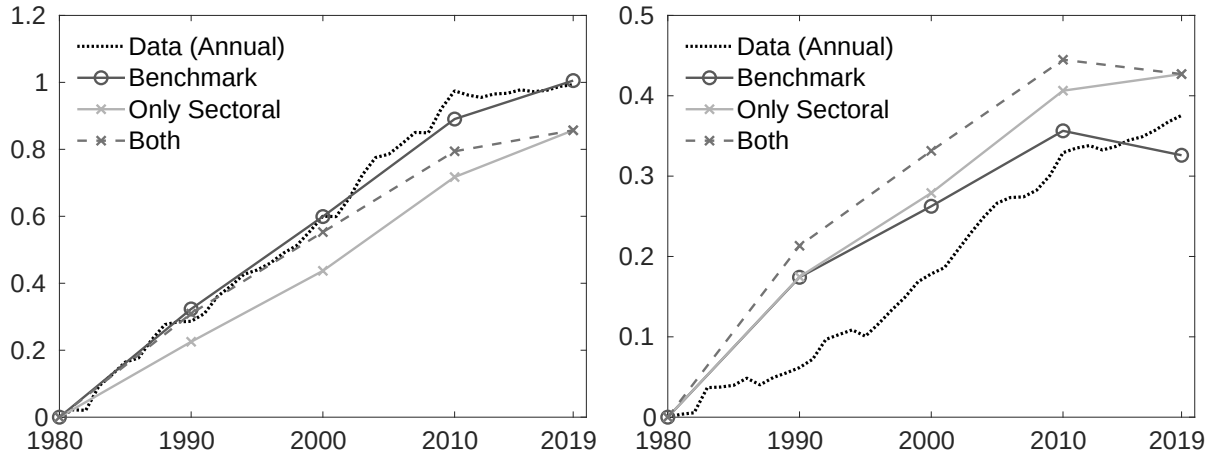


(a) Log TFP Growth

(b) Log GDP per Worker Growth

Fig. F.10: Aggregate Output and TFP Growth

Data: NIPA. Log 1980 levels are normalized to 0, so the slopes of the lines are the growth rates.

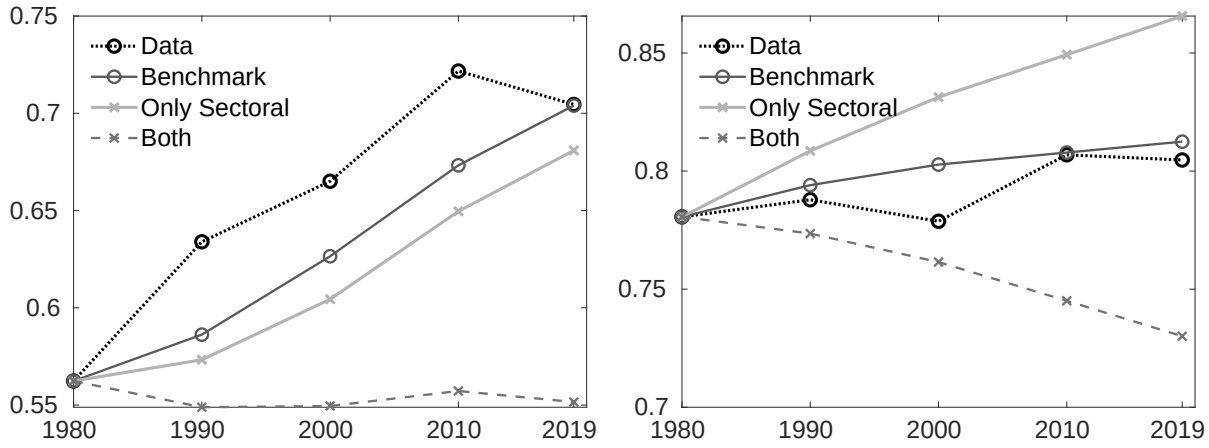


(a) Log GDP per worker, Manufacturing

(b) Log GDP per worker, Services

Fig. F.11: Benchmark vs. Counterfactuals, GDP per Worker

Data: NIPA. "Manufacturing" combines manufacturing, mining and construction, and services subsumes service and government. Log 1980 levels are normalized to 0, so the slopes of the lines are the growth rates.



(a) Services Employment Share, Routine Jobs

(b) Services Employment Share, Managers

Fig. F.12: Benchmark vs. Counterfactuals, Structural Change

Vertical axes are the fractions of routine jobs (left) and managers (right) working in services.

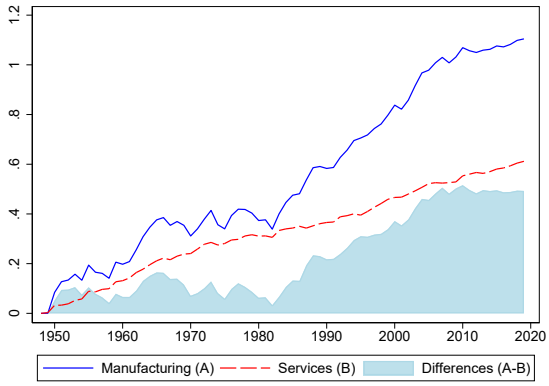
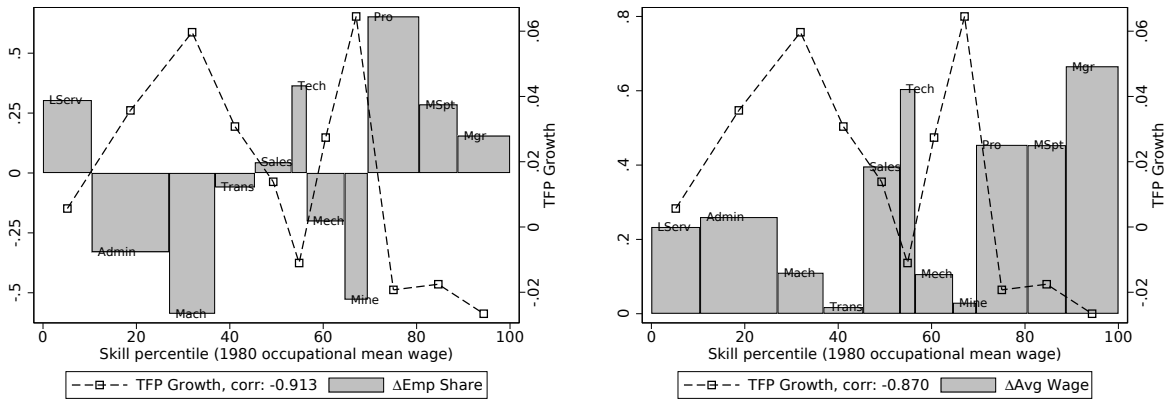


Fig. F.13: Log Sectoral TFP, 1947–2019.

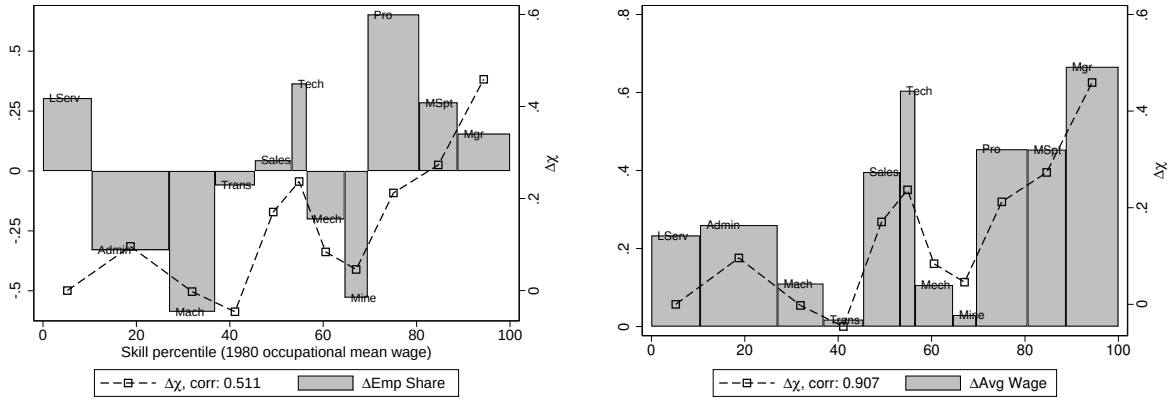
Source: BEA NIPA accounts. Sectoral output and capital are computed via cyclical expansion from the industry accounts as in Herrendorf et al. (2013). Employment is based on full-time equivalent number of persons in production in NIPA Table 6. Within each sector, TFP is measured as the Solow residual given a capital income share of 0.363, and log-TFP's are normalized to 0 in 1947.



(a) Employment

(b) Wages

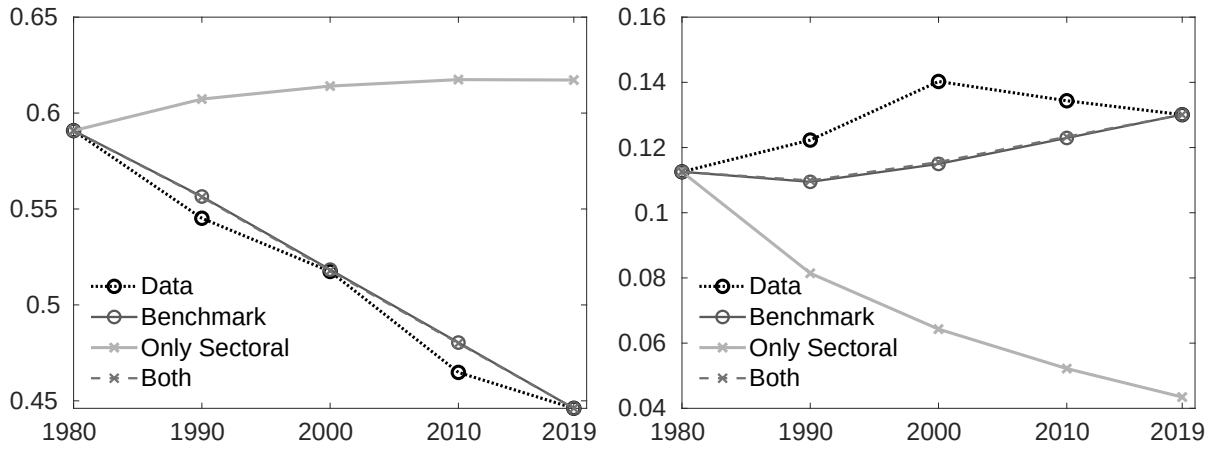
Fig. F.14: Task Productivity Growth, Employment Change, and Wage Growth.
The dashed line shows calibrated annual task-productivity growth on the right axis in both panels. The bars show the proportional change in occupational employment shares in panel (a) and the log change in occupational mean wages in panel (b), on the left axis. Occupations follow their 1980 wage ranking; the plotted correlations describe associations across occupations.



(a) Employment

(b) Wages

Fig. F.15: Skill-Requirement Changes, Employment Change, and Wage Growth.
The dashed line shows the calibrated requirement-change measure labelled $\Delta\chi$ on the right axis in both panels. The bars show the proportional change in occupational employment shares in panel (a) and the log change in occupational mean wages in panel (b), on the left axis. Occupations follow their 1980 wage ranking; the plotted correlations describe associations across occupations.



(a) Aggregate Employment Share, Routine Jobs

(b) Aggregate Employment Share, Managers

Fig. F.16: Benchmark vs. Counterfactuals, Polarization

Vertical axes are the aggregate employment shares of routine jobs (left) and managers (right).

Group	Raw exposure		Standardized		Rank	
	AIOI	β	AIOI	β	AIOI	β
Low Skill Services	0.641	0.165	-1.114	-0.736	9	7
Administrative Support	0.699	0.486	1.064	1.243	2	1
Machine Operators	0.652	0.075	-0.725	-1.291	8	11
Transportation	0.633	0.115	-1.418	-1.043	11	9
Sales	0.679	0.436	0.320	0.935	6	3
Technicians	0.690	0.396	0.734	0.689	4	4
Mechanics and Construction	0.641	0.096	-1.140	-1.156	10	10
Miners and Precision	0.657	0.150	-0.524	-0.829	7	8
Professionals	0.695	0.395	0.916	0.682	3	5
Management Support	0.705	0.456	1.299	1.058	1	2
Model managers	0.686	0.357	0.588	0.447	5	6

Table F.4: Broad-AI and LLM Exposure by Model Group.

AIOI is the retained Felten et al. measure; β is the human rating of Eloundou et al. Raw indices have different units. Standardized scores subtract the equal-group mean and divide by the sample standard deviation across all eleven groups, before the manual-loading override; rank 1 denotes greatest exposure. Professionals and management support are abstract-worker groups; model managers include self-employed people. The old raw means use the retained aggregates. New raw means use matched labor-supply weights after the official O*NET 2010–2019 crosswalk, with equal weights across unique rated descendant occupations within SOC and the established broad-code/family rules. Unmatched scores remain missing. Coverage in table order is 100.00, 98.31, 98.57, 99.67, 100.00, 100.00, 99.58, 100.00, 99.74, 100.00 and 99.90 percent, or 99.60 percent overall. On common matched support the largest shift in an old raw group mean is 0.000078; Pearson correlation changes from 0.937885 to 0.937652 and Spearman stays 0.845455.

F.1 AI exposure and scenario comparison

The comparison in Section 5.1 retains the broad-AI scenario and changes only the exposure measure and its required GDP-normalization intercept. The tables and figures below separate exposure diagnostics from conditional model outcomes.

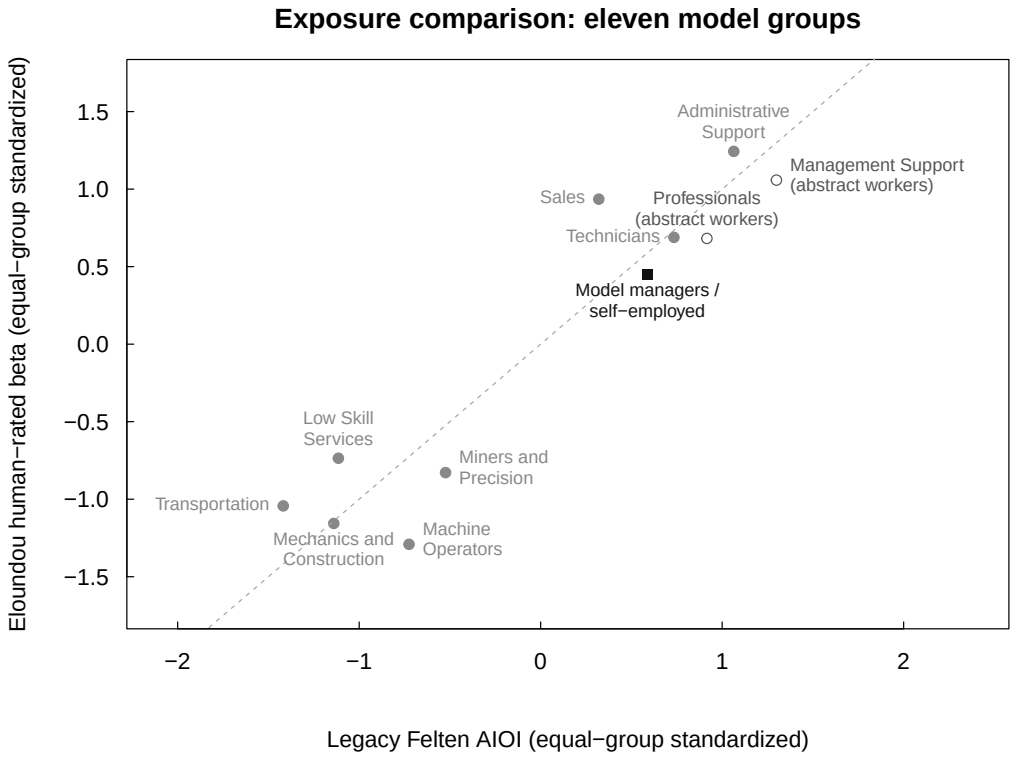


Fig. F.17: Comparison of Standardized AI Exposure Measures.

Eleven model groups, including the separate manager group, before the manual-loading override. The dashed line is $y = x$; equal standardized scales do not imply equal capability units. The group-level Pearson and Spearman correlations are 0.937885 and 0.845455; positive affine standardization leaves both unchanged. On a common sample of 471 Census-2010 occupations, before manager/self-employed reassignment, unweighted Pearson and Spearman correlations are 0.840048 and 0.849401. Using normalized 2010 labor-supply weights gives Pearson 0.837980 and weighted rank correlation 0.839163. Weighted Pearson is weighted covariance divided by weighted standard deviations; the rank statistic applies this formula to common-sample average-rank vectors, one convention for weighted rank correlation. Each occupation enters once. Detailed matched coverage is 99.60 percent; the by-group figures are in Table F.4.

Date	Path	GDP	Manual	Routine	Abstract	Manager
A. GDP (model units) and employment shares (percent)						
2030	Continuation	1.7031	13.23	39.16	33.81	13.80
2030	Broad AI (AIOI)	1.7031	13.35	39.90	31.63	15.12
2030	LLM exposure (β)	1.7031	13.17	38.48	33.09	15.26
2050	Continuation	1.5866	11.35	30.95	42.91	14.79
2050	Broad AI (AIOI)	1.5747	11.67	31.81	38.17	18.35
2050	LLM exposure (β)	1.5691	11.36	30.02	39.97	18.65
B. Occupational mean earnings relative to economy-wide mean						
2030	Continuation	—	0.529	0.726	1.136	1.897
2030	Broad AI (AIOI)	—	0.523	0.710	1.213	1.739
2030	LLM exposure (β)	—	0.530	0.734	1.151	1.748
2050	Continuation	—	0.548	0.682	0.946	2.169
2050	Broad AI (AIOI)	—	0.530	0.663	1.073	1.732
2050	LLM exposure (β)	—	0.542	0.697	1.003	1.760

Table F.5: Conditional AI Scenario Outcomes, 2030 and 2050.

Continuation and broad AI use the saved matching October implementation; LLM exposure uses the additional human-rated β scenario. GDP is aggregate model output. Employment shares refer to persons, while relative mean earnings equal each group's wage-bill share divided by its employment share. Routine combines the seven middle-skill worker occupations; abstract combines professionals and management support. Both exposure scenarios match continuation GDP in 2030, not in 2050. In 2050, new minus broad-AI employment-share differences are −0.31, −1.79, +1.81, +0.30 percentage points for manual, routine, abstract and managers; relative mean-earnings differences are +2.42, +4.97, −6.71, +1.60 log points. These are differences between conditional scenarios, not causal effects or adoption forecasts.

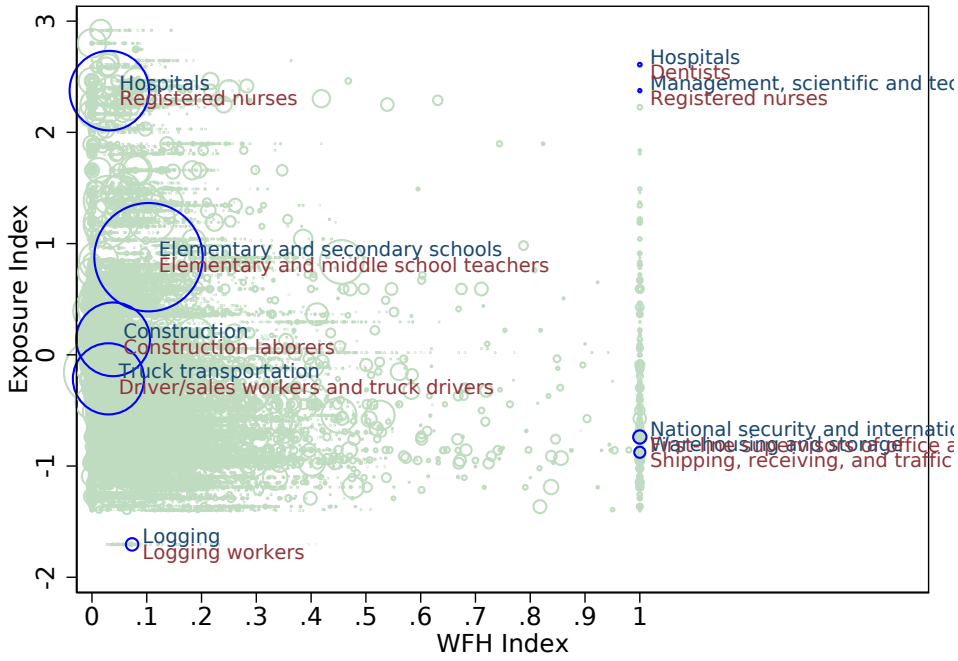


Fig. F.18: Infectious Exposure and Work-From-Home Indices.
Occupation and industry measures of infectious exposure and the feasibility of working from home, as used in the narrative slides. Circle sizes reflect employment. These indices motivate the counterfactual scenarios; they do not measure realized post-pandemic productivity growth.

F.2 Work-from-home illustration

The figures report the exposure measures and 2019–2050 scenarios summarized in Section 5.4. The comparison holds the capital path common across scenarios and matches GDP in 2030; the sectoral and occupational compositions may differ. The direction of the productivity shift follows infectious exposure or the negative of the WFH index. Skill distributions and requirements follow the continuation path.

F.3 Polarization or Structural Change?

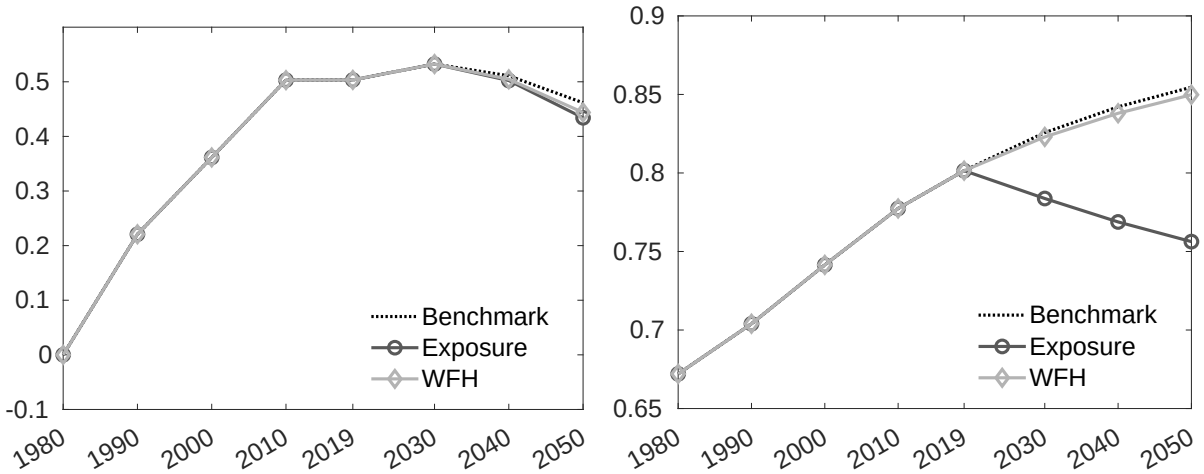
One may wonder whether it is not TSTC leading to structural change, but growth in sector-specific productivities leading to polarization, considering Lemma 3 in isolation.

To isolate the between-sector composition channel, hold within-sector occupational shares and relative average earnings fixed. Under this conditional calculation, a rise in manufacturing productivity A_m raises $\log \kappa$ at rate $1-\epsilon$. Thus the direct composition contribution is

$$\left.\frac{d\log L(j)}{d\log A_m}\right|_{\text{composition}} = (1-\epsilon)\left[\frac{L_s(j)}{L(j)}L_m - \frac{L_m(j)}{L(j)}L_s\right]. \tag{82}$$

It is negative for occupations with a higher within-sector employment share in manufacturing. This is the occupation-specific version of the composition term in (78); it is not the full equilibrium derivative when assignment and occupational selection respond.

For a finite comparison, let $s_{mR,1980}$ and $s_{sR,1980}$ denote routine employment shares within manufac-



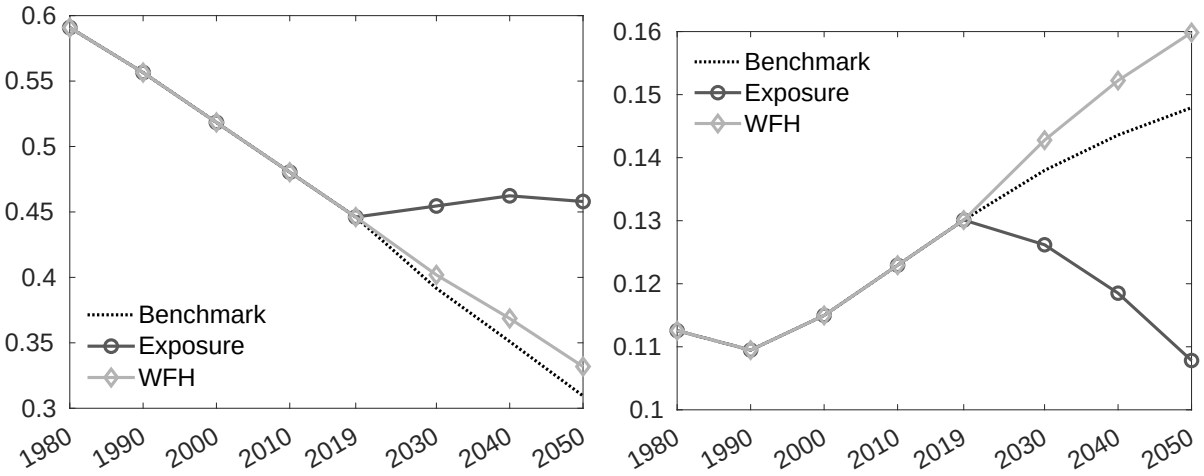
(a) log GDP (b) Structural Change

Fig. F.19: Post-Covid Counterfactuals: Output and Structural Change

turing and services. Holding those compositions fixed gives

$$\Delta L_R^{\text{composition}} = (L_{m,2019} - L_{m,1980})(s_{mR,1980} - s_{sR,1980}).$$

The 1980 shares in the occupational tabulation give $s_{mR,1980} = 25.86/32.79$ and $s_{sR,1980} = (59.09 - 25.86)/(100 - 32.79)$. Using the manufacturing shares 32.79 and 19.86 percent from the dated comparison in Table 10, the composition contribution is approximately -3.80 percentage points. The reported aggregate routine shares fall from 59.08 to 44.61 percent, or 14.47 percentage points. Thus this fixed-composition calculation accounts for about 26 percent of the change; rounding explains the 0.01-point difference in the 1980 routine totals. This is an accounting comparison, not a bound on the equilibrium effect of sector-specific technology.



(a) H. Polarization (b) V. Polarization

Fig. F.20: Infectious Exposure and Work-From-Home Counterfactuals